

Greedy Packing of Nested Rings: The Golden Threshold, Placement Rules, and a Tribonacci Floor

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Abstract

We study packings of annuli of a common width, allowing each ring to nest inside the hole of a larger one. The objectives of maximizing contact area and the number of placed rings diverge: area is superadditive in the radius, cardinality is not. Under superincreasing radii, every descending greedy maximizes every positive, strictly increasing, superadditive objective. More strongly, any choice among feasible containers yields the lexicographically maximal feasible set, for containers of arbitrary shape in every dimension. This placement irrelevance holds unconditionally for at most three rings and fails at four in disks and squares; twin instances exclude every universal rule based only on the observable state.

Write $\rho = \max_i(\sum_{j>i} r_j)/r_i$. The additive model has threshold exactly 1. For disks we prove the exact global threshold $\tau = \varphi$, with no failure at $\rho \leq \varphi$, for every finite inventory, even with independent hole radii. The key geometric theorem states that, under golden tail bounds, an entire disk list fits a circular container if and only if its three largest disks fit. It supplies a uniform exchange of parents, matching the four-ring golden counterexamples approaching φ from above. The Tribonacci constant $T \approx 1.83929$ remains the exact floor of a rigid subfamily.

A dimension-reduction lemma transfers spherical sharpness results to all dimensions $d \geq 2$, and a separate argument proves the golden threshold for at most five rings in those dimensions. For square pans, a Cartesian confinement criterion gives twins and a family proving $\tau_{\square} \leq Y \approx 1.684487745872346$; its optimality is open. For independent holes, the exact universal area guarantee under $\rho \leq \kappa < 1$ is $\min(1, \kappa^{-2} - 1)$, with optimality threshold $1/\sqrt{2}$. The repository contains 122 Lean algebraic and Cartesian theorems. Euclidean geometry, forest assembly and continuity remain written proofs; numerical checks do not substitute for them.

1 Introduction

Drop a batch of squid rings into a frying pan and a natural optimization problem appears: place as many rings as possible flat on the pan – or, if one cares about searing, maximize the total ring–pan contact area – exploiting the fact that a sufficiently small ring fits inside the hole of a larger one. Abstracted, this is the problem of packing annuli of common width into a disk with

*AGILabs. javiecija96@gmail.com. Computational verification scripts for every numerical claim, together with extended proofs and the verification reports cited in the appendix, are available in the accompanying repository, <https://github.com/JaviMaligno/calamares>. This version of the manuscript is the integrated v2 working revision of 15 September 2026, extending the archived release **v1-arxiv** of that repository; **python code/run_all.py --campaign** reproduces every verification in the map (band-by-band, several hours); **--full** covers the core-theorem scripts in about 35 minutes, and **lake build in lean/** rechecks the formalized certificates. **requirements.txt** pins the exact environment.

recursive nesting, a circular-container relative of the *Recursive Circle Packing Problem* (RCP) introduced by Pedrosa, Cunha and Tavares [2] to model the “telescoping” of tubes in shipping containers, and solved exactly by Gleixner, Maher, Müller and Pedrosa [4]. That literature is algorithmic: it packs a fixed set of rings into as few rectangular containers as possible, and its methods – like the greedy and GRASP procedures for RCP and the greedy heuristics for equal circles in a circular container [2, 3] – are *heuristics*: they propose placements without guarantees of optimality or of independence from the placement rule. Here we ask structural questions instead: which objectives the natural greedy algorithm *provably* optimizes, when the choice of *where* to put each ring is provably irrelevant, and which conditions on the radii govern the answers; to our knowledge, no comparable placement-independence guarantee appears in the circle-packing literature.

Our starting point is a classical phenomenon in one dimension. For bin packing with *divisible* item sizes, Coffman, Garey and Johnson [5] proved that First Fit Decreasing is optimal. We are not aware of an analogue for geometric packing of disks, let alone with nesting. We prove one, with *superincreasing* radii (each radius exceeding the sum of all smaller ones) playing the role of divisibility, and we determine exactly how far the hypothesis can be relaxed in the additive model, bracketing the geometric threshold. The answer is a surprise: the additive relaxation of the model has threshold exactly 1, while the disk model has threshold exactly the *golden ratio* – an explicit golden family fails at $\rho = \varphi + 3\varepsilon$. The Tribonacci constant – the natural conjectured threshold, which our own program long targeted – survives as the exact floor of the rigid nested family, the proved floor of a hierarchy of nested-exchange templates, and the conjectured universal floor of nested exchanges.

Contributions: main theorems. (1) A *superadditivity dichotomy*: under superincreasing radii the selection greedy is optimal for every positive, strictly increasing, superadditive value function of the radius – contact area is one – while cardinality (not superadditive) admits counterexamples even under superincreasing radii (Section 3). The selection statement is best seen as a general fact about downward-closed set systems with superincreasing weights, in the lineage of greedy analyses on independence systems [15, 16] and of the superincreasing knapsack [17]; the geometric content of this paper begins with the next item. (2) A *placement-obliviousness theorem*: under superincreasing radii, *any* rule for choosing among feasible containers yields the lexicographically maximal feasible set; the proof is container-shape- and dimension-agnostic (Section 4). (3) *Sharpness*: placement obliviousness holds unconditionally for $n \leq 3$ rings and fails at $n = 4$; twin instances with an identical prefix rule out every state-based placement rule, deterministic or randomized (Section 5). (The positive statement holds in every compact container and dimension; Section 11.1 transfers the spherical negative results to every $d \geq 2$.) (4) *Thresholds*: the additive model has universal threshold exactly $\rho = 1$; the geometric four-ring rigid subfamily has infimum exactly the Tribonacci constant T , proved with no idealization (Theorem 17, full proof in Appendix B); and an explicit *golden family* refutes the natural conjecture that the geometric threshold equals T : placement obliviousness fails at $\rho = \varphi + 3\varepsilon$ for every small $\varepsilon > 0$, which *proves* $\tau \leq \varphi < T$ (Theorem 19), and the family realizes every $\rho \in (\varphi, T)$ at a fixed width (Corollary 20). The matching universal guarantee is Theorem 22: every finite disk inventory with $\rho \leq \varphi$ is placement-oblivious, including independent holes. Its proof uses the three-largest-disks theorem, rather than a classification by inventory size or a computational coverage claim. The earlier four-ring proof is retained as a direct special-case analysis.

The status of the golden value, at a glance:

Statement	Status
$\tau \leq \varphi$	proved (golden family, Theorem 19)
$\tau_4 = \varphi$, infimum unattained	proved (Theorem 21)
Pan-exchange floor for pair profiles equals φ	proved (Theorem 86)
Universal $\tau = \varphi$, infimum unattained	proved (Theorem 22)

Specialized exchange floors. (5) *The width program*: the blocked-exchange infimum with relative width $\omega > 0$ is governed by the deformed Tribonacci cubic $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$; the full lower-bound curve is in closed form and its global minimum is exactly $13/7$, attained at a rational corner (Section 7); the combinatorial profile threshold is in closed form for every profile size, with golden plateau and exact Tribonacci crossing $\omega_T = 1/T - 1/2$. (6) *Generic containers*: an exact corona criterion for wall-tangent packings (with two corrective counterexamples and a $k = 5$ “pentagram” certificate), and a hierarchy of blocking walls – occupant holes, “blocking costs the slack”, and the double pocket – yielding metallic-mean floors (silver ratio $1 + \sqrt{2}$, golden line $\varphi^2 - (\varphi/2)\omega$) that push every blocked exchange with extra occupants above T – throughout the width convention $\omega \in (0, 1)$ of the program: for widths $\omega < 0.9626$ with one extra occupant, up to 0.624 and 0.896 with two and three, and for all $\omega \in (0, 1)$ with four or more (Section 8). The remaining gaps of the program are stated explicitly there and in Open Problem 43; they concern stronger template bounds, not the global golden guarantee. (7) A phase diagram for the divergence of the two objectives, and a decoupling of hardness into a geometric and a combinatorial layer, the latter with an explicit PARTITION reduction (Sections 9–10). (8) *Shape and dimension*: sibling queries involving k balls in a ball reduce to dimension $k - 1$. This preserves the sharpness results and the rigid Tribonacci floor for every spherical dimension $d \geq 2$. For square pans an exact quadrant-confinement criterion gives four-ring failures, state-indistinguishable twins, and a family proving $\tau_{\square} \leq Y \approx 1.684487745872346$ (Sections 11.1–11.3); these results make no claim of an exact global square threshold. (9) *Independent holes and area*: placement obliviousness under $\rho \leq 1$ remains valid for arbitrary compact containers, but contact area is no longer a common function of outer radius. We prove the exact factor $\min(1, \kappa^{-2} - 1)$ under $\rho \leq \kappa < 1$ and the sharp area-optimality threshold $1/\sqrt{2}$ (Section 11.5).

2 Model and preliminaries

Definition 1 (Rings and placements). Fix a width $w > 0$. A *ring* of outer radius $r > 0$ is an annulus with inner (hole) radius $\max(0, r - w)$; if $r \leq w$ it is a solid disk with no hole. Let $K \subset \mathbb{R}^d$ be a compact *container* (the *pan*; our motivating case is a disk of radius R in the plane, $d = 2$, but Theorems 6 and 9 hold for arbitrary K and d , with rings read as spherical shells). A *placement* of a subset S of the rings is a forest: each placed ring has as parent either the pan or a placed ring with $r_{\text{child}} \leq r_{\text{parent}} - w$ (it sits inside the parent’s hole); the children of a common parent must be packable as balls (by outer radius) with pairwise disjoint interiors inside the parent’s region — K itself, or the hole ball of radius $r - w$. Feasibility is a property of the assignment (siblings may be rearranged freely inside their container).

Definition 2 (Objectives; superincreasing radii). For radii $r_1 > r_2 > \dots > r_n$, the *contact area* of a placed ring is $a(r) = \pi(r^2 - \max(0, r - w)^2)$; the two objectives are $N = |S|$ and $A = \sum_{i \in S} a(r_i)$ over feasible S . The radii are *superincreasing* if $r_i > \sum_{j>i} r_j$ for all i ; the violation is measured by $\rho = \max_i (\sum_{j>i} r_j) / r_i$. We call $(\sum_{j>i} r_j) / r_i$ the *tail* (ratio) of ring i , so that ρ is the largest tail; “the tail of x ” always refers to this ratio. The *lex-max* set L is defined

greedily: scanning $i = 1, \dots, n$, include i whenever $(L \cap \{1, \dots, i-1\}) \cup \{i\}$ is feasible. Feasibility is *downward closed*: removing a ring from a feasible placement leaves a feasible placement of the remaining set — the removed ring’s disk becomes free space, and its nested children (each of radius at most $r - w$, less than the freed disk) stay where they are, re-parented to the removed ring’s container, so no piece moves. Hence L is the unique lexicographically maximal feasible set, and the *selection greedy* (add ring i if the resulting *set* remains feasible) computes it. Note that the selection greedy queries a *set-feasibility oracle*; it is an information bound, not an efficient algorithm — deciding sibling packability is itself hard (Section 10). The placement greedies below, by contrast, only ever attempt concrete insertions. A *descending greedy with placement rule* processes rings in decreasing radius and places each into *some* feasible container (the pan or the hole of an already placed ring), skipping it only if no container admits it.

Lemma 3 (Row lemma). *Balls with radii $x_1, \dots, x_k$ satisfying $\sum_j x_j \leq C$ pack inside a ball of radius C : place them tangent in a row along a diameter. Consequently, if a ball of radius m is removed from any packing, any collection of balls of total radius at most m can be inserted into the vacated ball without disturbing the rest.*

Proof. With centers on a diameter at coordinates $p_j = -C + 2\sum_{l < j} x_l + x_j$, each satisfies $|p_j| \leq C - x_j$. $\square$

3 The superadditivity dichotomy

Lemma 4. *The contact area a is strictly increasing and superadditive on $(0, \infty)$: $a(x) + a(y) \leq a(x + y)$.*

Proof. Monotonicity is clear. For $x, y \geq w$: $\pi w(2x - w) + \pi w(2y - w) = \pi w(2(x+y) - 2w) \leq \pi w(2(x+y) - w)$. For $x, y \leq w$ with $x + y \leq w$: $x^2 + y^2 \leq (x + y)^2$. For $x, y \leq w < x + y$: $x^2 + y^2 \leq w(x + y) \leq 2w(x + y) - w^2$, using $x^2 \leq wx$, $y^2 \leq wy$ and $x + y \geq w$. For $x \leq w \leq y$: $\pi x^2 + \pi w(2y - w) \leq \pi w(2x + 2y - w)$ since $x^2 \leq 2wx$. $\square$

Lemma 5 (Lexicographic dominance). *Let the radii be superincreasing and let v be any positive, increasing, superadditive function of the radius. If S, T are feasible sets and the first index (in decreasing radius order) at which they differ belongs to S , then $\sum_S v(r_i) \geq \sum_T v(r_i)$, with equality only if $T \subseteq S$.*

Proof. Let i be the first differing index and $P = S \cap \{1, \dots, i-1\} = T \cap \{1, \dots, i-1\}$. If $T \setminus P = \emptyset$ then $T \subseteq S$ and monotonicity of the sum under inclusion suffices. Otherwise $T \setminus P \subseteq \{i+1, \dots, n\}$, so $\sum_{T \setminus P} r_j < r_i$ by superincreasingness, and by iterated superadditivity and monotonicity $\sum_{T \setminus P} v(r_j) \leq v(\sum_{T \setminus P} r_j) < v(r_i)$. Hence $\sum_T v = \sum_P v + \sum_{T \setminus P} v < \sum_P v + v(r_i) \leq \sum_S v$. $\square$

Theorem 6 (Selection greedy; arbitrary container, any dimension). *With superincreasing radii, the selection greedy computes the lex-max feasible set and therefore maximizes $\sum_{i \in S} v(r_i)$ for every positive, strictly increasing, superadditive v ; in particular it maximizes the contact area A .*

Proof. Feasibility is downward closed, so the selection greedy computes L (if it could add $i \notin L$, any maximal feasible extension would lexicographically beat L). Lemma 5 shows L dominates every feasible set. $\square$

Remark 7 (Positioning). Theorem 6 is at heart a statement about arbitrary downward-closed set systems with superincreasing weights, and as such it belongs to the classical circle of ideas around greedy algorithms on independence systems [15, 16] and the superincreasing knapsack, whose lexicographic structure was analyzed by Gupte [17]. The hypotheses on v are all needed: positivity rules out $v \equiv -1$ (superadditive and non-decreasing, yet maximized by the empty set), and strict monotonicity is what makes the dominance in Lemma 5 strict. The geometric novelty of this paper lies in Theorem 9, where the exchange argument must respect the geometry of nested containers.

Proposition 8 (Cardinality is not rescued). *There are superincreasing instances on which every descending greedy is suboptimal for N . Example (verified; script `superinc.py`): pan $R = 10$, $w = 4.8$, radii $\{9.95, 5.0, 4.3, 0.6\}$. Greedy places $\{9.95, 5.0\}$ ($N = 2$, optimal area; no step offers a choice of container, so all placement rules coincide), while $\{5.0, 4.3, 0.6\}$ packs in a row in the pan ($N = 3$). The obstruction is structural: N corresponds to $v \equiv 1$, which is not superadditive, and Lemma 5 fails for it.*

4 Placement obliviousness under superincreasing radii

Theorem 9 (Placement rule irrelevance). *Let the radii be superincreasing (weakly: $r_i \geq \sum_{j>i} r_j$ suffices; under the weak hypothesis the dominance of Lemma 5 holds with non-strict inequality, which is enough for optimality) and let $K \subset \mathbb{R}^d$ be an arbitrary container. Every descending greedy, with an arbitrary rule for choosing among feasible containers, places exactly the lex-max feasible set. In particular best fit (place in the feasible container of smallest capacity, defined for a general container as the radius of its largest inscribed ball), worst fit (largest capacity) and random placement are all optimal for every positive, strictly increasing, superadditive objective.*

Proof. Let F denote the greedy’s placement just before processing ring i , with placed set G , and suppose $G \cup \{i\}$ is feasible via a witness placement P ; it suffices to show some container of F admits i (the converse direction is immediate, and then greedy’s set equals L by induction as in Theorem 6). The available containers – the pan plus one hole per ring of G – coincide in F and P , and in P the smallest ring i parents nobody.

Let m be the largest ring of G assigned different containers by F and P (if none, P restricted to G equals F and i ’s container in P witnesses admission). Let u be F ’s container for m and v be P ’s. Since the greedy processes rings in decreasing order, when F placed m the occupants of u were exactly the rings larger than m that F assigns to u ; by maximality of m these coincide with the rings larger than m that P assigns to u , and F certified that this set together with m packs in u . Build P' from P : (i) every ring smaller than m that P kept in u is moved into the ball of radius r_m vacated by m inside v – they fit in a row by Lemma 3, because their total radius is at most $\sum_{j:r_j < r_m} r_j \leq r_m$, and they touch nothing else in v ; (ii) m moves to u , joining exactly the larger-than- m rings whose compatibility F certified. Nesting constraints persist: u and v are the pan or holes of rings larger than m , which do not move, and every ring moved into v is smaller than r_m , which fitted in v . Thus P' is a feasible witness agreeing with F on all rings of radius $\geq r_m$: the first disagreement strictly shrinks. Iterating at most $|G|$ times yields a witness P^* agreeing with F on all of G and placing i in some container c^* ; since the occupants of c^* in P^* are exactly those in F , the greedy admits i there. $\square$

Remark 10. The proof uses only (a) downward closure of packings, (b) the row lemma applied inside the vacated ball, and (c) the total-radius bound from superincreasingness. Nothing refers

to the shape of K or the dimension: the theorem covers rectangular pans (griddles) and spherical shells nested in arbitrary containers in $\mathbb{R}^d$ verbatim. Computational corroboration of the strongest consequence: on 100 random superincreasing instances, best-fit, worst-fit and random placement produced optimal (hence, by Lemma 5, identical) outcomes without exception (script `test_oblivious.py`; beyond the exact one-circle, two-circle and row criteria its feasibility oracle is a heuristic packer with restarts – a sampling caveat of the corroboration, not an ingredient of the proof).

5 Sharpness: the $n = 4$ transition and twin instances

Proposition 11 ($n \leq 3$, arbitrary container). *For any compact container $K \subset \mathbb{R}^d$, $d \geq 1$, and arbitrary strictly decreasing radii of common positive width, every descending greedy on at most three rings places the lex-max set.*

Proof. A ring skipped before any has been admitted cannot occur in any feasible set: its outer ball would have to lie in K , even if nested. Discard such rings. With at most two remaining there is no previous assignment choice that can obstruct admission. With three, if the second ring is skipped, the prefix pair is infeasible because its only possible parents are the root and the first ring, and both were tested. The last decision then involves only one previously admitted ring.

It remains to consider $r_1 > r_2 > r_3$ with the first two admitted. Let F be their greedy assignment and suppose a witness P places all three. If the parent of r_2 agrees in F and P , the latter's home for r_3 works in F , since siblings may be rearranged. If F nests r_2 but P keeps it at the root, shrink that root ball concentrically to radius r_3 : it fits beside r_1 in K , so F admits r_3 at the root. If F keeps r_2 at the root but P nests it, then $r_3 < r_2 \leq r_1 - w$, so the empty hole of r_1 in F admits r_3 . Thus the third ring is admitted whenever the full prefix is feasible. Conversely every legal admission is a feasibility witness. No convexity or special shape of K was used. $\square$

Lemma 12 (Cap confinement). *Circles of radii 10, 4.9, 4.8 do not pack in a disk of radius 15.*

Proof. Write c_A, c_X, c_Y for the centers, measured from the pan center. Then $|c_A| \leq 5$, $|c_X| \leq 10.1$, $|c_Y| \leq 10.2$, $|c_A - c_X| \geq 14.9$, $|c_A - c_Y| \geq 14.8$, $|c_X - c_Y| \geq 9.7$. From $|c_A| + |c_X| \geq 14.9$, $a := |c_A| \in [4.8, 5]$. Let $u = -c_A/|c_A|$. Then $c_X \cdot u = (|c_X - c_A|^2 - |c_X|^2 - a^2)/(2a) \geq (120 - a^2)/(2a) \geq 9.5$ (decreasing in a), so the transverse part of c_X has norm at most $\sqrt{10.1^2 - 9.5^2} \leq 3.43$; similarly $c_Y \cdot u \geq (115 - a^2)/(2a) \geq 9.0$ with transverse norm at most $\sqrt{10.2^2 - 9^2} = 4.8$. Hence $|c_X - c_Y|^2 \leq 10.1^2 + 10.2^2 - 2(9.5 \cdot 9.0 - 3.43 \cdot 4.8) \leq 67.98$, i.e. $|c_X - c_Y| \leq 8.25 < 9.7$, a contradiction. $\square$

Theorem 13 (Failure at $n = 4$). *Placement obliviousness fails for four rings: with $R = 15$, $w = 0.3$ and radii $\{10, 5, 4.9, 4.8\}$ (all four form the lex-max set, witnessed by the pan pair $\{10, 5\}$ at exact tangency and the hole pair $\{4.9, 4.8\}$ filling the hole 9.7 exactly), best fit nests the 5, forces the 4.9 into the pan, and blocks the 4.8 everywhere by Lemma 12 and the exact two-circle conditions; worst fit places all four (Figure 1). (The worst-fit run also uses that $\{10, 5, z\}$ does not pack in the pan for $z \in \{4.9, 4.8\}$: the pan pair is diametrically rigid and the rescaled pocket of Proposition 51 is $10b(1/2) = 30/7 \approx 4.286 < z$; the count of four is in any case robust to either oracle answer.)*

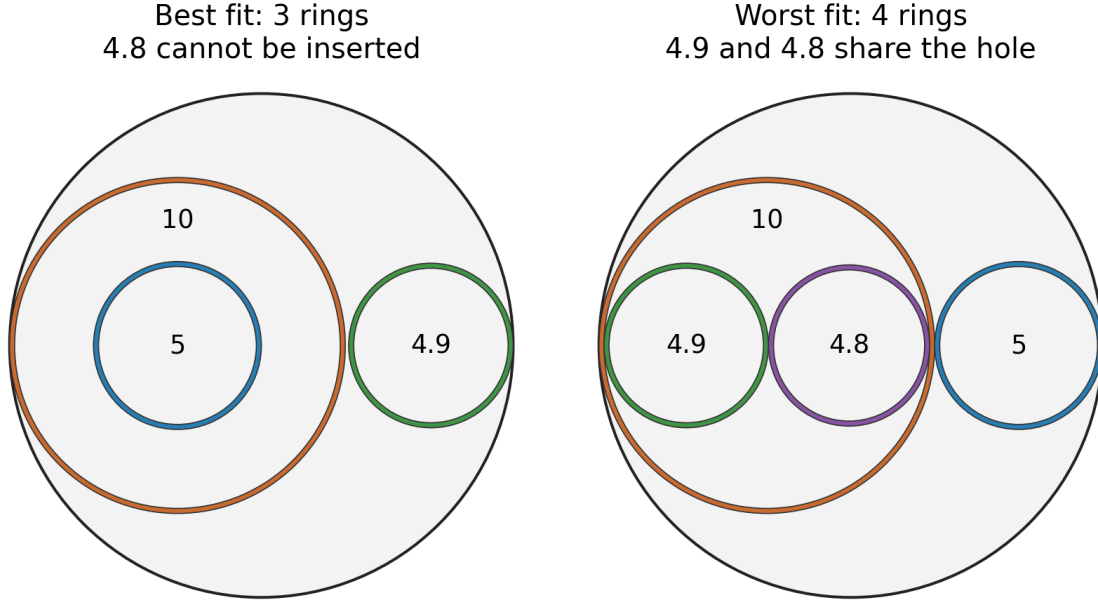


Figure 1: The $n = 4$ counterexample of Theorem 13: best fit (left) nests the 5 and loses the 4.8; the witness (right), realized by worst fit, places all four rings.

Theorem 14 (Twin instances; no state-based rule). *Let $R = 15$, $r_1 = 10$, $r_2 = 5$, $w = 0.505$ (hole 9.495), and*

$$I_1 = \{10, 5, 4.99, 4.50\}, \quad I_2 = \{10, 5, 4.76, 4.74\}.$$

In I_1 the correct placement of the 5 is the pan (best fit fails, worst fit succeeds); in I_2 it is the hole (worst fit fails, best fit succeeds). At the decisive step the observable state – containers, capacities, occupants, incoming ring, R and w – is identical in I_1 and I_2 . Consequently no deterministic placement rule that is a function of the state attains the lex-max set on all instances, and for every randomized rule some instance is failed with probability at least $1/2$.

Proof ingredients. Feasibilities are two-circle-exact except for two triples, both settled rigorously: $\{10, 4.99, 4.50\}$ does not pack in the disk of radius 15 (cap confinement as in Lemma 12 yields $|c_X - c_Y| \leq 8.80 < 9.49$), and the pan pair $\{10, 5\}$ is diametrically rigid, whence no third circle of radius ≥ 4.7 fits (confinement again: for $z = 4.76$, $|c_z - c_B| \leq \sqrt{204.86 - 20 \cdot 8.8} \leq 5.38 < 9.76$; in general $|c_z - c_B|^2 \leq z^2 - 130z + 625 < (5+z)^2$ for every $z > 30/7$, so the claim holds for all $z \geq 4.7$). Packability of $\{10, 4.76, 4.74\}$ in radius 15 is exhibited by an explicit boundary configuration (angles 180° , 30.8° , -32.0° ; all pairwise constraints verified). The key phenomenon enabling the twins is that triple-packability is *not monotone in the sum*: the unbalanced pair $\{4.99, 4.50\}$ (sum 9.49) fails beside the 10 while the balanced pair $\{4.76, 4.74\}$ (sum 9.50) succeeds. The four executions, every step after the decisive one forced (a single feasible container or none): in I_1 , nesting the 5 sends the 4.99 to the pan ($5 + 4.99 > 9.495$) and blocks the 4.50 everywhere (pan by the triple fact, hole by $5 + 4.50 > 9.495$, nesting by capacity) – three rings – while the 5 in the pan sends both 4.99 and 4.50 to the hole ($4.99 + 4.50 = 9.49 \leq 9.495$) – all four; in I_2 , nesting the 5 sends 4.76 and 4.74 to the pan (the explicit boundary configuration) – all four –

while the 5 in the pan blocks the pan for both (≥ 4.7), nests the 4.76 alone and blocks the 4.74 ($4.76 + 4.74 = 9.50 > 9.495$) – three rings. See Appendix F. $\square$

Remark 15. The obstruction is informational, not existential: following any witness of the lex-max set is itself a legal greedy execution (a ring outside the lex-max set is never admissible), so some execution always succeeds; what cannot exist is a state-based rule that finds it. For rules with access to the full input the question becomes one of computational complexity and remains open (Section 13).

6 Thresholds: the Tribonacci floor and the golden counterexample

Theorem 16 (Additive model: threshold exactly 1). *Replace sibling feasibility by its additive surrogate (a set of siblings fits iff their radii sum to at most the container capacity; exact for at most two circles and sufficient in general by Lemma 3). Then the universal threshold for placement obliviousness is exactly 1: obliviousness holds for all instances with $\rho \leq 1$ (the proof of Theorem 9 applies verbatim), while for every $\varepsilon > 0$ some instance with $\rho \leq 1 + \varepsilon$ fails – individual instances with larger ρ may of course still succeed. Indeed, fix r_1 , set $s = r_1/2$ and $R = r_1 + s = \frac{3}{2}r_1$, pick any $w \in (\frac{r_1}{4}, \frac{r_1}{2} - \delta)$, and take $r_2 = s$ and the exact pair $r_3 = \frac{r_1}{4} + \frac{2\delta}{3}$, $r_4 = \frac{r_1}{4} + \frac{\delta}{3}$ (so $r_3 + r_4 = s + \delta$ and $r_4 < r_3 < r_2 < r_1$ strictly), with any $0 < \delta < \min(\frac{r_1}{4}, \frac{\varepsilon r_1}{2})$: the witness fills the pan with $\{r_1, r_2\}$ and the hole of r_1 with the pair ($r_3 + r_4 = s + \delta \leq r_1 - w$), while best fit nests r_2 ($r_2 = s \leq r_1 - w$ since $w \leq r_1/2$), sends r_3 to the pan, and blocks r_4 additively ($r_1 + r_3 + r_4 > R$ since $r_3 + r_4 > s$; $r_2 + r_4 > r_1 - w$ since $w > r_1/4$; $r_4 > r_2 - w$ likewise). The three tail ratios are $(2s + \delta)/r_1 = 1 + \delta/r_1$, $(s + \delta)/s = 1 + 2\delta/r_1$ and $r_4/r_3 < 1$, so $\rho = 1 + 2\delta/r_1 \leq 1 + \varepsilon$; verified instances reach $\rho = 1.03$ and 1.02 (script `umbral.py`).*

The rigid four-ring counterexample family has its floor at the Tribonacci constant. An earlier version of this statement relied on a tangency idealization ($r_3 \rightarrow r_2$, $w \rightarrow 0$); the following theorem removes it entirely: the rigid configuration is not assumed but *emerges* as the extremal case of a concavity argument.

Theorem 17 (Rigid-subfamily Tribonacci floor, no idealization). *Normalize $r_1 = 1$ and write $t = r_2$, $p = r_3$, $q = r_4$, $\omega = w > 0$. Let $\mathcal{F}$ be the family of four-ring instances with (F1) diametral pan $R = 1 + t$, (F2) $p + q \leq 1 - \omega$ (the witness pair fits the hole of r_1), and (F3) the triple $\{1, p, q\}$ does not pack in the disk of radius $1 + t$. Then every instance of $\mathcal{F}$ satisfies $\rho > T$, strictly, for every width $\omega > 0$ and all strict radii $q < p < t < 1$; moreover $\inf_{\mathcal{F}} \rho = T$ and the infimum is not attained. The family is nonempty only for $t < t^* = 1/T$, and it contains the $n = 4$ counterexample and the twin I_1 of Theorem 14.*

Proof sketch (complete proof in Appendix B). For two circles of radii a, b internally tangent to the wall of a disk of radius R , the minimal angular separation $\theta(a, b)$ obeys the half-angle identity $\sin^2(\theta/2) = f(a)f(b)$ with $f(x) = x/(R - x)$. Placing the triple $\{1, p, q\}$ wall-tangent with the unit ring in the middle packs it whenever $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$ (the two non-adjacent arguments are separated by monotonicity of θ), and this angular condition reduces *algebraically* to

$$\psi(p) + \psi(q) \geq \tau_t, \quad \psi(x) = \sqrt{\frac{t-x}{x}}, \quad \tau_t = \frac{t}{\sqrt{1+t}},$$

(an algebraic *sufficient* condition for packing, which is all the contrapositive needs: (F3) forces $\psi(p) + \psi(q) < \tau_t$). In the coordinates $\alpha = \psi(p)$, $\beta = \psi(q)$ the sum $p + q = U(\alpha) + U(\beta)$ with $U(z) = t/(1 + z^2)$, and U is concave on $[0, \tau_t]$ exactly when $\tau_t \leq 1/\sqrt{3}$, i.e. $t \leq (1 + \sqrt{13})/6 = 0.7676\dots$, which covers the relevant range with room to spare; hence the infimum of $p + q$ over the blocked (open) region sits at the *corner* of its closure, $p = t$, $q = b(t)$, where $b(t) = t(1 + t)/(1 + t + t^2)$ is the Descartes pocket of the rigid pair. This gives the two pressures $t + b(t) < p + q \leq 1 - \omega < 1$ (the first one strict, whence the strict floor and the non-attainment), whose compatibility is precisely the cubic $t^3 + t^2 + t < 1$, and then $\rho \geq (p + q)/t > 1 + (1 + t)/(1 + t + t^2) =: L(t)$ with L strictly decreasing and $L(t^*) = T$. Exactness of the pocket at the rigid pair (needed only for the direction $\inf \leq T$) is a rigidity computation: the constraint left-hand side factors exactly as $4(t^2 + t + 1)(b(t) - q)$; and the approximating family lives genuinely inside $\mathcal{F}$ by a closure-and-monotonicity lemma for the packing region (compactness of witnesses), which shows the blocked interval in p is half-open with an attained endpoint. $\square$

Remark 18 (Slack pans). The hypothesis $R = r_1 + r_2$ can be relaxed to $R \geq r_1 + r_2$ at no cost: non-packability is antitone in the container (a packing in a smaller disk is a packing in a larger one after translation), so (F3) at radius R implies (F3) at radius $r_1 + r_2$ and Theorem 17 applies verbatim; the infimum remains T . Slack in the pan never helps the blocker — what it changes is the set of possible *occupants*, the subject of Section 8.

Define the *geometric threshold* as

$$\tau := \inf\{\rho(I) : \text{some greedy execution fails to place the lex-max set on } I\},$$

so that placement obliviousness holds for every instance with $\rho < \tau$ and fails at values of ρ arbitrarily close to τ (possibly at τ itself, if the infimum is attained); “floor” always refers to an infimum of ρ over a family of blocked exchanges. The natural conjecture – which was tenable until late in this project – is that $\tau = T$. It is *false*. The witness capacity that lifts the floor from φ to T in Theorem 17 exists only when the greedy’s container for the pivotal ring is a *hole*; when it is the pan itself, the floor drops to the first rung of the ladder, the Descartes pocket, and the golden ratio appears:

Theorem 19 (The golden family: the threshold is at most φ). *For every $\omega \in (1 - \varphi/2, \varphi - 1) = (0.1910, 0.6180)$ and every*

$$0 < \varepsilon < \varepsilon^* := \min\left(\frac{T - \varphi}{3}, \frac{\varphi/2 - \omega}{2}, \frac{4\sqrt{5} - 8 - \varphi/2}{2}\right)$$

(all three bounds positive on the window; the middle one enforces $s_1 \leq \varphi - \omega$), the instance

$$R = \varphi + 1, \quad w = \omega, \quad r = \{\varphi, 1, \frac{\varphi}{2} + 2\varepsilon, \frac{\varphi}{2} + \varepsilon\}$$

has $\rho = \varphi + 3\varepsilon < T$, its lex-max set is all four rings, and the descending greedy under worst fit places only three: placement obliviousness fails at $\rho = \varphi + 3\varepsilon$.

Proof. Write $s_1 = \frac{\varphi}{2} + 2\varepsilon$, $s_2 = \frac{\varphi}{2} + \varepsilon$. Every feasibility below is exact. The value of ρ : the tails are $(1 + s_1 + s_2)/\varphi = \varphi + 3\varepsilon/\varphi$ (by the golden identity $(1 + \varphi)/\varphi = \varphi$), $s_1 + s_2 = \varphi + 3\varepsilon$ (dominant) and $s_2/s_1 < 1$, and $\varphi + 3\varepsilon < T$ for $\varepsilon < (T - \varphi)/3$. The witness: place 1 in the hole of φ (capacity $\varphi - \omega \geq 1 \iff \omega \leq \varphi - 1$) and $\{\varphi, s_1, s_2\}$ as a corona in the pan; the corona

exists by an *exact* certificate. By monotonicity of θ it suffices that the symmetric trio $\{\varphi, \sigma, \sigma\}$ with $\sigma = s_1$ has angular sum below 2π , i.e. $2A + B < \pi$ with $A = \theta(\varphi, \sigma)/2$, $B = \theta(\sigma, \sigma)/2$. Here $f(\varphi) = \varphi$, and when $2A > \pi/2$ (otherwise the claim is immediate from $B < \pi/2$),

$$2A + B < \pi \iff \sin B < \sin 2A \iff f(\sigma)^2 < 4\varphi f(\sigma)(1 - \varphi f(\sigma)) \iff f(\sigma)(1 + 4\varphi^2) < 4\varphi,$$

which with $f(\sigma) = \sigma/(\varphi^2 - \sigma)$, $8\varphi + 5 = (2 + \sqrt{5})^2$ and $\varphi^3 = 2 + \sqrt{5}$ reduces to $\sigma(8\varphi + 5) < 4\varphi^3$, i.e. $\sigma < 4(\sqrt{5} - 2) = 0.9442\dots$; this holds since $s_1 = \varphi/2 + 2\varepsilon < 4(\sqrt{5} - 2)$ by the third bound in ε^* . (At $\varepsilon = 0$ the certificate is the exact identity $4\varphi(\sqrt{5} - \varphi) = 4 > 1$.) So all four rings are placeable. *The greedy under worst fit*: $\varphi \rightarrow \text{pan}$; the ring 1 fits both the pan ($\varphi + 1 \leq R$, exact tangency, as in Theorem 13) and the hole of φ , and worst fit – the container of largest capacity, $R = \varphi + 1 > \varphi - \omega$ – chooses the *pan*. Now $\{\varphi, 1\}$ fills the pan diametrically: the pair is rigid and, by the exact necessity of Proposition 51 (rescaled), any third circle has radius at most $b_2(\varphi, 1) = \varphi/2$ – the golden pocket identity. Hence neither s_1 nor s_2 fits the pan; the hole of φ admits one of them but not both ($s_1 + s_2 = \varphi + 3\varepsilon > \varphi - \omega$, two-circle exact); the hole of the ring 1 admits neither ($s_2 > 1 - \omega \iff \omega > 1 - \frac{\varphi}{2} - \varepsilon$); and s_2 cannot nest in s_1 (nesting would require $\omega \leq s_1 - s_2 = \varepsilon$, while $\omega > 1 - \frac{\varphi}{2} > \varepsilon^*$); and s_1 fits alone in the hole of φ ($s_1 \leq \varphi - \omega$, via ε^*), which is where the greedy puts it before s_2 is stranded. An exhaustive placement tree confirms the count: three rings with 1 in the pan, four with 1 in the hole (`code/aureo.py`). $\square$

Corollary 20 (The family realizes every $\rho \in (\varphi, T)$). *For $\rho_0 \in (\varphi, T)$ fix the width $\omega = 0.3$ (any value in the window works) and take $s_2 = \varphi/2 + \varepsilon$, $s_1 = s_2 + \eta$ with $2\varepsilon + \eta = \rho_0 - \varphi$ and $0 < \eta < \rho_0 - \varphi$ small. The dominant tail is $s_1 + s_2 = \rho_0$ exactly. All walls hold explicitly: $s_1 < 1$ and the corona certificate $s_1 < 4(\sqrt{5} - 2)$, since $s_1 = \varphi/2 + \frac{\rho_0 - \varphi}{2} + \frac{\eta}{2} < \varphi/2 + \frac{T - \varphi}{2} + \frac{\eta}{2} = 0.9196\dots + \frac{\eta}{2} < 0.9443$ for $\eta < 0.049$; $s_2 > \varphi/2 = 0.809 > 0.7 = 1 - \omega$ (H_m blocked); $s_1 + s_2 = \rho_0 > \varphi > \varphi - \omega$ (the hole of φ rejects the pair); $s_2 > s_1 - \omega$ (their gap is $\eta < 0.3 = \omega$, so no nesting); and $s_1 \leq \varphi - \omega = 1.318$ (s_1 fits alone in the hole, where the greedy strands it). $\square$*

The instance is golden in four distinct ways: the occupant is φ (the unique positive fixed point of $2b(A)A = 1 + 2b(A)$), the pocket is $b_2(\varphi, 1) = \varphi/2$, the tail of φ is $(1 + \varphi)/\varphi = \varphi$, and $\rho \rightarrow \varphi$. The family is not degenerate: the ω -window has width 0.427, ε ranges over an interval, and the diametral tangency survives pan slack up to $\delta^* = 0.0248$ (numerically, via the angular criterion). It does not appear in structured searches over generic radii because it lives at a codimension-one corner (pan tangency) that random radii never visit.

Theorem 21 (Exact threshold for four rings). *Let τ_4 denote the infimum defining τ restricted to inventories of at most four rings in a disk. Every failing execution on such an inventory has $\rho > \varphi$. Consequently $\tau_4 = \varphi$, and this infimum is not attained at positive width and strictly decreasing radii.*

Proof. We first isolate the combinatorial reduction. Consider the first index where a greedy execution differs from the lex-max set. It skips a ring that is feasible together with its previously admitted set. By Proposition 11, there must be at least three previously admitted rings: otherwise the same execution on the subinventory of these rings and the skipped ring would contradict that proposition. Previously omitted rings leave no occupants or containers in F , so the restricted execution is legal and its full inventory is feasible. Thus all four rings are feasible, and the execution admits the first three and rejects the fourth. Normalize the second radius to

1, and write the radii as $A > 1 > p > q > 0$, with scaled width $w > 0$. Let F be the greedy assignment of $A, 1, p$ and P a full witness.

If F and a full witness agree on the parent of 1, they can also be made to agree on the parent of p : use the exchange step of Theorem 9 at p . The only smaller ring is $q < p$, so any displaced child fits in the vacated outer ball of p ; if q is a child of p , carry it with p . The larger parents and their assignments stay fixed, and the destination was certified when the greedy placed p . The resulting witness admits q in F , a contradiction. Therefore every full witness places 1 in the opposite parent from F : the only choices are the root and the hole of A .

Apply the exchange step at 1, and let S be the immediate children smaller than 1 in its greedy destination, as assigned by P . If S has at most one member, its subtree fits in the vacated unit ball. If $S = \{p, q\}$ and $p + q \leq 1$, the row lemma does the same. In either operation 1 travels to its destination with its own subtree. This gives a full witness agreeing with F on 1, which was just ruled out. Hence $S = \{p, q\}$ and $p + q > 1$. There are now exactly two cases.

The greedy nests 1 in A . The witness has root pair $\{A, 1\}$ and hole pair $\{p, q\}$ in A , so $R \geq A + 1$ and $p + q \leq A - w$. If the root triple $\{A, p, q\}$ were packable as root children, nesting 1 in A (legal since F certified $1 \leq A - w$) would give a full witness with the required parent of 1. Thus that triple is infeasible. Divide all lengths by A : $t = 1/A$, $\omega = w/A$ and $R/A \geq 1 + t$, while ρ is unchanged. These are the hypotheses of Theorem 17, with pan slack allowed by Remark 18. Therefore $\rho > T > \varphi$.

The greedy places 1 at the root. Now P nests 1 in A , and p, q are root children. In particular $A - w \geq 1$. In F , the ring p must lie in the hole of A : if it were at the root or inside 1, the empty hole of A would admit $q < 1$. Since the root pair $\{A, 1\}$ is feasible, $R \geq A + 1$. The exact diametral pocket of this pair is

$$g(A) = \frac{A(A+1)}{A^2 + A + 1}.$$

The sufficiency direction of Proposition 51, rescaled, supplies a packing of $\{A, 1, q\}$ at radius $A + 1$ whenever $q \leq g(A)$; such a packing also works in the actual pan and is compatible with p inside A . Since q is rejected, $q > g(A)$. Consequently

$$\rho \geq p + q > 2g(A), \quad \rho A \geq 1 + p + q > 1 + 2g(A).$$

Put $D = A^2 + A + 1 > 0$. The identity $\varphi^2 = \varphi + 1$ gives

$$\begin{aligned} D(2g(A) - \varphi) &= (A - \varphi)((2 - \varphi)A + 1), \\ D(1 + 2g(A) - \varphi A) &= (\varphi - A)(\varphi A^2 + (2\varphi - 2)A + \varphi - 1). \end{aligned}$$

For $A \geq \varphi$, the first identity implies $2g(A) \geq \varphi$; for $1 < A \leq \varphi$, the second implies $1 + 2g(A) \geq \varphi A$. In both cases the strict inequalities above yield $\rho > \varphi$. Finally Theorem 19 supplies four-ring failures with $\rho \downarrow \varphi$, establishing the infimum and non-attainment. $\square$

The two algebraic identities and the ordered-ring implication used in the second case are kernel-checked in `Calamares/FourRing.lean`. The forest reduction and the Euclidean pocket remain written proofs. The preceding four-ring proof does not by itself cover arbitrary inventories. The following uniform argument closes that gap.

6.1 The uniform golden guarantee

Theorem 22 (Global golden guarantee). *For every finite inventory of strictly decreasing positive outer radii in a disk, every descending greedy execution returns the lexicographically maximum feasible subinventory when $\rho \leq \varphi$. The assertion also holds with independent hole radii $0 \leq h_i < r_i$.*

The greedy semantics are those of Section 2: previously chosen parents stay fixed, but sibling coordinates may be rearranged. A rejection means that no container admits the extended forest. The proof works for an arbitrary finite inventory; it does not enumerate successive sizes. Write $\beta(a, b) = ab(a + b)/(a^2 + ab + b^2)$.

6.2 Tail bounds and a geometric pocket

A list of disks whose radii sum to s fits in a disk of radius s : place them consecutively along a diameter. This row construction may replace a disk envelope without changing its surrounding packing.

Lemma 23 (Two tail bounds). *If $p + U \leq \varphi m$ and $U \leq \varphi p$, then $U \leq m$. If $a > b > 0$, $T \leq \varphi b$, and $b + T \leq \varphi a$, then $T \leq 2\beta(a, b)$.*

Proof. For the first assertion, $(1 + \varphi)U \leq \varphi p + \varphi U \leq \varphi^2 m = (1 + \varphi)m$. For the second, normalize $b = 1$, put $x = a/b > 1$, $g = x(x + 1)/(x^2 + x + 1)$, and write $t = T/b$. We have $t \leq \min(\varphi, \varphi x - 1)$. The identities

$$\begin{aligned} 2x(x + 1) - \varphi(x^2 + x + 1) &= (x - \varphi)((2 - \varphi)x + 1), \\ (x^2 + x + 1) + 2x(x + 1) - \varphi x(x^2 + x + 1) &= (\varphi - x)(\varphi x^2 + (2\varphi - 2)x + \varphi - 1) \end{aligned}$$

give $2g \geq \varphi$ when $x \geq \varphi$, and $1 + 2g \geq \varphi x$ when $1 < x \leq \varphi$. Thus $t \leq 2g$ in both cases. Rescale by b . $\square$

Lemma 24 (Balanced rows). *Let a nonempty finite list of positive radii have total T and largest member p . After reserving p , its remaining members can be divided into two groups, each of total at most $T/2$.*

Proof. Assign each remaining member to the currently lighter group. Both load differences remain at most p , since the added member is at most p . At the end the loads u, v satisfy $u + v = T - p$ and $|u - v| \leq p$; hence $2u, 2v \leq T$. Each group fits its row envelope. $\square$

Place disks $a \geq b > 0$ in a disk of radius $a + b$, with centers $(-b, 0)$ and $(a, 0)$. The upper and lower pockets have inscribed disks of radius $\beta(a, b)$, each tangent to the two disks and the wall. Their centers are

$$\left(\frac{(a - b)(a + b)^2}{a^2 + ab + b^2}, \pm 2\beta(a, b) \right).$$

These coordinates verify all three tangencies and separation of the two pockets directly. We need a stronger fact about a single pocket.

Lemma 25 (Two unequal disks in one pocket). *If $0 < r, s \leq \beta(a, b)$ and $r + s \leq b$, the two disks of radii r, s fit simultaneously in the upper pocket. The lower pocket remains available. A zero radius may be omitted.*

Proof. Put $L = a + b$. A disk of radius $x \leq \beta(a, b)$ tangent to the wall has its center in the upper half-plane at an admissible angle exactly in

$$\left[2 \arcsin \sqrt{\frac{bx}{a(L-x)}}, \quad \pi - 2 \arcsin \sqrt{\frac{ax}{b(L-x)}} \right].$$

The law of cosines gives the two endpoints. The interval is nonempty exactly when $x(a^2 + b^2) \leq ab(L - x)$, equivalent to $x \leq \beta(a, b)$. Place r at its lower endpoint and s at its upper endpoint, and set

$$A^2 = \frac{as}{b(L-s)}, \quad B^2 = \frac{br}{a(L-r)}, \quad A, B \geq 0.$$

Their angular separation is $\pi - 2 \arcsin A - 2 \arcsin B$; the required separation is $2 \arcsin(AB)$. It suffices that $A^2 + B^2 + 3A^2B^2 \leq 1$: this implies $\sqrt{1-A^2}\sqrt{1-B^2} \geq 2AB$ and therefore $\arcsin A + \arcsin B + \arcsin(AB) \leq \pi/2$. After multiplication by the positive denominator $ab(L-r)(L-s)$, the margin is

$$L^2[a(b-r-s) + (a-b)r] + (a-b)^2rs \geq 0.$$

The disks lie wholly above the diameter: its intersection with the container is covered by a, b , so a disk disjoint from them cannot cross it. The lower pocket is consequently free. $\square$

6.3 The minimal triple and its opposite gap

Lemma 26 (A fully tangent minimal triple). *For $a \geq b \geq c > \beta(a, b)$, the three disks admit a smallest containing disk of radius $R_0 > a + b$, in which they are pairwise tangent and all tangent to the wall. With*

$$A = 1/a, \quad B = 1/b, \quad C = 1/c, \quad t = \sqrt{AB + AC + BC},$$

its radius satisfies $1/R_0 = 2t - A - B - C$.

Proof. Existence of a minimum follows by compactness, using a row as an upper bound. In any feasible placement of three disks, each disk can be moved to the wall while the other two are fixed. For its center x , choose a nonzero direction v such that $(x - x_j) \cdot v \geq 0$ for the other two centers. Two closed half-planes through the origin have such a common direction in the plane. Distances to both other centers are nondecreasing along the ray. Move to the wall and repeat for the others.

For wall-tangent disks with pair sums at most R , let $\theta_{ij} \in [0, \pi]$ denote their minimum angular separation. Three such positions exist exactly when

$$\theta_{ab} + \theta_{ac} + \theta_{bc} \leq 2\pi.$$

Necessity follows by summing the three directed cyclic gaps, each at least its corresponding minimum. Conversely choose gaps in $[\theta_{ij}, \pi]$ summing to 2π ; their endpoints have sums at most 2π and 3π , so this is possible. Each gap is then the smaller angle for its pair.

Two-disk necessity gives $R_0 \geq a + b$. At equality the two largest are forced into the diametral arrangement. Move c to the wall while fixing them and use the interval in Lemma 25; it would give $c \leq \beta(a, b)$, a contradiction. Thus $R_0 > a + b$. All three minimum angles are continuous and strictly between 0 and π near R_0 . Their sum equals 2π , since a strict inequality would permit decreasing R_0 . Taking these gaps realizes every tangency. Descartes' equation for the oriented curvatures $A, B, C, -1/R_0$ gives the stated formula; its other root has $1/R_0 < 0$. $\square$

Lemma 27 (Compensation by the other gap). *In the placement of Lemma 26, there is a further disk of radius $d > 0$, disjoint from all three disks and contained in the same container, with $c + d \geq 2\beta(a, b)$.*

Proof. First justify the geometric existence. Invert about the tangency point of a, b . Their boundaries become parallel lines and their interiors become the exterior half-planes of a strip. The container interior becomes the exterior of a circle inscribed in the strip. A disk inside the strip tangent to both lines has radius equal to half its width; its center lies on the midline. Tangency to the image of the container leaves exactly two such disks, on opposite sides of that circle. They have disjoint interiors. The inversion center lies inside the image of the container circle, hence outside both solution disks; their inverse images are bounded disks. Inverting back gives two admissible disks in the original container, one being c and the other denoted d . This argument identifies the two components of the free region; a height inequality alone would not establish their disjointness.

Set $S = A + B$, $P = AB$, and $Q = S^2 - P$. Then $t^2 = P + SC$ and $c > \beta(a, b) = S/Q$ imply $SC < Q$, hence $t < S$ and $C < S$. The classical Descartes reflection formula for the two solutions tangent to a, b and the wall [1, Section 3.3] gives

$$D = \frac{1}{d} = 2(A + B - 1/R_0) - C = 4S + C - 4t > C > 0.$$

The roots are distinct because $t < S$, and inversion has exhibited exactly two admissible disks, so the reflection identifies the other one. Using $t^2 = P + SC$ and the displayed expression for D yields

$$Q(C + D) - 2SCD = 2(S - t)^2(2S + 2t - C) \geq 0.$$

Divide by $QCD > 0$ to obtain $c + d \geq 2S/Q = 2\beta(a, b)$. □

6.4 Three largest disks determine local feasibility

Theorem 28 (Three largest disks). *Let $a > b \geq c \geq r_4 \geq \dots \geq r_n > 0$, $n \geq 3$, and suppose every tail is at most φ times its corresponding radius. The entire list fits a disk of radius R if and only if a, b, c fit. Equality between the smaller radii is permitted.*

Proof. Only sufficiency needs proof. Let $U = \sum_{j \geq 4} r_j$. Lemma 23, applied to a, b , gives $c + U \leq 2\beta(a, b)$.

If $c > \beta(a, b)$, place the three largest disks in their minimum container $R_0 \leq R$. Lemma 27 provides a disk of radius $d \geq 2\beta(a, b) - c \geq U$ in the remaining gap. Replace it by a row containing all the remaining disks.

If $c \leq \beta(a, b)$, place a, b diametrically in radius $a + b \leq R$ and place c in the lower pocket. For a nonempty remaining list, write $e = r_4$ and $V = \sum_{j \geq 5} r_j$. The two-tail assertion of Lemma 23, first at b, c and then at c, e , gives

$$e + V \leq b, \quad V \leq c, \quad e \leq c \leq \beta(a, b).$$

Thus Lemma 25 places the two envelope disks e, V in the upper pocket. Replace V by a row of its actual members; omit it when $V = 0$. The empty remaining list requires no extra step. These envelopes construct positions; they do not replace the inventory in the definition of ρ or in a greedy execution. □

6.5 A uniform exchange of parents

Suppose a descending execution differs from the lexicographic maximum. At the first differing radius it must omit a ring of that maximum: the opposite difference would contradict lexicographic maximality of the latter. Restrict to the already admitted set G and this omitted ring. This restricted inventory is feasible and still has $\rho \leq \varphi$. Let F be its greedy forest on G . Among all complete witness forests P , choose one having the longest prefix of parents in agreement with F ; this maximum exists because there are finitely many parent assignments. If all parents agree, the omitted ring can be accepted in F , contradicting rejection.

Otherwise let m be the largest disagreement, with destination u in F and origin $v \neq u$ in P . Both containers are the root or holes of rings larger than m . All larger parents agree. All smaller rings may be reconstructed as leaves, since we only need to align the prefix through m .

Write T for the total radius below m . If $T \leq m$, remove the smaller rings, put m in u using the prefix certificate from F , and place all smaller rings in a row inside the outer disk vacated by m in v , preserving the larger siblings there. This aligns m . Hence assume $T > m$. Let $p < m$ be the next radius and $U = T - p$. The first assertion of Lemma 23 gives $U \leq m$.

Consider the rooted tree formed by the larger rings and the virtual root. The degree of a container is its number of immediate larger children. The following three cases cover every such tree.

A branching container with at least three larger children. Let w have children $a_1 > \dots > a_k > m$, $k \geq 3$. The auxiliary disk list

$$a_1, \dots, a_k, m, m$$

satisfies all golden tail bounds: for each a_i its auxiliary tail is at most its original tail, because $2m < m + T$; the last two tail ratios are 1, 0. Its three largest disks already coexist in w . Theorem 28 therefore gives two radius- m slots alongside all the larger children.

If $w \neq u$, put p in one slot, a row of total U in the other, and m in u using F . This includes $w = v$; no third slot in w is needed. If $w = u$, put m, p in its slots and the row of total U in the outer disk vacated by m in v , using P there.

A container with exactly two larger children. Let its children be $a > b > m$. Their original tails imply $m + T \leq \varphi b$ and $b + m + T \leq \varphi a$; thus $m + T \leq 2\beta(a, b)$ by Lemma 23. Since $T > m$, $\beta(a, b) > m$. The container has radius at least $a + b$, so a diametral placement of a, b provides two radius- m slots in its opposite pockets. Apply the preceding slot assignment. This also proves the degree-two criterion without a separate case enumeration.

No branching container. The larger tree is a chain with exactly one leaf container. Since $u \neq v$, at least one of u, v has exactly one larger child a . If it is u , F certifies the pair a, m there; if it is v , P does so. In either case that container has radius at least $a + m$. The original tails give $T \leq \varphi m$ and $m + T \leq \varphi a$, hence $T \leq 2\beta(a, m)$. Reserve p ; Lemma 24 divides the others into two rows of total at most $T/2 \leq \beta(a, m)$.

When using u , put a, m there diametrically, the two rows in its pockets, and p in the outer disk vacated by m in v . When using v , keep a there, put p inside the envelope reserved for a phantom disk m , and put the two rows in the pockets; the real m goes to u using F .

In every case the parents of all larger rings are unchanged. Local interior packings move with their owners. Those owners have radius larger than m , so the assembly creates no cycle

even when one chosen container is an ancestor of another. The complete new witness agrees with F through m , contrary to the maximal choice of P . This proves Theorem 22.

Corollary 29 (Exact global threshold). *The disk threshold is $\tau = \varphi$, and its infimum is not attained. This holds both for common positive width and for independent positive widths (equivalently, independent holes smaller than the outer radii).*

Proof. Theorem 22 rules out every failure at $\rho \leq \varphi$. The common-width four-ring family of Theorem 19 has failures tending to φ from above and belongs to both classes. $\square$

The algebra of the angular margin, the opposite-gap identity and the row-envelope bounds is certified in `Calamares/ThreeCore.lean`. Inversion, compactness, Descartes and forest assembly remain written proofs. The specialized exchange program that follows supplies stronger information about particular templates; its coverage limits are not premises of this global result.

7 The width program: profile thresholds and the 13/7 corner

For quantitative bounds within restricted exchange templates, the proof of Theorem 9 localizes all difficulty in a single *exchange step*: m is the largest ring that the greedy F and a witness P place in different containers (so F and P agree on the placement of every ring larger than m) $u = c_F(m)$, $v = c_P(m)$, and the set S of rings smaller than m that P keeps in u must be reinserted using the resources freed by moving m to u : the vacated disk D_m (radius r_m), the hole H_m (capacity $r_m - w$, travelling with m), nesting, and the geometry of v . Normalize $r_m = 1$ and $\omega = w/r_m$.

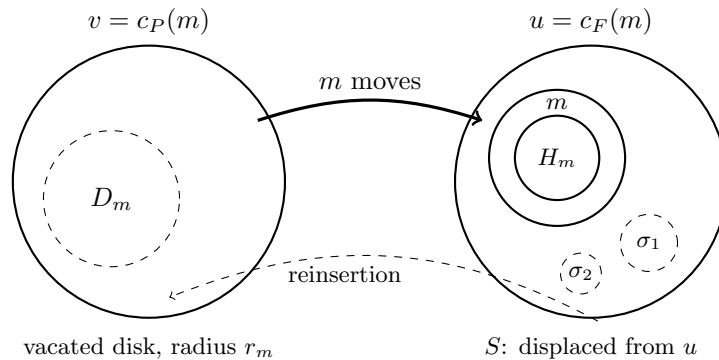


Figure 2: The exchange step of Theorem 9: m is the largest ring placed differently by the greedy F and a witness P . Moving m to u frees the vacated disk D_m (radius r_m) in v and brings along the hole H_m (capacity $r_m - w$); the set S of smaller rings that P keeps in u must be reinserted using D_m , H_m , nesting, and the geometry of v .

Width convention: unless a result explicitly states otherwise, throughout the exchange-step program (this section, the next, and Appendices C and D) we assume $0 < \omega < 1$, so that the pivot m is a genuine ring with a nonempty hole H_m . The model does admit $\omega \geq 1$ (a solid-disk pivot); that regime is outside this specialized program except in results that state “every $\omega > 0$ ” explicitly: Theorem 30, Corollary 61’s Tribonacci closure, Corollary 79, and cases (i)–(ii) of Theorem 86. Their proofs route the $\omega \geq 1$ regime through walls that do not use H_m (the trio frontier and the witness capacity) and so remain valid for a solid pivot. Everything

ρ imposes on S is the pair of inequalities $\sum S \leq \rho$ and $\sum_{l>j} x_l \leq \rho x_j$; a lower bound for ρ over all *blocked* exchanges is therefore a lower bound for any failure of placement obliviousness. We call a *wall* any such necessary condition for blocking: the contrapositive of an explicit unblocking placement, which every blocked exchange must therefore satisfy. Two regimes must be distinguished. When u is the *hole* of a ring (the *nested* exchange), the witness placement supplies the capacity wall $\sum S \leq \text{cap}(u)$, and this section and the next prove floors that all exceed T for a hierarchy of nested templates: the Tribonacci constant is the exact floor of the rigid family and the conjectured universal floor of nested exchanges. When u is the *pan*, that wall is absent, the floor drops to the Descartes pocket, and the golden family of Theorem 19 realizes it: the *pan* exchange governs the global threshold of Theorem 22. Complete proofs for every statement of this section are given in Appendix C.

The combinatorial layer is closed. For two-ring profiles the blocked threshold (a *floor* in the terminology of Section 6: an infimum over blocked exchanges) is exactly $\rho_2^*(\omega) = \max(1, 2(1 - \omega))$; in particular no two-ring blocking is compatible with $\rho < T$ for $\omega \leq 1 - T/2 = 0.0804$. For three rings the threshold is closed,

$$\rho_3^*(\omega) = \max\left(1, \min(2(1 - \omega), \max(\varphi, \frac{2}{1+2\omega}))\right),$$

with a purely additive proof and a *golden plateau* $\rho_3^* \equiv \varphi$ on $\omega \in [\frac{\sqrt{5}-2}{2}, 1 - \frac{\varphi}{2}]$; and a general additive tree argument shows $\rho_k^* = \rho_3^*$ for every $k \geq 3$. Consequently the combinatorial threshold crosses T exactly at

$$\omega_T = \frac{1}{T} - \frac{1}{2} = T^2 - T - \frac{3}{2} = 0.043689 \dots$$

below ω_T the exchange step never blocks with $\rho < T$, with no geometry at all (Propositions 54–57, Theorem 56, Corollary 58).

The canonical template with positive width. For $\omega > \omega_T$ geometry enters through the canonical template (v a diametral pan with one big neighbor α ; u the hole of α , capacity $\alpha - \omega$; $S = \{\sigma_1 \geq \sigma_2\}$ placed there by the witness). Blocking must defeat four resources, giving the walls $\sigma_2 > 1 - \omega$ (H_m), $\sigma_2 > \alpha - \omega - 1$ (σ_2 next to m inside u), $\sigma_1 + \sigma_2 \leq \alpha - \omega$ (the witness), and non-packability of the triple $\{\alpha, \sigma_1, \sigma_2\}$ in the pan (Figure 3).

Two structural facts organize the optimum. First, on the blocking frontier of the triple one has the closed form $t(\sigma_1) + t(\sigma_2) = t(b(\alpha))$ with $t(s) = \sqrt{(1-s)/s}$ (the Descartes pocket read in the t -coordinate), and the frontier slope satisfies the α -independent identity

$$\kappa = -\frac{\partial \sigma_2}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \quad g(s) = s^3(1-s), \quad \kappa \geq 1 \text{ for all } \alpha > 1,$$

so the cheapest blocked pair sits at $\sigma_1 \rightarrow 1, \sigma_2 \rightarrow b(\alpha)$; moreover $1 + b(\alpha) \geq \alpha$ iff $\alpha \leq T$, so no blocking exists at all for $\alpha \leq T$. Second, the witness capacity deforms the cubic: with T_c the positive root of $\alpha^3 = c(\alpha^2 + \alpha + 1)$, the witness branch has infimum $\Phi(\omega) = T_{1+\omega} - \omega$, strictly increasing and concave with $\Phi'(\omega) = (2\alpha + 1)/(\alpha^2(\alpha^2 + 2\alpha + 3))$ at $\alpha = T_{1+\omega}$. The full curve of the blocked infimum $T_{\text{can}}(\omega)$ is obtained in closed form *as a proved lower bound* — the matching upper bound is exact on the witness branch and at the corner below, and inherits the angular-criterion caveat (the criterion is proved sufficient, not necessary, so the matching

the pan (v): diametral pair $\{\alpha, m\}$

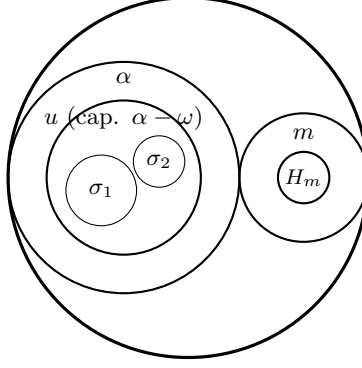


Figure 3: The canonical template: v is a diametral pan with one big neighbor α ; u is the hole of α (capacity $\alpha - \omega$), where the witness keeps $S = \{\sigma_1 \geq \sigma_2\}$. Blocking must defeat four resources, giving the walls $\sigma_2 > 1 - \omega$ (the hole H_m), $\sigma_2 > \alpha - \omega - 1$ (σ_2 next to m inside u), $\sigma_1 + \sigma_2 \leq \alpha - \omega$ (the witness), and non-packability of the triple $\{\alpha, \sigma_1, \sigma_2\}$ in the pan.

upper bound is conditional there) on the other two branches: $2(1 - \omega)$ on $(0, \omega_1]$, a sextic-algebraic middle branch on $[\omega_1, 1/7]$, and Φ on $[1/7, 0.30]$, where $\omega_1 = 0.0413570\dots$ is the root of $4\omega^3 - 20\omega^2 + 25\omega - 1$ in $(1/25, 1/14)$ (Lemma 59, Theorem 60, Corollary 61, Propositions 62 and 64).

Theorem 30 (The 13/7 corner). *For all $\omega > 0$ (including the solid-pivot regime $\omega \geq 1$: for $\omega \geq \frac{1}{7}$ the proof below uses only (B1) and (W), which do not use H_m and remain valid for a solid pivot),*

$$T_{\text{can}}(\omega) \geq \frac{13}{7} = T + 0.017856\dots,$$

with equality exactly in the limit at the rational corner $(\omega, \alpha, \sigma_2) = (\frac{1}{7}, 2, \frac{6}{7})$, where all four walls saturate simultaneously (note $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7} \cdot 7$). The bound is unconditional – in particular it does not depend on the exactness of any angular feasibility criterion – and the approximating family is genuine. Thus positive width only raises the Tribonacci floor, uniformly.

The fine structure is curious: the curve is *not* monotone on $(0, 1/7]$ – ω_1 is a local minimum, followed by a bump of height $1.1 \cdot 10^{-4}$ at $\omega_{\text{peak}} \approx 0.0445$ (a root of an explicit degree-8 polynomial) before descending to the corner.

8 Generic containers: coronas, walls, and metallic floors

The canonical template assumes v contains only the big neighbor and m . Generic occupied containers require additional walls within this specialized template analysis. The following toolkit quantifies those obstructions; it is not needed for the universal golden proof. Throughout, v is a pan of radius R with occupants $\{\alpha\} \cup O \cup \{m\}$, $O = \{o_1 \geq \dots \geq o_j\}$, $o_i \geq m = 1$, u is the hole of α , $S = \{\sigma_1 \geq \sigma_2\}$, and holes may be occupied arbitrarily, at any nesting depth. Complete proofs for every statement of this section are given in Appendix D.

The universal trio frontier. For a trio $\{A, x, y\}$ wall-tangent in a disk of *arbitrary* radius R (interior domain, and under the necessary hypothesis $A \geq \min(x, y)$), the angular blocking

condition is *linear* in the coordinates $T_c(x) = \sqrt{(c-x)/x}$, $c = R - A$: θ -infeasibility $\iff T_c(x) + T_c(y) \leq \tau_R = c/\sqrt{AR}$. The pocket generalizes to $b_R(A) = ARc/(AR + c^2)$, increasing in R , and the frontier slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$, $g_c(s) = s^3(c-s)$, is uniform in R . Without the hypothesis $A \geq \min(x, y)$ the equivalence is false (an explicit pointwise counterexample witnesses the failure; the failure region is swept by script), a fact found by adversarial verification (Lemma 66).

An exact corona criterion. Call a *corona* a packing of k circles all wall-tangent. For a fixed cyclic order the corona constraints form a linear feasibility system in the angular gaps, so coronas are decidable exactly; every subset of a corona yields a necessary *certificate* (its induced cyclic sum of pairwise angles is at most 2π). For $k = 4$ the certificates are exactly sufficient – for arbitrary symmetric separations in $(0, \pi]$, by a negative-cycle argument: a violating cycle with U ascents needs more than $2U$ edges, so on four nodes only $U = 1$ cycles can violate, and those are exactly the subset certificates – and the global criterion collapses to *two* inequalities: the top trio, plus the total of the *zigzag* order (a_1, a_3, a_2, a_4) , which minimizes the cyclic total by a convexity/majorization argument ($\theta = g(\log f(a) + \log f(b))$ with g convex). Two natural guesses are refuted by explicit counterexamples: the consecutive-arc condition alone is *not* sufficient for $k \geq 4$, and the sorted order is *not* optimal (the zigzag is: large circles must not be angular neighbors). For $k = 5$ sufficiency requires exactly one additional certificate, the *pentagram* (the five-diagonal star $\{5/2\}$, with combinatorially sharp example $\theta = \pi$ on the diagonals), conjecturally redundant for separations of geometric origin (Lemmas 67–68, Theorems 69–71).

Occupant walls and metallic floors. Each extra occupant's hole is itself a reinsertion resource, and blocking it is *taxed*. The basic count: with free holes, blocking forces every extra occupant to within one width of m ($o_i < \sigma_2 + \omega$), and the largest occupant's tail gives $\rho > (j+2)/(1+\omega)$ – each occupant pays $1/(1+\omega)$ – whence, with a refined branch $\rho > 4/(1+2\omega)$ for $\omega \geq \frac{1}{2}$, $\rho > T$ for $\omega < 2/T - \frac{1}{2} = 0.5874$ when $j = 1$ and for *all* ω when $j \geq 2$; adding occupants is strictly suboptimal for the blocker, since the canonical curve satisfies $\Phi(\omega) < 2$ identically (the polynomial identity $(2+\omega)^3 - (1+\omega)((2+\omega)^2 + (2+\omega) + 1) = 1$). For occupied holes the tax is quantified by the *opposite-disk lemma*: in a disk of capacity c with largest piece σ , if the remaining mass is at most $c - \sigma$ the whole family packs (the wall-tangent σ leaves a free disk of radius exactly $c - \sigma$ opposite, filled in a row); hence blocking a hole costs at least its slack: every *node* y – an extra occupant o_i , or any ring of radius $\geq r_m$ nested at any depth inside one – satisfies $y < \sigma_2 + \omega + X_y$, where X_y is the total radius of y 's hole content (α and m themselves are excluded: α 's hole is u , governed by its own walls, and excluding m is essential). Following the *minimal* such ring and balancing two tails yields, for arbitrary hole occupancy at any depth,

$$\rho > \Psi(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + 1},$$

the metallic mean of $2(1 - \omega)$ (we write $\Psi \equiv \Psi_1$ interchangeably: it is the $j = 1$ member of the ladder Ψ_j below): $\Psi(0) = 1 + \sqrt{2}$ is the silver ratio, $\Psi(1/4) = 2$ exactly, and $\Psi > T$ iff $\omega < (T - 1)^2/2 = 0.3522$. Allowing m itself to carry children splits by an evacuation dichotomy into the same bound and a second metallic mean, the root of $u^2 - (2 - \omega)u - 1$, which dominates automatically; its own crossing is at $(T - 1)^2$ exactly – twice the previous one, and the certifying identity is the Tribonacci polynomial: $(T - 1)^2 T - (2T - T^2 + 1) = T^3 - T^2 - T - 1$ (Lemmas 72 and 75, Proposition 73, Theorems 77 and 78).

The double-pocket wall and the golden line. The geometric wall needs no angular criterion at all: blocking implies the four rings $\{\alpha, o_1, \sigma_1, \sigma_2\}$ do not pack in the pan, hence not in the disk of radius $\bar{R} = \alpha + o_1$ (containment), where the pair $\{\alpha, o_1\}$ is diametrically rigid and leaves exactly *two* Descartes pockets $b_2(\alpha, o_1) = \alpha o_1(\alpha + o_1)/(\alpha^2 + \alpha o_1 + o_1^2)$, one on each side (their mirror centers are $4b_2$ apart: the exact identity $y_0 = 2b_2$). Thus blocking forces $\sigma_1 > b_2(\alpha, o_1)$. Feeding this wall into the program of the previous walls produces a *golden* optimum: the binding corner is $\alpha = 2$, $o_1 = \sqrt{5} - 1$ – the pair with unit pocket, $b_2(2, \sqrt{5} - 1) = 1$ exactly, self-dual in the sense $A_{\max}(\sqrt{5} - 1) = 2$, $A_{\max}(2) = \sqrt{5} - 1$ – and, for one extra occupant and *any* hole occupancy, both branches of the analysis meet the same line: $\rho > T$ unconditionally for all $\omega < \omega_A$, via

$$\rho > \varphi^2 - \frac{\varphi}{2}\omega, \quad \omega_A = 2 - 2(T - 1)(\varphi - 1) = 0.962585\dots$$

(the golden line itself is proved in all but one corner configuration – the corner with a nested node inside the occupant and $\alpha < o_1$, called branch B in the appendix, possible only for $\omega < \frac{1}{2}$ – where the proved bound is $\geq 2 > T$, via $\Psi_B(\frac{1}{2}) = 2$ exactly). The proof combines two exact facts – b_2 is concave in α ($\partial^2 b_2 / \partial \alpha^2 = -6\alpha o^3(\alpha + o)/D^3$) and $D^2 - \alpha o^2(o + 2\alpha) = (\alpha + o)(\alpha^3 + \alpha^2 o + o^3) > 0$ – with one-variable boundary certificates anchored at the identity $f_1 \equiv G_\omega$ at $o_1 = \sqrt{5} - 1$, where $G_\omega := \varphi^2 - (\varphi/2)\omega$ is the golden-line bound. For $j \geq 2$ occupants the ladder $\Psi_j = (1 - \omega) + \sqrt{(1 - \omega)^2 + j}$ (crossing T at $1 - (T^2 - j)/(2T)$: 0.624 for $j = 2$, 0.896 for $j = 3$, all ω for $j \geq 4$) covers the corresponding ranges unconditionally, for arbitrary nested occupancy of the holes: a *leaf argument* closes the case where occupants' holes contain further rings of radius $\geq r_m$. Every occupant subtree contains a leaf (a node none of whose hole contents reach r_m); taking L the largest of the j disjoint leaves, its tail collects the other $j - 1$ leaves together with m , σ_1 , σ_2 and all small mass W , while the blocking wall at a leaf reads $L < \sigma_2 + \omega + W$; optimizing the resulting two-branch program returns exactly Ψ_j , with the crossing at $\sigma_2 = 1 - \omega$ – the same metallic-mean mechanism as Ψ . Small rings anywhere compatible with each theorem's template – and with S still a pair – are free for all the combinatorial walls, whose unblocking placements are local to D_m , H_m , the holes, and the slot next to m inside u (Lemma 80, Theorem 81, Proposition 82, Corollary 83).

Three-piece profiles: the zigzag pays. Larger reinsertion profiles are *cheaper* for the adversary in pure nest accounting ($\rho_3^* \leq \rho_2^*$), but in the canonical template the geometry reverses the sign: blocking with $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\} \subset (\omega, 1)$ and arbitrary occupancy of H_m forces $\rho > \max(\Phi(\omega), \frac{13}{7})$ for *all* ω , with no exceptional branch. If $\sigma_3 \leq \sigma_1 - \omega$ then σ_3 (the *dust*) rides inside σ_1 and the pair program is inherited wall by wall (the 13/7 corner bound applies); if nothing nests and the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona, the witness chain gives $\sigma_1 + \sigma_2 \geq 1 + b(\alpha) \geq \Phi(\omega)$; and if the trio *does* pack, the U_4 criterion read backwards forces the *zigzag* certificate to fail, and the zigzag implies exactly $\sigma_2 > b(\alpha)$ – the Descartes pocket by a third route, via $\sin^2 A + \sin^2 B > 1$. Crossing that wall with the witness capacity lands on the *golden deformation* γ_ω , the root of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$ with $\gamma_0 = \varphi$, and the tail bounds meet at another rational corner: $\gamma_{2/7} = 2$ exactly ($8 = 4 + 2 + 2$) with value $\frac{17}{7}$ – the sister of the 13/7 corner (Theorem 84, Proposition 85).

Status. Together with Theorem 30, every blocked *nested* exchange in the templates above satisfies $\rho > T$: for all ω in the canonical template – with S a pair or a triple – for $\omega < 0.9626$ with one extra occupant under arbitrary hole occupancy, for $\omega < 0.624$ and $\omega < 0.896$ with

two and three, and for all $\omega \in (0, 1)$ with four or more. All these floors exceed $T > \varphi$. The global golden guarantee is already established by Theorem 22; the following questions concern the stronger nested-template threshold. Of the remaining nested gaps, the width tips at $j = 1$ ($\omega \geq 0.9626$, where the golden line still gives $\varphi^2 - \frac{\varphi}{2}\omega \geq \varphi^2 - \frac{\varphi}{2} = 1.809 > \varphi$) and at $j = 3$ ($\omega \geq 0.896$, where $\Psi_3(\omega) > \Psi_3(1) = \sqrt{3} > \varphi$) matter only for pinning the nested threshold at exactly T . At the golden level itself, the remaining nested configurations – the $j = 2$ tip for $\omega \in [\varphi/2, 1)$, the quantitative “gap lemma” regime (small rings against the geometric walls; tiny σ_2), and profiles of four or more pieces beyond the reduction branch – have written proofs (Theorems 93 and 94: the entire nested template, every j), with duality-certificate backing and independent verification, in Appendix E. Each statement in this section is backed by a verification script and a verification report in the repository (Appendix A).

9 The two objectives diverge: a phase diagram

Contact area behaves like $2\pi w \sum r_i$ minus a per-ring penalty πw^2 ; cardinality counts rings. *Divergence* means the area optimum has strictly smaller cardinality than the cardinality maximum. Two rings never diverge: if both fit together the full set is optimal for both objectives (area is strictly increasing under inclusion), and otherwise every feasible set has at most one ring, a cardinality the area optimum attains. Three rings can already diverge: at $R = 10$, $w = 9/2$, radii $\{8, 101/20, 99/20\}$ (rational-exact; script `divergencia3`, 5/5), the small pair is exactly diametral ($101/20 + 99/20 = 10$), nothing coexists with or nests in the 8 (its hole has radius $7/2 < 99/20$), and the areas compare as $a(8) = \frac{207}{4}\pi > \frac{198}{4}\pi = a(101/20) + a(99/20)$: the area optimum is the single 8 while cardinality prefers the pair. The mechanism is pure superadditivity at large width, where all holes are too small to permit nesting and the problem degenerates to circle packing; sampling places the onset of three-ring divergence near $w/R \approx 0.26$ for this family (swept, not a proved threshold). In the *nesting-active* regime the divergence survives only marginally: small rings that fit k times in the pan but at most $k - 2$ times in the large ring’s hole (with $k - 1$ in the hole the two objectives tie), and the minimal instance of this hole mechanism has four rings: $R = 10$, $w = 1$, radii $\{9.0, 4.2, 4.2, 4.2\}$ — the area optimum is the 9.0 with one nested 4.2 ($N = 2$, $A \approx 76.7$), the cardinality optimum the three 4.2’s ($N = 3$, $A \approx 69.7$). Figure 4 maps the divergence region for the family “one large ring b plus unlimited equal small rings of radius s ” at width $w = 1$, $R = 10$ (the regime of the hole mechanism; script `franja`, which generates the figure): a staircase governed by the proved optimal thresholds for n equal circles in a disk [10, 11, 12, 13, 14]; its upper edge is exactly the three-circle threshold $0.4641R$.

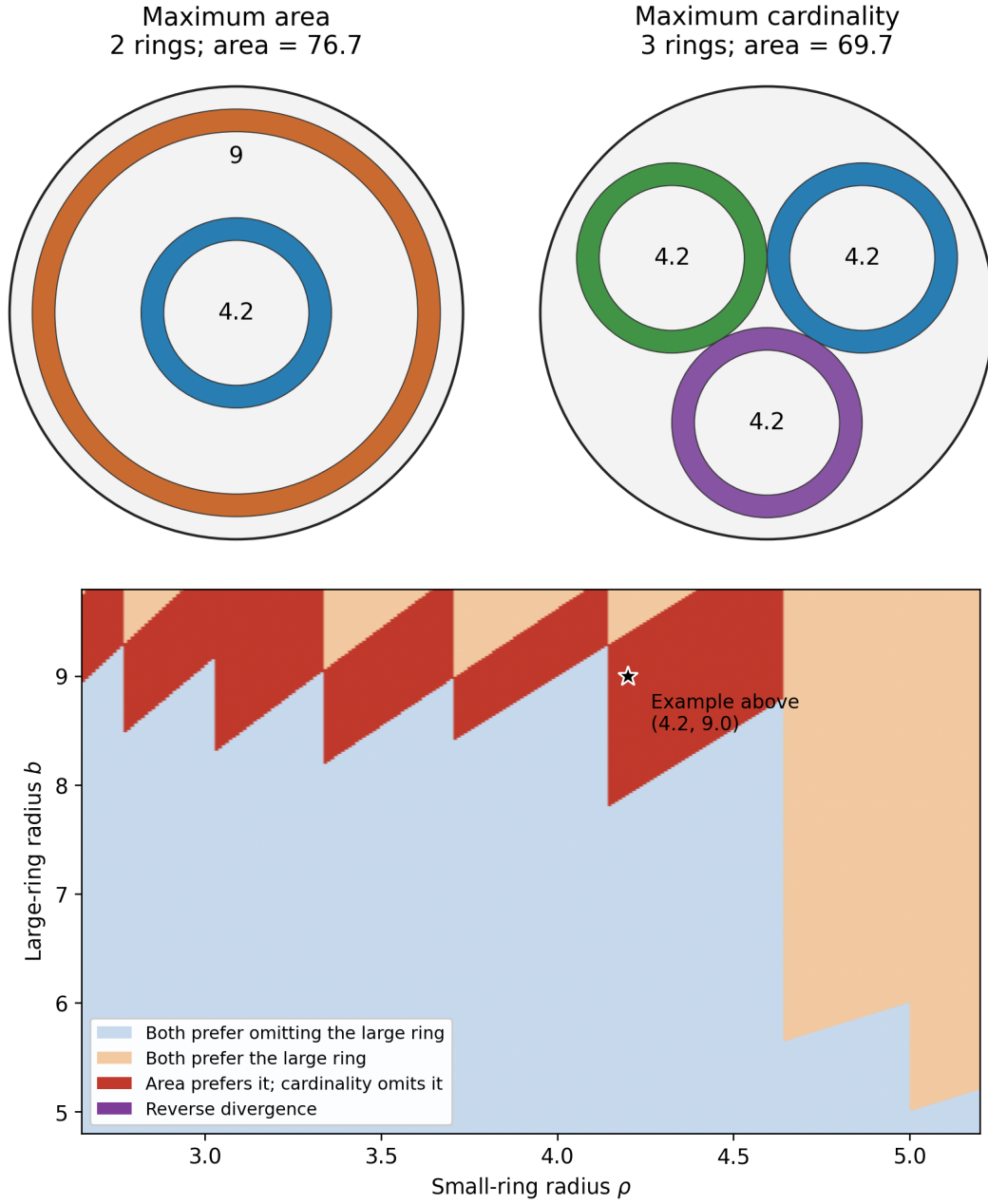


Figure 4: Top: the minimal divergence instance; area optimum (left) versus cardinality optimum (right). Bottom: phase diagram of the divergence band for uniform width.

10 Hardness decouples into two layers

If all radii lie within a band of width less than w , no nesting is possible and the decision problem “do all rings fit?” is exactly circle packing of the outer radii into the pan.

The geometric layer. Since Theorems 6 and 9 hold for arbitrary containers, our model contains, as the special case of a *square* pan with a no-nesting band, the problem of deciding whether given circles fit into a square, which is NP-hard by reduction from 3-PARTITION [6]; hence the geometric layer of the model is NP-hard. Membership in NP is itself delicate, because tight packings may require coordinates without short descriptions – a phenomenon made precise by the $\exists\mathbb{R}$ -completeness framework for two-dimensional packing [8]. For the disk pan specifically, NP-hardness of deciding whether given disks fit into a circular container is established in the journal version of Fekete–Keldenich–Scheffer [7, Theorem 5.1] (the conference version does not contain this result – we thank the referee for the precise pointer). Integer-programming formulations for circle packing in rectangular containers are studied in [18].

The combinatorial layer. Independently of the geometry, the *additive surrogate* (Theorem 16) is weakly NP-hard, and here the reduction can be made exact despite the per-ring penalty in the area.

Proposition 31 (The additive layer is weakly NP-hard). *Deciding whether an instance of the additive surrogate admits a feasible set of contact area at least a given threshold is weakly NP-hard, already for instances with a no-nesting radius band.*

Proof. Reduce from EQUAL-CARDINALITY PARTITION: given positive integers $a_1, \dots, a_n$, n even, with $\sum_i a_i = 2B$, decide whether some S with $|S| = n/2$ has $\sum_S a_i = B$. (This variant is NP-hard: from a PARTITION instance $b_1, \dots, b_m$ with $\sum_i b_i = 2B_0$, take any integer $K \geq 1$ and the $2m$ numbers $b_1 + K, \dots, b_m + K, K, \dots, K$ (m copies of K); an m -subset with t items of the first kind has sum $mK + \sum_{\text{chosen}} b_i$, so m -subsets of sum $mK + B_0$ correspond exactly to subsets of the original instance of sum B_0 , of any size, padded to cardinality m with K -items.) Fix any width $w > 0$, set $M = 3B$, pick $\delta \in (\frac{w}{3B}, \frac{w}{2B})$, and take radii $r_i = \delta(a_i + M)$ with pan capacity $R^+ = \delta(\frac{n}{2}M + B)$. Dividing the area inequality by π and choosing δ rational in the interval makes every datum of the instance rational, so the reduction is fully formal. All radii lie in $[3B\delta, 5B\delta]$, so $r_i > w$ (each is a genuine ring, $a(r) = 2\pi wr - \pi w^2$) and $r_{\max} - r_{\min} = 2B\delta < w$ (no nesting): feasibility is exactly $\sum_S r_i \leq R^+$. Write $s = |S|$ and $t = \sum_S a_i \leq 2B$; the area is $A(S) = 2\pi w\delta(sM + t) - \pi w^2 s$, and we ask whether $A(S) \geq \Theta := 2\pi w\delta(\frac{n}{2}M + B) - \pi w^2 \frac{n}{2}$. If a balanced partition exists, taking S with $s = n/2$, $t = B$ attains Θ exactly. Conversely: $s > n/2$ is infeasible, since it would force $(s - \frac{n}{2})M \leq B - t \leq B < M$. For $s = \frac{n}{2} - q$ with $q \geq 1$, the requirement $A(S) \geq \Theta$ rearranges to $wq \geq 2\delta(qM + B - t)$, but

$$2\delta(qM + B - t) \geq 2\delta(qM - B) \geq 2\delta q(M - B) = 4\delta qB > wq$$

using $t \leq 2B$, $q \geq 1$, $M = 3B$ and $\delta > \frac{w}{4B}$ (implied by $\delta > \frac{w}{3B}$): impossible. Finally $s = n/2$ forces $t \geq B$ from $A \geq \Theta$ and $t \leq B$ from feasibility, i.e. $t = B$: a balanced partition. $\square$

Superincreasing radii eliminate exactly this combinatorial layer – precisely as in the superincreasing knapsack [17] – leaving only the geometric one: by Theorem 9, the area optimum then reduces to at most $O(n^2)$ sibling-packing feasibility queries (per ring, one per currently open

container), or n calls to the stronger oracle “does the current placement admit ring i ”. Useful black-box tools for the remaining geometry include the critical-density theorem for disks in a disk (any family of total area at most half the container’s packs, and $1/2$ is tight) [7] and Split Packing [9].

11 Generalizations

The extensions of container shape and dimension below have complete written proofs. Other changes of the model remain research directions.

11.1 Container shape and dimension

Theorems 6 and 9, and now Proposition 11, hold for arbitrary compact containers in $\mathbb{R}^d$. For spherical containers the negative results also transfer to higher dimensions, for a different reason: each relevant sibling query involves at most three balls.

Lemma 32 (Dimension reduction for sibling queries). *Let $k \geq 2$ and $d \geq k - 1$. Balls of positive radii $a_1, \dots, a_k$ pack in a ball of radius R in $\mathbb{R}^d$ if and only if they pack in one of the same radius in $\mathbb{R}^{k-1}$. A single ball fits if and only if its radius is at most R , in every positive dimension.*

Proof. The forward embedding from lower to higher dimension preserves all center distances and norms. Conversely, center the container at zero and let $c_1, \dots, c_k$ be the centers of a packing. Their affine hull A has dimension at most $k - 1$. Let q be the orthogonal projection of zero onto A . Since q is orthogonal to $A - q$,

$$\|(c_i - q) - (c_j - q)\| = \|c_i - c_j\| \geq a_i + a_j, \quad \|c_i - q\|^2 = \|c_i\|^2 - \|q\|^2 \leq (R - a_i)^2.$$

Here $R - a_i \geq 0$ by the original containment. A common translation of the centers therefore preserves separation and does not increase their norms. Identify $A - q$ isometrically with a subspace of $\mathbb{R}^{k-1}$. This translates the whole affine hull; it does not project the individual centers onto a predetermined lower-dimensional space. $\square$

Corollary 33 (Transfer of the sharpness results). *For spherical pans in every dimension $d \geq 2$, the following hold with the same radii and widths as in the plane:*

1. *placement obliviousness holds for at most three pieces and fails at four, as in Theorem 13;*
2. *the twin instances of Theorem 14 exclude every deterministic state-based rule, and every randomized state-based rule fails on some instance with probability at least $1/2$;*
3. *the rigid family of Theorem 17 has infimum exactly T , not attained at positive width;*
4. *the dimension-dependent geometric threshold satisfies $1 \leq \tau_d \leq \varphi$, and its restriction to inventories of at most four rings satisfies $\tau_{4,d} = \varphi$, unattained.*

Proof. By Lemma 32, every query of at most three siblings in a ball has the same answer in all dimensions $d \geq 2$. A fixed assignment forest whose containers have at most three children can be realized by assembling these local packings from the leaves: each child’s entire subtree stays inside its outer ball.

The positive part of (1) is Proposition 11. Its negative part uses the root pair $\{10, 5\}$, the hole pair $\{4.9, 4.8\}$, and the forbidden root triple $\{10, 4.9, 4.8\}$. All are preserved. The four executions and the two witnesses of Theorem 14 also use only individual, pair and triple queries. The state at the arrival of the 5 is identical in the two inventories, and the necessary choices are still opposite. If the probability of choosing the root is p , the failure probabilities are $1 - p$ and p .

In (3), conditions (F1) and (F2) are scalar, and (F3) is precisely a triple query. Thus the entire admissible parameter family and its function ρ are unchanged, including the approximating sequence. Finally, the witness of Theorem 19 has the root triple $\{\varphi, s_1, s_2\}$ and the nested pivot 1. Its failing execution only queries pairs and triples as well. It gives the same failures with $\rho = \varphi + 3\varepsilon \rightarrow \varphi$. The lower bound in (4) is Theorem 9 under weak superincreasingness. For the four-ring lower bound, the forest reduction in Theorem 21 is dimension-independent. Each of its two remaining cases uses only pairs and triples in spherical containers, so the rigid floor and the diametral-pocket argument apply unchanged. Every four-ring failure therefore still has $\rho > \varphi$; the golden family gives the matching infimum. $\square$

The corollary does not imply $\tau_d = \tau_2$ or $\tau_d = \varphi$: other counterexamples may involve four or more siblings. Nor does monotonicity of packing feasibility in the dimension, by itself, give monotonicity of the threshold for failure of a greedy execution.

Proposition 34 (The first dimensional separation). *There are four distinct sibling radii that fit in the unit ball in $\mathbb{R}^3$ but not in the unit disk in $\mathbb{R}^2$, namely $\{441, 440, 439, 438\}/1000$.*

Proof. Set $u = 8/25$ and use the tetrahedral centers

$$(u, u, u), \quad (u, -u, -u), \quad (-u, u, -u), \quad (-u, -u, u).$$

Each norm squared is $3u^2$ and each pair distance squared is $8u^2$. Even with all radii enlarged to $441/1000$, containment and separation follow from

$$(1 - 441/1000)^2 - 3u^2 = 5281/10^6 > 0, \quad 8u^2 - (882/1000)^2 = 10319/250000 > 0.$$

For planar infeasibility, shrink all radii to $r = 438/1000$. The centers would lie in radius $1 - r$. A center at zero is impossible since $1 - r < 2r$. Otherwise two consecutive centers in angular order have angular separation at most $\pi/2$, so their squared distance is at most $2(1 - r)^2 < (2r)^2$; the difference is $16961/125000$. This contradicts separation. This is a difference of sibling feasibility, not a new failure of placement obliviousness. $\square$

11.2 Square pans: confinement and twin instances

Only the root changes to $[0, s]^2$; every ring's hole is still a disk. The capacity used by best fit at the root is its inradius $s/2$. The following sufficient criterion excludes all placements of a trio, without assuming an optimal tangency pattern.

Lemma 35 (Quadrant confinement). *Let $a \geq b \geq c > 0$, $a \leq s/2$, and $p \geq q \geq 0$. If*

$$p \geq s/2 - b, \quad p^2 + (s - a - b)^2 < (a + b)^2, \quad (1)$$

$$q^2 + (s - a - c)^2 < (a + c)^2, \quad p + q \geq s - b - c, \quad (2)$$

$$L := s - c - a - q \geq 0, \quad 2L^2 < (b + c)^2, \quad (3)$$

then disks of radii a, b, c do not pack in the square of side s .

Proof. Reflect the whole packing about either midline as needed, so the largest center A has both coordinates in $[a, s/2]$. The coordinates of B, C lie in $[b, s - b]$ and $[c, s - c]$, respectively. Each absolute coordinate difference between A and B is at most $s - a - b$. If $|B_x - A_x| \leq p$, the first quadratic inequality contradicts separation. The negative alternative $B_x - A_x < -p$ is excluded by $A_x - B_x \leq s/2 - b \leq p$. Thus $B_x > A_x + p$, and likewise $B_y > A_y + p$, whence $A_x, A_y < s - b - p$.

The same argument with the other quadratic inequality gives $|C_x - A_x| > q$. Its negative alternative is impossible because $A_x - C_x < s - b - p - c \leq q$. Therefore $C_x > A_x + q$ and $C_y > A_y + q$. Since $p \geq q$ and $b \geq c$, both small centers lie in the box $[a + q, s - c]^2$. Their squared distance is at most $2L^2$, contradicting (3). All separations used for packings are non-strict; tangencies have not been excluded. $\square$

Proposition 36 (An exact square counterexample). *With side $s = 4.8568$, width $w = 0.16$ and radii $\{a, 1, b, c\} = \{1.845, 1, 0.844, 0.841\}$, all four rings fit but best fit places exactly three. The dominant tail is $\rho = 337/200$. Every decimal here and below denotes the exact terminating rational.*

Proof. Apply Lemma 35 with $p = 1.591$, $q = 1.581$. The margins in its five inequalities are, in their displayed order,

$$\frac{33}{5000}, \quad \frac{2079}{25000000}, \quad \frac{66559}{25000000}, \quad \frac{1}{5000}, \quad \frac{53587423}{25000000}, \quad L = \frac{2949}{5000}.$$

The witness places a at (a, a) and 1 at $(s - 1, s - 1)$: $2(s - a - 1)^2 - (a + 1)^2 = 16337/25000000 > 0$. In the hole of a , of radius $H = a - w = b + c = 1.685$, place the disk of radius b at relative center $(-c, 0)$ and that of radius c at $(b, 0)$.

Best fit instead nests 1 in a , since $1 \leq H < s/2$. The disk b then fits only at the root: $1 + b > H$ and $b > 1 - w$. Root feasibility follows by shrinking the disk 1 in the witness. At c , the root triple is excluded by the lemma, and all holes are excluded by $1 + c > H$, $c > 1 - w$ and $c > b - w$. The tails are $179/123$, $337/200$, and $841/844$, so the middle one dominates. Together with Proposition 11, this establishes the same three/four-piece transition for a square. $\square$

Theorem 37 (Square twins). *Let $s = 5.1214$, $w = 0.301$, and*

$$I_1 = \{2, 1, 0.850, 0.849\}, \quad I_2 = \{2, 1, 0.950, 0.750\}.$$

Both full inventories are feasible. In I_1 the pivot 1 must be placed at the root for the greedy to succeed; in I_2 it must be nested in 2. Hence no state-based rule succeeds universally, and every randomized state-based rule has failure probability at least $1/2$ on one of these instances.

Proof. The hole of 2 has radius $H = 1.699$. Lemma 35 excludes $\{2, 0.850, 0.849\}$ using $(p, q) = (1.72, 1.718)$, and excludes $\{2, 1, 0.750\}$ using $(p, q) = (2.12, 1.39)$; direct rational substitution verifies all its inequalities. Shrinking the last disk therefore excludes $\{2, 1, z\}$ for every $z \geq 0.750$. On the other hand $\{2, 0.950, 0.750\}$ has the explicit centers

$$(2.96, 2), \quad (0.95, 4.1714), \quad (4.3714, 4.3714).$$

Containment is immediate. The three positive separation margins are

$$\frac{1314449}{25000000}, \quad \frac{663599}{12500000}, \quad \frac{221399449}{25000000}.$$

The root pair $\{2, 1\}$ fits in opposite corners since $2(s - 3)^2 > 9$. Shrinking its unit disk at the same center also packs $\{2, z\}$ for every $0 < z < 1$, justifying admission of the first small ring when the pivot is nested. This pair, with the two small rings inside 2, witnesses I_1 , as $0.850 + 0.849 = H$. For I_2 , use the displayed root triple and nest 1 in 2. Every small ring fits alone in H , but none fits there beside 1, since even $1 + 0.750 > H$. None fits in the pivot's hole of radius 0.699, and the last cannot nest in the preceding small ring: $0.849 > 0.850 - w$ and $0.750 > 0.950 - w$.

If the pivot is nested, both small rings are forced to the root, where the first triple is infeasible and the second feasible. If the pivot is at the root, both are excluded from the root and must use H , where only the first pair fits ($0.950 + 0.750 = 1.700 > H$). All subsequent decisions are therefore forced. At the pivot, the state, side, width and inventory size are identical. If the root is chosen with probability p , the failure probabilities are $1 - p$ and p . The dominant tails are 1.699 and 1.700, respectively. $\square$

11.3 An algebraic upper bound for the square threshold

Define $\tau_\square$ as the infimum of ρ over square-pan instances admitting a failing greedy execution, with side and positive width allowed to vary, as in the definition of τ .

Theorem 38 (An approximating square family). *Let Y be the unique positive root of*

$$(17 + 10\sqrt{2})Y^2 + (72 + 16\sqrt{2})Y - (112 + 96\sqrt{2}) = 0.$$

Then

$$1 \leq \tau_\square \leq Y = 1.684487745872346\dots$$

There are four-ring counterexamples to best fit with strictly decreasing rational radii, positive rational width and rational side, whose values of ρ tend to Y from above. No optimality of Y is asserted, either globally or for all parameters of the confinement criterion.

Proof. Put $k = \sqrt{2}$, $C = 1 + k/2$, and consider first the balanced data $a_0 = 1 + t$, $b_0 = c_0 = t$, $s_0 = C(2 + t)$, $p_0 = q_0 = s_0/2 - t$. The common quadratic margin is

$$G(t) = (a_0 + t)^2 - p_0^2 - (s_0 - a_0 - t)^2 = \frac{(17 + 10k)t^2 + (36 + 8k)t - (28 + 24k)}{8}.$$

It is strictly increasing for $t \geq 0$ and has exactly one positive root $t_0 = Y/2$. Indeed,

$$G(21/25) = \frac{8897}{5000} - \frac{639k}{500} < 0, \quad G(17/20) = \frac{5953}{3200} - \frac{399k}{320} > 0,$$

using $7/5 < k < 10/7$. For $21/25 < t < 17/20$,

$$L_0 = s_0/2 - a_0 = k/2 + (k - 2)t/4, \quad \frac{1}{2} < \frac{229}{400} < L_0 < \frac{5}{7}, \quad 2L_0^2 < \frac{50}{49} < 4t^2.$$

Thus containment of a_0 , positivity of p_0 , and the final confinement inequality have strict margins.

Fix any $t_0 < t < 17/20$ and perturb by $\eta > 0$:

$$\begin{aligned} a &= 1 + t + 2\eta, & m &= 1, & b &= t + \eta, & c &= t - \eta, \\ w &= 1 - t + 2\eta, & H &= a - w = 2t, & s &= C(a + 1), & p &= q = s/2 - t. \end{aligned}$$

At $\eta = 0$ both quadratic margins equal $G(t) > 0$. The strict inequalities just established, including $a < s/2$ and $p > 0$, are continuous in η ; finitely many such inequalities remain strict on some interval $0 < \eta < \eta_0(t)$. The two remaining walls hold identically:

$$p - (s/2 - b) = \eta > 0, \quad p + q - (s - b - c) = 0.$$

Choose also $\eta < t$, $\eta < 1 - t$, and $3\eta < 2t - 1$. Then $a > 1 > b > c > w > 0$, and Lemma 35 excludes the root triple $\{a, b, c\}$.

The witness places $\{a, 1\}$ in opposite corners and $\{b, c\}$ on a diameter of the large hole. Both are legal because $2(s - a - 1)^2 = (a + 1)^2$, both root radii are at most $s/2$, and $b + c = H$. Best fit nests 1, since $1 < H < a < s/2$. The differences excluding the alternative holes for the two small rings are

$$\begin{aligned} 1 + b - H &= 1 - t + \eta > 0, & 1 + c - H &= 1 - t - \eta > 0, \\ b - (1 - w) &= 3\eta > 0, & c - (1 - w) &= \eta > 0, & c - (b - w) &= 1 - t > 0. \end{aligned}$$

The first small ring fits at the root by shrinking the pivot in the witness; the last is excluded everywhere. Best fit places exactly three rings. Moreover

$$\frac{1 + b + c}{a} = \frac{1 + 2t}{1 + t + 2\eta} < 2t, \quad \frac{c}{b} < 1 < 2t,$$

where the first inequality uses $1 < 2t^2 + 4t\eta$ and $t > 21/25$. Thus $\rho = 2t \rightarrow 2t_0 = Y$ from above, proving the upper bound. The lower bound follows from Theorem 9.

For entirely rational instances choose rational $t \downarrow t_0$ and then rational η in the nonempty interval above. Replace $s = C(a + 1)$ by a slightly larger rational and reset $p = q = s/2 - t$. All strict inequalities persist by continuity in s , the two displayed wall identities persist exactly, and the root pair gains strict separation. Density of the rationals completes the construction. The unperturbed balanced limit itself is not a strict-radius counterexample and is not used as one. $\square$

Eliminating the radical shows that Y also satisfies

$$89Y^4 + 1808Y^3 + 4704Y^2 - 9984Y - 5888 = 0.$$

The positive branch is specified by the quadratic over $\mathbb{Q}(\sqrt{2})$, not by an arbitrary choice of a positive root of this quartic. Exact rational examples in the accompanying code attain $\rho = 1.684488$, 1.684487746 , and 1.6844877458724 . These finite checks certify only those instances; the limiting bound uses the continuity proof above.

Formal verification scope. `Square.lean` proves Lemma 35 as a universal Cartesian exclusion over linearly ordered commutative rings, including the reflections. `SquareTwins.lean` proves the required twin exclusions and the explicit feasible triple with the same Cartesian predicate. `SquareLimit.lean` checks the balanced-gap and norm polynomial identities. These files contain 24 theorems, in addition to the 57 earlier algebraic certificates and the three in `FourRing.lean`. The Euclidean interpretation, assignment forests, algorithmic arguments, dimension reduction and continuity remain written proofs; they are not claimed as fully formalized Lean theorems. All new numerical checks use exact rational arithmetic, and all displayed decimals specifying instances are exact.

11.4 Up to five rings in every spherical dimension

The global proof above is stated for the plane. The following separately verified extension covers up to five rings in every spherical dimension; it uses explicit pocket constructions, not a dimension reduction for four or five arbitrary siblings.

Theorem 39 (Five-ring golden threshold). *In a spherical pan of any dimension $d \geq 2$, with strictly decreasing positive outer radii and independent holes $0 \leq h_i < r_i$, every descending greedy execution on at most five rings returns the lex-max set when $\rho \leq \varphi$. Thus $\tau_{\leq 5, d} = \varphi$, with unattained infimum. The threshold is also φ for exactly five rings.*

Proof. The diametral pair and both pocket centers preceding Lemma 25 embed in every dimension $d \geq 2$: all norms and pairwise distances remain unchanged. Two scalar consequences of Lemma 23 suffice. First, for $a > m > p \geq q > 0$, the tail bounds $p + q \leq \varphi m$ and $m + p + q \leq \varphi a$ give $p + q \leq 2\beta(a, m)$, hence $q \leq \beta(a, m)$. Second, for $A > B > m$ and $T > m$, the bounds $m + T \leq \varphi B$ and $B + m + T \leq \varphi A$ give $m + T \leq 2\beta(A, B)$, hence $\beta(A, B) > m$.

Restrict a first disagreement with the lex-max set to the admitted prefix and the omitted ring, as in the proof of Theorem 22, and align a complete witness with the greedy forest. A disagreement at a ring m with at most one smaller ring is resolved by the row exchange. Only the second and third rings can therefore remain. Write u for the destination of m in the greedy forest and $v \neq u$ for its parent in the witness.

At the second ring, write the inventory as $A > m > p > q > t$, omitting absent terms. The disagreement certifies both $R \geq A + m$ and $h_A \geq m$. If there are fewer than two smaller rings, use the row exchange. Otherwise the first scalar consequence above gives $q, t \leq \beta(A, m)$. If u is the root, place the diametral pair A, m there, p in the hole of A , and q, t in its two pockets. If u is the hole of A , use a phantom pair A, m at the root: put p in the phantom m disk and q, t in the pockets, while the real m goes inside A . All smaller rings may be made leaves.

At the third ring, a blocked row exchange requires exactly five rings $A > B > m > p > q$, with p, q both immediate children of u in the witness and $p + q > m$. In particular m has no smaller descendants. The parents of A, B already agree. If both are at the root, then $R \geq A + B$ and $\beta(A, B) > m$. When u is the root, place m, p in its two pockets and q in the disk vacated by m in v . Otherwise put p, q in the pockets and m in its certified hole u .

It remains that B is inside A . If u is the root, its certificate admits the pair A, m ; put q in one pocket and p in the disk vacated in v , keeping B inside A . If u is the hole of A , do the same with the pair B, m inside A . In both cases the first scalar consequence bounds q by the relevant pocket radius. If u is the hole of B , then v is either the root or the hole of A . Use respectively the phantom pair A, m in the root or B, m in the hole of A , put p in the phantom disk and q in a pocket, and place the real m in B . These cases exhaust $u \neq v$. All constructions preserve the larger parents and assemble locally, so each step increases the aligned prefix. This proves the guarantee.

The four-ring family of Theorem 19 supplies the upper bound in every $d \geq 2$, by Corollary 33. For exactly five rings, add a radius $\delta > 0$ in the empty hole of the radius-1 ring in its witness, with $\delta < \min(w, 1 - w, s_2)$. The fifth ring is processed after the existing first failure, which therefore persists. Its addition raises each old tail ratio by at most δ/s_2 . Let δ tend to zero together with the golden-family parameter. The common-width example belongs also to the independent-hole class. The lower bound rules out attainment in either class. $\square$

11.5 Independent holes and the exact area guarantee

Let ring i have outer radius r_i and inner radius h_i , with $0 \leq h_i < r_i$. Its area divided by π is $a_i = r_i^2 - h_i^2 > 0$. There need not be a single value function of outer radius representing these areas.

Proposition 40 (Independent-hole placement). *For arbitrary compact pans in any positive dimension, independent holes preserve placement obliviousness under $\rho \leq 1$, including equality. The unconditional guarantee for at most three rings also remains valid.*

Proof. The parent-alignment proof of Theorem 9 uses only outer radii in the row exchange. At a pivot m , all displaced smaller children have total outer radius at most r_m , so they and their subtrees fit in the outer disk or ball vacated by m . A hole that previously contained m certifies enough capacity for the replacement; no comparison of two widths is needed. The chosen destination admits m with its larger siblings under the greedy certificate. Parent radii strictly increase, so the assembly is acyclic. The same downward closure and maximal-prefix argument applies at equality. For at most three rings, the proof of Proposition 11 similarly only uses $r_3 < r_2 \leq h_1$ when replacing one child by another. $\square$

In a disk, Theorem 22 strengthens the placement guarantee to $\rho \leq \varphi$. Neither placement statement by itself implies optimality of the independent-hole contact areas.

Lemma 41 (Tail capture). *Let L be the lex-max feasible set and $T_i = \sum_{j>i} r_j$. If $i \in L$ and $h_i \geq T_i$, then every $j > i$ belongs to L .*

Proof. In a witness of the selected prefix through i , its hole is empty. Place all later rings there as a row of outer disks. This produces a feasible completion containing every later index, without adding any previously omitted index. Lexicographic maximality gives the claim. $\square$

Theorem 42 (Sharp area factor). *Fix $0 < \kappa < 1$. For independent holes in any compact planar pan and $\rho \leq \kappa$, every descending greedy execution satisfies*

$$A_{\text{greedy}} \geq c(\kappa) A_{\text{opt}}, \quad c(\kappa) = \min(1, \kappa^{-2} - 1).$$

For $\kappa \leq 1/\sqrt{2}$, the lex-max set is the unique area-optimal set. For $1/\sqrt{2} < \kappa < 1$, the constant is best possible even for two rings in a disk; its infimum over positive-optimum instances is not attained.

Proof. The greedy set is L by Proposition 40. Compare it with any distinct feasible set S . Their first difference is an index $i \in L \setminus S$. Let C be the normalized area of their common prefix and B the normalized area of members of S after i . If $h_i \geq T_i$, Lemma 41 gives $S \subsetneq L$, so positivity gives $A(L) > A(S)$. Otherwise $0 \leq h_i < T_i \leq \kappa r_i$, and

$$B \leq \sum_{j \in S, j>i} r_j^2 \leq T_i^2, \quad a_i > r_i^2 - T_i^2 \geq (\kappa^{-2} - 1)T_i^2 \geq c(\kappa)B.$$

Since $C \geq 0$ and $0 < c(\kappa) \leq 1$, this yields $A(L) \geq C + a_i > c(\kappa)(C + B) = c(\kappa)A(S)$, with the common factor π restored if desired. For $c = 1$ this proves uniqueness; otherwise compare with an area optimum. If that optimum is L , the guarantee is immediate and its ratio is $1 > c$.

For sharpness when $\kappa > 1/\sqrt{2}$, take a radius-1 pan and

$$(r_1, h_1) = (1, \kappa - \varepsilon), \quad (r_2, h_2) = (\kappa, 0), \quad 0 < \varepsilon < \kappa - \sqrt{1 - \kappa^2}.$$

The first outer disk fills the root, and its hole is too small for the second ring; no set containing both is feasible. The greedy takes the first, while the optimum is the second. Here $\rho = \kappa$ and

$$\frac{A_{\text{greedy}}}{A_{\text{opt}}} = \kappa^{-2} - 1 + \frac{2\varepsilon}{\kappa} - \frac{\varepsilon^2}{\kappa^2} \longrightarrow \kappa^{-2} - 1$$

from above. Both widths are positive. The strict comparison already proved excludes attainment of this infimum. $\square$

Consequently the universal area-optimality threshold for independent holes is exactly $1/\sqrt{2}$, including its good endpoint. Under strict superincreasing radii alone there is no positive uniform area factor: with $0 < e < 1/4$, use pan radius $1 + e$ and $(r_1, h_1) = (1 + e, 1 - e)$, $(r_2, h_2) = (1, 0)$. The greedy-to-optimum ratio is $4e$, whereas $\rho = 1/(1 + e) < 1$. The seven theorems of `Calamares/VariableWidth.lean` certify the algebra, not the capture lemma or the geometric assembly. The five-ring proof uses the pocket certificates of `FiveRing.lean`; all its placements above are explicit.

11.6 Further model variations

Flexible rings. Real squid rings bend: a ring partially resting on another still touches the pan outside the overlap, up to a lifted ramp of width δ around each overlap. The idealization $\delta = 0$ (contact area equals the annulus minus the overlapped region) defines a continuous relaxation in which partial placements trade contact for cardinality; $\delta > 0$ penalizes overlaps by a dead band proportional to the overlap perimeter.

Infinite inventories. Unlimited supplies (already used in the phase diagram) suggest degenerate limits: maximal nested chains $r, r - w, r - 2w, \dots$, density questions, and Apollonian-type limit configurations.

Other objectives. Theorem 6 applies to positive, strictly increasing, superadditive values. Weighted counts and concave values lie outside this class. Proposition 8 shows that the selection guarantee fails for the constant value $v \equiv 1$, which belongs to both categories.

12 Conclusions

For disk containers, the exact placement threshold is the golden ratio: every descending greedy execution reaches the lex-max feasible set at $\rho \leq \varphi$, whereas four-ring failures approach that value from above. The proof is uniform in the finite inventory size. Golden tail pressure reduces local disk feasibility to the three largest members, and two auxiliary slots resolve every branching exchange while the single-child construction resolves chains. These arguments allow independent hole radii and do not depend on the historical numerical coverage of exchange templates.

Placement independence and area optimality are different properties. With common width, superadditivity supplies the earlier selection guarantees. With independent holes, the sharp area factor is $\min(1, \kappa^{-2} - 1)$ under $\rho \leq \kappa < 1$, and its optimality threshold is $1/\sqrt{2}$. The square bound Y , the rigid-family floor T , and the higher-dimensional five-ring theorem retain their separate scopes. The remaining questions below concern stronger or different claims, rather than a missing case in the global disk proof.

13 Open problems

The universal disk threshold is settled by Theorem 22, including the endpoint and independent holes. The following questions are not consequences of that theorem.

Open Problem 43. Determine the sharp lower floors of the specialized nested-exchange families beyond the rigid family, whose floor is exactly T . Replace the remaining bounded computational certificates by analytic proofs where useful, or exhibit limits of those particular constructions. The historical coverage limits in Appendix E are not unresolved cases of the global golden threshold.

Open Problem 44. Determine the exact square threshold, currently bounded above by Y , and audit the extension of the new global disk theorem to spherical containers in higher dimensions. The independently reviewed dimensional claims in this version remain those stated in Section 11.1 and Theorem 39.

Open Problem 45. Beyond the regimes settled here – including $\rho \leq \varphi$ for disk containers – can a placement rule with access to the full input attain the lex-max set with polynomially many calls to a sibling-packing oracle, or is there a complexity-theoretic obstruction?

Open Problem 46. Characterize the divergence band of Section 9 exactly; determine the approximability of the cardinality objective; decide whether the additive surrogate admits a pseudopolynomial algorithm for integer radii; and settle membership in NP (versus $\exists\mathbb{R}$ -completeness) of disk-into-disk packability, whose NP-hardness is [7, Theorem 5.1].

Acknowledgements

The results of Sections 6–8 were developed with extensive computer assistance (Anthropic’s Claude) under a draft–verify workflow: every result was independently rederived from its bare statement by an adversarial verification pass before being accepted, and several statements were corrected or sharpened by that pass; the reports are part of the repository. To be explicit about epistemic status: the mathematical guarantee for every claim is the written proof and, where a proof delegates an identity, an exact symbolic computation; the verification workflow and the numerical sweeps are quality control and evidence, never a substitute for proof.

The v2 extensions include Theorem 22, the five-ring dimensional guarantee and the area theorem for independent holes. They were developed with OpenAI Codex and reviewed independently using Claude Code/Fable. The dated review payload manifests and verbatim reports are in `docs/reviews/`. Those reviews are static; local execution of Lean and the rational certificates is recorded separately. The new golden proof and its exchange dependencies received two static Fable reviews; their requested clarifications are incorporated. The integrated English exposition and this complete PDF received a local editorial and rendering review, not an additional full-manuscript Fable review.

A Verification map

Every theorem of Sections 6–8 is backed in the accompanying repository by (i) a self-contained draft with complete proofs, (ii) a verification script whose blocks separate exact symbolic identities (`sympy`) from numerical sweeps and whose checks all pass (a few blocks are declared

explorations or statements of convention and cannot fail; the scripts label them as such), and (iii) an *adversarial verification report*: an independent rederivation of each result from its statement alone, followed by an audit and directed attacks. The correspondence is tabulated below; per-round narratives (each refutation, repair and re-verification) are recorded in the repository reports (`docs/drafts/VEREDICTOS.md` and the per-script reports); the Notes column keeps only the headline. Unqualified fractions count script blocks; fractions marked “claims” count verified claims.

Result	Script(s)	Notes
Uniform golden theorem (Thm. 22)	<code>ThreeCore.lean</code> ; <code>code/tres_mayores.py</code>	Seven algebraic certificates; 10 006 placement controls, not a universal numerical proof; two static Fable reviews of the source proof
Five-ring spherical guarantee (Thm. 39)	<code>FiveRing.lean</code>	Eleven certificates; explicit pockets and parent-alignment proof
Independent-hole area factor (Thm. 42)	<code>VariableWidth.lean</code> ; <code>code/grosor_variable.py</code>	Seven certificates; exact examples and negative controls; tail capture written
Arbitrary-container $n \leq 3$; dimension reduction and transfer	<code>docs/drafts/generalizacion_dimensional.md</code>	Written proofs; no packing heuristic used
Four-ring threshold (Thm. 21)	<code>lean/Calamares/FourRing.lean</code>	Two identities and ordered-ring bound; forest reduction written
Square confinement and D (Lem. 35, Prop. 36)	<code>code/cuadrado_certificado.py</code>	17 exact gates; universal Cartesian exclusion in Lean
Square twins and finite improvement	<code>code/cuadrado_gemelas.py</code> , <code>cuadrado_optimizado.py</code>	17+13 exact gates; Cartesian exclusions and witness in Lean
Square limiting bound (Thm. 38)	<code>code/cuadrado_limite.py</code>	39 exact gates on three rational instances; continuity written; three polynomial identities in Lean
Additive threshold (Thm. 16)	<code>code/umbral.py</code>	reproduces the $\rho = 1.03/1.02$ instances
Rigid floor (Thm. 17)	<code>code/rigido.py</code>	10/10 claims; draft <code>suelo_rigido</code>
Slack pans; universal trio frontier	<code>code/universal.py</code>	5/5
κ identity	<code>code/h1.py</code>	6/6 claims
Width curve; Thm. 30	<code>code/grosor.py</code> , <code>code/esquina.py</code>	8/8 and 6/6 claims
Profile thresholds	<code>code/tresk.py</code> , <code>code/cuatrok.py</code>	drafts <code>perfil_tres</code> , <code>cuatro</code>
Corona criterion	<code>code/corona.py</code>	5/5
Occupant walls; opposite disk; metallic floors	<code>code/ocupantes.py</code> , <code>code/bloqueadores.py</code>	5/5 and 6/6
Double pocket; golden line	<code>code/bolsillo.py</code>	6/6
Three-piece profiles; golden deformation	<code>code/striple.py</code>	5/5
Golden family (Thm. 19)	<code>code/aureo.py</code>	5/5

Result	Script(s)	Notes
Pan-exchange golden floor	<code>code/batalla2.py</code> , <code>code/microcelda.py</code>	6/6 and 5/5
Larger profiles	<code>code/perfilp.py</code> , <code>code/rstar.py</code>	5/5 and 6/6
Corona-versus-tails campaign	<code>code/coronacolas.py</code> , <code>code/coronanidada.py</code>	5/5 each
Duality certificates	<code>code/zigzag.py</code>	5/5
Exchange assembly; container port; scaling	<code>code/ensamblaje.py</code> , <code>code/puertocii.py</code> , <code>code/escala.py</code>	6/6, 7/7, 5/5
Written-proof layer: compaction, insertion, gap lemma, hole coronas, geometric tail, optimization, repack	<code>code/compactacion.py</code> , <code>code/insercion.py</code> , <code>code/insercion- anidada.py</code> , <code>code/gaplemma.py</code> , <code>code/coronaagujero.py</code> , <code>code/colageometrica.py</code> , <code>code/optimizacion.py</code> , <code>code/repack.py</code>	5/5–7/7 each
Exact-arithmetic and LP layers	<code>code/r2bcert.py</code> , <code>code/arcolp.py</code> , <code>code/bolsillos.py</code>	5/5 each
Conditional duality lemma; arc-LP to $k = 6$	<code>code/f3cierre.py</code>	5/5; round refuted a stronger closure claim
Multipiece certificates	<code>code/r2bmulti.py</code>	5/5; domain corrected to the sweep box
Golden-threshold reduction; heavy branches	<code>code/areduccion.py</code>	5/5; algebra confirmed by hand
Diametral-pocket geometry ($X_Y > 0$)	<code>code/espxy.py</code>	5/5; an over-claimed legality was recorded as errata
Sliver vacuity; floor rigidity	<code>code/espvals.py</code>	5/5; round supplied a missing wall
Global-tail audit	<code>code/auditcolas.py</code>	5/5; first version refuted, repaired
Forced sub-pocket (near-equal tops)	<code>code/f3vacio.py</code>	5/5
Light mirrored $X_Y > 0$ (k pieces)	<code>code/espkp.py</code>	5/5; dust convention made explicit
Heavy mirrored $X_Y > 0$	<code>code/esppesada.py</code> , <code>code/espfinal.py</code>	5/5 and 6/6; one surgical gap repaired
Occupant channel (light; heavy)	<code>code/espcanal.py</code> , <code>code/espcanalp.py</code>	5/5 each; a residual sheet dissolved, a tower cut repaired
Exact duality-gap converse	<code>code/f3converso.py</code>	5/5; first round refuted the draft, all repairs re-verified
C3.3 corona wall (box certificate)	<code>code/rstarcert.py</code>	5/5; 71 boxes
$2 \rightarrow 3$ divergence	<code>code/divergencia3.py</code>	5/5; rational-exact counterexample
Phase-diagram figure	<code>code/franja.py</code>	generator
D1 shadow budget (box certificate)	<code>code/insercioncert.py</code>	5/5; 149+173 boxes, all occupant counts
Branch-B golden-line patches	<code>code/goldencert.py</code>	5/5; exact Sturm isolation
Gap-lemma cross-derivation	<code>code/gaplemmacert.py</code>	5/5; contrast only

Result	Script(s)	Notes
Width tails $\omega > 1.6$; $Y > 6.6$ tails	<code>code/espomegacola.py</code> , <code>code/r2bcolas.py</code>	5/5 each
Deep-tower mid band (light; heavy)	<code>code/esptorre.py</code> , <code>code/esptorrep.py</code>	5/5 each
Width tail of all six channel cells	<code>code/espomegacanal.py</code>	first round refuted two coverage gaps; repaired, re-confirmed
Slot-reduction lemma; placement engine	<code>code/lemaA.py</code>	5/5; engine rebuilt after refutation, confirmed against an LP oracle
G-b' over all Y , mass, piece counts	<code>code/lemaA2.py</code>	6/6; two unsound caps repaired before refereeing
Heavy assembly murals	<code>code/lemaA3.py</code>	5/5; a $k \geq 6$ block cap repaired
G-e/G-g		
Light channel, $k \geq 2$ extras	<code>code/lemaA4.py</code>	domain $\omega \leq 1.25$ (leaf extras, full nested-mass range), parents on $\omega \leq 1.05$, $W_v \leq 8$, and the $j_v \leq 1$ slice of $[1.6, 2]$; nine rounds, four intermediate refutations repaired
Heavy profile, $k \geq 2$ extras	<code>code/lemaA5.py</code>	$\omega \leq 1.25$ after the certified-corner extension; leaf extras, $W_z \leq 34$, full W_v range
Quintet certificate of Theorem 94	<code>code/quintetocert.py</code>	4/4; whole domain, 11,973 boxes, tolerance-free rational acceptance, no platform hypotheses; supersedes <code>code/bolsillos.py</code> and <code>code/arcolp.py</code> for the $j = 1$ quintet

Several statements were corrected or sharpened by the adversarial rounds (a missing hypothesis in the trio frontier, a refuted “leak” family, the $\alpha \geq o_1$ case split, a refuted induction in the duality layer, a repaired extension lemma and case split in the assembly); the reports record each incident.

A fourth layer is machine-checked. The repository contains a Lean 4 development (`lean/`, core only, no `mathlib`) that formalizes the *exact-certificate layer* on which the proofs rest: arithmetic in $\mathbb{Q}$ and $\mathbb{Q}[\sqrt{5}]$ with a decidable real order, and polynomial identities over $\mathbb{Z}[A]$, $(\mathbb{Q}[e])[d]$ and $(\mathbb{Q}[\sqrt{5}])[r]$. Its original fifty-seven theorems cover the golden arithmetic of Theorem 19 ($\varphi^2 = \varphi + 1$, $b_2(\varphi, 1) = \varphi/2$, the fixed point and its factorization, the witness certificate at $s^* = 4(\sqrt{5} - 2)$, the ε_0 margin and the width window), the metallic-mean certificates of Theorem 86 (including $\Psi(1/2) = \Psi_B(1) = \Psi_2(\varphi/2) = \varphi$ and the monotonicity certificate for g), the Tribonacci floor ($P(\varphi) = -1$, the bracket $1.8392 < T < 1.8393$ and a certificate that P increases strictly on $[1, \infty)$), the 13/7 corner, the additive threshold family, the pin-ner and boundary-corner certificates of Theorem 88, and the duality-certificate identities of Appendix E (the vanishing Descartes discriminant and pocket $\varphi/2$ of the golden pair, the critical margin identity, the (N) -heritage golden line, the mirror corner $b_2(2, \sqrt{5} - 1) = 1$, the strong-breathing chain $\varphi(3 - \varphi) = 2\varphi - 1 = \sqrt{5}$, $\sqrt{5} - (\varphi - 1) = \varphi$, the golden π -trio identity behind $\theta(\varphi, 1/\varphi) + \theta(1/\varphi, 1) + \theta(1, \varphi) = \pi$ at $R = 2\varphi$, the reduction-lemma thresholds $t_0 = (\varphi - 1)/4 = 1/(4\varphi)$ with $5t_0 = \varphi - \beta^*$ and $4t_0 = \varphi - 1$ exact at $\beta^* = (9 - \sqrt{5})/8$, the

diametral-pocket identities $\varphi^2 + \varphi + 1 = 2\varphi^2$, $x^*(\varphi) = \varphi/2$ and $\varphi(4 - \varphi) = 3\varphi - 1$, and the forbidden-triple and golden-cubic certificates of the near-equal-tops vacuity $-2 \cdot (\varphi/2) = \varphi$, the polynomial identity $r(1 + r + r^2) + r(1 + r) - \varphi(1 + r + r^2) = r^3 + (2 - \varphi)r^2 + (2 - \varphi)r - \varphi$ in $(\mathbb{Q}[\sqrt{5}])[r]$, $q(9/10) = 171/271$ and the ceiling inequality $\varphi - 9/10 - 171/271 < 25/83$; and, from the extra-ring campaign, the tail-slope identities $2/\varphi = \sqrt{5} - 1 > 1$ and $(2/\varphi)(2/\varphi - 1) = 7 - 3\sqrt{5}$ of the W_z -tail's upper piece with the numerator coefficient $(6 + 2\sqrt{5})(7 - 3\sqrt{5}) = 12 - 4\sqrt{5}$, the heavy-mass bound $2\varphi/3 < 1.079$ with its genuine extremum at $b = \varphi/3$, and the spillover-bound identities of Theorem 92 with $\varphi - 1 = 1/\varphi$; and the exact anchors of the quintet certificate — the stacked-witness margin $2\varphi^4 - (\varphi + 2)^2 = \varphi - 1$ at the golden point, $2\varphi - 1 = \sqrt{5}$ with $\sqrt{5} \geq 2$ for the mid-arc criterion, and the semiarc threshold collapse $1 + f(2/\varphi) + f(2) = 1 + \sqrt{5}$ at $R = 1 + \sqrt{5}$. The v2 adds 24 square theorems, described in Section 11.3, and three four-ring algebraic theorems, bringing that stage to 84. Later modules add 18 certificates for five rings and variable width, three for balanced partitions, three for two-tail pressure, seven for reservoir bounds, and seven for the three-largest-disks proof: 122 in the repository. The reservoir bounds and the historical adjacent-swap classification are not premises of Theorem 22. There are no `sorry`s and no new axioms (`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`); `native_decide` is not used, so every proof is checked by the Lean kernel. The square's Cartesian inequalities are formalized. Corona rigidity, angular criteria, placement trees, evacuations, dimension reduction and continuity remain written proofs, with supporting scripts and reports.

B Complete proof of Theorem 17

Throughout, work in the normalization of Theorem 17: $r_1 = 1$, $t = r_2$, $p = r_3$, $q = r_4$, width $\omega > 0$, pan radius $R = 1 + t$, and recall $b(t) = t(1 + t)/(1 + t + t^2)$, $T^3 = T^2 + T + 1$, $t^* = 1/T$ (equivalently $t^{*3} + t^{*2} + t^* = 1$), $\varphi = (1 + \sqrt{5})/2$. All exact algebraic identities used below are additionally verified in `code/rigido.py` (block V5).

Lemma 47 (Half-angle identity). *Let two circles of radii a, b be internally tangent to the wall of a disk of radius R (centers at distances $R - a$, $R - b$ from the center). If $a + b \leq R$, the minimal angular separation $\theta(a, b)$ guaranteeing disjoint interiors is well defined, lies in $[0, \pi]$, satisfies*

$$\sin^2 \frac{\theta(a, b)}{2} = f(a)f(b), \quad f(x) = \frac{x}{R - x},$$

and is increasing in each argument; two wall-tangent circles at angular separation γ have disjoint interiors iff $\gamma \geq \theta(a, b)$.

Proof. By the law of cosines the center distance $D(\gamma)^2 = (R - a)^2 + (R - b)^2 - 2(R - a)(R - b)\cos \gamma$ is increasing on $[0, \pi]$, and disjointness is $D \geq a + b$. At the threshold, $1 - \cos \theta = [(a + b)^2 - (a - b)^2] / (2(R - a)(R - b)) = 2ab / ((R - a)(R - b))$, i.e. $\sin^2(\theta/2) = f(a)f(b)$; and $f(a)f(b) \leq 1$ is exactly $a + b \leq R$. Monotonicity is that of f . $\square$

Lemma 48 (Construction). *Let $q \leq p \leq t < 1$ and $R = 1 + t$. If $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$, then $\{1, p, q\}$ packs in the disk of radius R .*

Proof. All three angles are defined ($1 + p \leq R \iff p \leq t$, and $p + q \leq 2t < R$). Place the three circles wall-tangent with the unit circle in the middle: centers at angles 0 , $+\theta(1, p)$, $-\theta(1, q)$. Each circle is contained in the disk (internal wall tangency). The pairs $(1, p)$ and $(1, q)$ are

separated by exactly their thresholds. For (p, q) , the angular difference is $\Delta = \theta(1, p) + \theta(1, q)$ and the separation is $\gamma = \min(\Delta, 2\pi - \Delta)$. If $\gamma = \Delta$: by monotonicity (Lemma 47, $1 \geq p$), $\theta(1, q) \geq \theta(p, q)$, so $\Delta \geq \theta(p, q)$. If $\gamma = 2\pi - \Delta$: the hypothesis gives $2\pi - \Delta \geq \theta(p, q)$. $\square$

Lemma 49 (Algebraic reduction). *Define $\psi(x) = \sqrt{(t-x)/x}$ (decreasing, $\psi(t) = 0$) and $\tau_t = t/\sqrt{1+t}$; note the exact identities $\psi(b(t)) = \tau_t$ and $t/(1+\tau_t^2) = b(t)$. If $q \leq p \leq t < 1$ and $\psi(p) + \psi(q) \geq \tau_t$, then $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$, and hence $\{1, p, q\}$ packs in the disk $R = 1+t$ by Lemma 48.*

Proof. Let $A = \sqrt{f(1)f(p)}$, $B = \sqrt{f(1)f(q)}$, $C = \sqrt{f(p)f(q)}$ (the sines of the half-angles); the goal is $\arcsin A + \arcsin B + \arcsin C \leq \pi$. If $\arcsin A + \arcsin B \leq \pi/2$ there is nothing to prove, since $\arcsin C \leq \pi/2$. Otherwise the goal is equivalent, both sides lying in $[0, \pi/2]$, to $C \leq \sin(\arcsin A + \arcsin B) = A\sqrt{1-B^2} + B\sqrt{1-A^2}$. Dividing by $C > 0$ and writing $G(x) = f(1)/f(x) - f(1)^2$,

$$\frac{A\sqrt{1-B^2}}{C} = \sqrt{G(q)}, \quad \frac{B\sqrt{1-A^2}}{C} = \sqrt{G(p)}, \quad G(x) = \frac{(1+t)(t-x)}{t^2x},$$

the last identity by direct computation with $f(1) = 1/t$, $f(x) = x/(1+t-x)$; the radicals are real because $f(1)f(x) \leq 1 \iff x \leq t$. Thus $\sqrt{G(x)} = (\sqrt{1+t}/t)\psi(x)$ and, by hypothesis, $\sqrt{G(p)} + \sqrt{G(q)} = (\sqrt{1+t}/t)(\psi(p) + \psi(q)) \geq (\sqrt{1+t}/t)\tau_t = 1$, which is the required inequality. $\square$

The contrapositive used below: hypothesis (F3) implies $\psi(p) + \psi(q) < \tau_t$. This is the only direction ever used – the *constructive* one – so no exactness of any angular criterion is assumed anywhere in this appendix.

Lemma 50 (Corner minimization). *Let $0 < q \leq p \leq t < 1$ with $\psi(p) + \psi(q) < \tau_t$. Then: (1) $p, q > b(t)$; (2) if $t \geq 1/\varphi$ then $p+q > 1$; (3) if $t < 1/\varphi$ then $p+q > t+b(t)$.*

Proof. (1) Each summand is $< \tau_t$, ψ is decreasing and $\psi(b(t)) = \tau_t$. (2) $p+q > 2b(t)$ and $2b(t)-1 = (t^2+t-1)/(t^2+t+1) \geq 0$ exactly when $t \geq 1/\varphi$. (3) Substitute $\alpha = \psi(p)$, $\beta = \psi(q)$, so $p = U(\alpha)$, $q = U(\beta)$ with $U(z) = t/(1+z^2)$ strictly decreasing; the region is $\alpha, \beta \geq 0$, $\alpha + \beta < \tau_t$. Increasing α to $\tau_t - \beta$ decreases p , so $p+q > U(\tau_t - \beta) + U(\beta) =: W(\tau_t - \beta)$. Now $U''(z) = 2t(3z^2 - 1)/(1+z^2)^3 \leq 0$ for $z \leq 1/\sqrt{3}$, and $\tau_t \leq 1/\sqrt{3} \iff 3t^2 - t - 1 \leq 0 \iff t \leq (1 + \sqrt{13})/6 = 0.7676\dots$, which covers $t < 1/\varphi = 0.618\dots$ with room to spare. Hence U is concave on $[0, \tau_t]$, W is concave, and its minimum over $[0, \tau_t]$ is at the endpoints: $W \geq W(0) = U(0) + U(\tau_t) = t + t/(1+\tau_t^2) = t + b(t)$, by the identity $t/(1+\tau_t^2) = b(t)$. $\square$

Proof of the strict floor in Theorem 17. Let $I \in \mathcal{F}$, normalized. By (F3) and the contrapositive of Lemmas 48–49, $\psi(p) + \psi(q) < \tau_t$. If $t \geq 1/\varphi$, Lemma 50(2) gives $p+q > 1$, contradicting (F2) ($p+q \leq 1-\omega < 1$); so $t < 1/\varphi$ and Lemma 50(3) gives the blocking pressure $p+q > t+b(t)$. Chaining with the witness pressure (F2),

$$t + b(t) < p + q \leq 1 - \omega < 1,$$

and $t + b(t) < 1 \iff t(1+t) < (1-t)(1+t+t^2) = 1-t^3 \iff t^3 + t^2 + t < 1 \iff t < t^*$ (which also proves that $\mathcal{F}$ forces $t < t^*$). Finally

$$\rho \geq \rho_2 = \frac{p+q}{t} > \frac{t+b(t)}{t} = 1 + \frac{1+t}{1+t+t^2} =: L(t),$$

and L is strictly decreasing ($L' = -(t^2+2t)/(1+t+t^2)^2 < 0$) with $L(t^*) = 1/t^* = T$ (an identity modulo $t^{*3} + t^{*2} + t^* = 1$). Since $t < t^*$: $\rho > L(t) > L(t^*) = T$. $\square$

It remains to prove exactness of the infimum. Two ingredients: the rigid pocket is exact, and infeasibility survives small perturbations.

Proposition 51 (Exact rigid pocket). *For $0 < q \leq t < 1$, the triple $\{1, t, q\}$ packs in the disk of radius $R = 1 + t$ if and only if $q \leq b(t)$.*

Proof. Rigidity. Centers satisfy $|c_1| \leq t$, $|c_t| \leq 1$ and $|c_1 - c_t| \geq 1 + t$; the triangle inequality forces equality throughout, so up to rotation $c_1 = (-t, 0)$, $c_t = (1, 0)$. *Sufficiency.* The pocket circle tangent to the wall and to both has radius $b(t)$; any $q \leq b(t)$ placed concentrically inside it works. *Necessity.* Let X be the center of q and $d = |X|$; if $d = 0$ then $|X - c_1| = t < 1 + q$, infeasible, so $d > 0$. Let γ be the angle between X and $(1, 0)$. Disjointness reads $d^2 + t^2 + 2dt \cos \gamma \geq (1 + q)^2$ and $d^2 + 1 - 2d \cos \gamma \geq (t + q)^2$, which bound $\cos \gamma$ from below and above; compatibility of the two bounds requires $(1 + t)d^2 \geq (1 + q)^2 + t(t + q)^2 - t(1 + t)$, and with $d \leq R - q = 1 + t - q$:

$$(1 + t)(1 + t - q)^2 - (1 + q)^2 - t(t + q)^2 + t(1 + t) \geq 0.$$

The left-hand side factors *exactly* as $4(t^2 + t + 1)(b(t) - q)$ (symbolic identity, V5), whence $q \leq b(t)$. $\square$

Lemma 52 (Closure and monotonicity of packability). *Let $D = \{(t, p, q) : 0 < q \leq p \leq t < 1\}$ and say $z = (t, p, q)$ packs if there are centers with $|c_1| \leq t$, $|c_2| \leq 1 + t - p$, $|c_3| \leq 1 + t - q$, $|c_1 - c_2| \geq 1 + p$, $|c_1 - c_3| \geq 1 + q$, $|c_2 - c_3| \geq p + q$. Let $E \subseteq D$ be the packable set. Then: (1) E is downward closed in (p, q) ; (2) E is closed in D ; (3) for fixed t, q , the set $\{p \in [q, t] : (t, p, q) \in E\}$ is empty or a closed interval $[q, p_{\max}]$ with the maximum attained; in particular, if $(t, t, q) \notin E$, then (when the interval is nonempty; if it is empty every p is infeasible and any δ serves) $(t, p, q) \notin E$ for all $p \in (p_{\max}, t]$ – the infeasibility of the rigid corner propagates to $\delta := t - p \in [0, \delta_0)$, $\delta_0 = t - p_{\max} > 0$.*

Proof. (1) The same centers witness (all six constraints relax). (2) Witness centers live in a fixed compact ball; take a convergent subsequence of witnesses along $z_k \rightarrow z$ and pass to the limit in the six non-strict continuous constraints. (3) By (1) the set is an initial segment, by (2) it is closed. $\square$

Proposition 53 (The infimum is T and is not attained). $\inf\{\rho(I) : I \in \mathcal{F}\} = T$.

Proof. $\geq$ and non-attainment: the strict floor above. $\leq$: fix $t < t^*$, set $\varepsilon_t = (1 - t - b(t))/2 > 0$ and $q_t = b(t) + \min(\varepsilon_t, (t - b(t))/2)$. By Proposition 51, (t, t, q_t) is infeasible; by Lemma 52(3) there is $\delta \in (0, \min(\varepsilon_t, t - q_t))$ with $(t, t - \delta, q_t)$ infeasible. (Write $\rho_i = (\sum_{j>i} r_j)/r_i$ for the i th tail ratio.) The instance $I_t = (1, t, t - \delta, q_t)$ with $\omega = 1 - (t - \delta + q_t) > 0$ satisfies (F1)–(F3) and the strict orderings, so $I_t \in \mathcal{F}$, and $\rho(I_t) = \rho_2 \leq (t + b(t) + \varepsilon_t)/t = L(t) + \varepsilon_t/t \rightarrow L(t^*) = T$ as $t \rightarrow t^*$ (the other tails are smaller: $\rho_1 = t + p + q < 3t < T$ and $\rho_3 = q/p < 1$). $\square$

Proof of Remark 18. A packing in a disk of radius $R' \leq R$ is a packing in the disk of radius R (translate to concentric position; containment is preserved, pairwise constraints are unchanged). Hence non-packability at radius R implies non-packability at radius $r_1 + r_2 \leq R$, so an instance with a slack pan satisfying (F2) and (F3) at its own radius satisfies (F2)–(F3) at radius $r_1 + r_2$, and the floor applies verbatim; the approximating family of Proposition 53 has $R = r_1 + r_2$, which is admissible, so the infimum is unchanged. $\square$

C Complete proofs for the width program

This appendix proves every statement of Section 7. Throughout, normalize $r_m = 1$ and $\omega = w/r_m \in (0, 1)$, and write $\beta := 1 - \omega$. The reinsertion resources of the exchange step are the two root containers A (the vacated disk D_m , capacity 1) and B (the hole H_m , capacity β), plus recursive nesting inside S itself: a ring x fits in the hole of a ring y iff $x \leq y - \omega$. For siblings in a disk of capacity c we use: one circle, $s \leq c$; two circles, $s + s' \leq c$, which is *exact* (sufficiency by tangent placement on a diameter; necessity from $s + s' \leq |c_1 - c_2| \leq |c_1| + |c_2| \leq (c - s) + (c - s')$); and for three or more only *constructive* certificates – the row of Lemma 3 and explicit placements – so that every lower bound below is unconditional. For a profile $S = \{s_1 \geq \dots \geq s_k\} \subset (0, 1)$,

$$\rho_{\text{needed}}(S) = \max\left(\sum S, \max_j (\sum_{l>j} s_l) / s_j\right),$$

and $\rho_k^*(\omega) = \inf\{\rho_{\text{needed}}(S) : S \text{ not reinsertable in } A \uplus B\}$. Two elementary facts are used without mention: ρ_{needed} is monotone under adding smaller rings, and if a prefix $\{s_1, \dots, s_j\}$ is already non-reinsertable then so is S (every placement restricts).

C.1 Profile thresholds

Proposition 54 (Two rings). *$S = \{\sigma_1 \geq \sigma_2\}$ is non-reinsertable iff $\sigma_1 + \sigma_2 > 1$ and $\sigma_2 > 1 - \omega$. Hence $\rho_2^*(\omega) = \max(1, 2(1 - \omega))$, approached by $\sigma_1 = \sigma_2 \rightarrow \max(\frac{1}{2}, 1 - \omega)^+$.*

Proof. The possible placements are: both in A (iff $\sigma_1 + \sigma_2 \leq 1$, two-circle exact); one in each nest (the one in B measures $\geq \sigma_2$, so it requires $\sigma_2 \leq \beta$); σ_2 nested in σ_1 (requires $\sigma_2 \leq \sigma_1 - \omega \leq \beta$); both in B (stronger than one in each). Both stated conditions kill all four; failure of either condition enables the first or second placement. The infimum: blocking forces $\rho \geq \sigma_1 + \sigma_2 > \max(1, 2\beta)$, and the twin family attains the bound in the limit (its tail ratios tend to 1 and 0). $\square$

Proposition 55 (Three rings: characterization). *Let $s_1 \geq s_2 \geq s_3$ in $(0, 1)$. S is non-reinsertable iff exactly one of the following holds (the cases are disjoint by the position of s_1, s_2 relative to β):*

- (i) $s_2 > \beta$ and $s_1 + s_2 > 1$ (then s_3 is arbitrary);
- (ii) $s_2 > \beta$, $s_1 + s_2 \leq 1$, $s_3 > \beta$, and the trio does not pack in the unit disk (possible only for $\omega > \frac{1}{2}$);
- (iii) $s_2 \leq \beta < s_1$, $s_1 + s_3 > 1$, $s_2 + s_3 > \beta$, and $s_3 > s_1 - \omega$;
- (iv) $s_1 \leq \beta$, $s_2 + s_3 > 1$, and $s_3 > s_1 - \omega$.

Proof. The nesting forests on three rings are finite: no edge; one edge (s_3 in s_1 , s_3 in s_2 , s_2 in s_1); the chain $s_3 \subset s_2 \subset s_1$; the star $\{s_2, s_3\}$ in the hole of s_1 . Top-level rings are distributed among A and B under the sibling rules above; S is reinsertable iff some combination is feasible.

(i). Neither s_1 nor s_2 fits in B ($> \beta$) and s_2 cannot nest in s_1 ($s_2 \leq s_1 - \omega < \beta < s_2$ is absurd), so both must go to A , and $s_1 + s_2 > 1$ forbids it: blocked for every s_3 . Conversely, with $s_2 > \beta$ and $s_1 + s_2 \leq 1$: if $s_3 \leq \beta$, place $\{s_1, s_2\}$ in a row in A and s_3 in B ; if $s_3 > \beta$, all three live in the band $(\beta, 1)$, B is useless, nobody nests ($s_i - \omega < \beta < s_3$), so all three must go to A and reinsertability is exactly packability of the trio in the unit disk – case (ii), whose condition

is also sufficient by the same argument. Habitability of (ii): $s_1 + s_2 > 2\beta$ and $s_1 + s_2 \leq 1$ force $\omega > \frac{1}{2}$.

(iii). ($\Leftarrow$) $s_1 > \beta$ has no parent and does not fit in B : $s_1 \in A$. No edge survives: $s_3 > s_1 - \omega \geq s_2 - \omega$ kills all three. A companion of s_1 in A needs $s_1 + s_i \geq s_1 + s_3 > 1$: impossible. So s_2, s_3 must both go to B , and $s_2 + s_3 > \beta$ forbids it. ($\Rightarrow$) If $s_1 + s_3 \leq 1$: $\{s_1, s_3\}$ in A , s_2 in B . If $s_2 + s_3 \leq \beta$: $\{s_2, s_3\}$ in B , s_1 in A . If $s_3 \leq s_1 - \omega$: nest s_3 in s_1 , $s_1 \rightarrow A$, $s_2 \rightarrow B$.

(iv). ($\Leftarrow$) $s_2 + s_3 > 1$ makes *every* pair infeasible in any root ($s_i + s_j \geq s_2 + s_3 > 1 > \beta$), so each root hosts at most one top-level ring; two roots cannot host three rings unless someone nests, and $s_3 > s_1 - \omega$ kills all edges as in (iii). ($\Rightarrow$) If $s_2 + s_3 \leq 1$: $\{s_2, s_3\}$ in A , s_1 in B ($s_1 \leq \beta$). If $s_3 \leq s_1 - \omega$: nest s_3 in s_1 , $s_1 \rightarrow A$, $s_2 \rightarrow B$. $\square$

Theorem 56 (Three rings: closed formula). *For all $\omega \in (0, 1)$,*

$$\rho_3^*(\omega) = \max\left(1, \min(2(1 - \omega), \max(\varphi, \frac{2}{1+2\omega}))\right),$$

with extremal families (as $\varepsilon \rightarrow 0^+$): $\{\frac{1}{2} + \omega, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ on the hyperbola, $\{1/\varphi, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ on the golden plateau $\omega \in [\frac{\sqrt{5}-2}{2}, 1 - \frac{\varphi}{2}]$, $\{\beta + \varepsilon, \beta + \varepsilon, \delta\}$ with $\beta = 1 - \omega$ on $\omega \in [1 - \frac{\varphi}{2}, \frac{1}{2}]$ (there $2(\beta + \varepsilon) > 1$: the pair cannot row), and $\{\frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon, \delta\}$ for $\omega \geq \frac{1}{2}$ (where $\rho_3^* = 1$; the family $\{\beta + \varepsilon, \beta + \varepsilon, \delta\}$ rows into the unit disk there and no longer blocks). The proof is purely additive: neither case (ii) nor any three-circle geometry enters.

Proof. We bound the infimum of ρ_{needed} over each case of Proposition 55 and take the minimum.

Case (i). $\rho \geq s_1 + s_2 > \max(1, 2\beta)$, and the family $s_1 = s_2 = \max(\beta, \frac{1}{2}) + \varepsilon$, $s_3 = \delta \rightarrow 0$ is blocked with $\rho_{\text{needed}} \rightarrow \max(1, 2\beta) = \rho_2^*$. (This “pair plus dust” argument also shows $\rho_{k+1}^* \leq \rho_k^*$ for every k .)

Case (ii). Any blocked profile has $\sum S > 1$ (otherwise the row of Lemma 3 places the trio in A), so its infimum is ≥ 1 , the formula’s value on $\omega \geq \frac{1}{2}$.

Case (iii). From $s_3 > \max(1 - s_1, s_1 - \omega)$ and $s_2 \geq s_3$: $\sum S \geq g(s_1) := s_1 + 2\max(1 - s_1, s_1 - \omega)$, which decreases until $s_1 = (1 + \omega)/2$ and increases after, with minimum $(3 - \omega)/2$; and $s_1 > \beta$. If $\omega \leq \frac{1}{3}$ then $\beta \geq (1 + \omega)/2$ and $\sum S > g(\beta) = 3 - 5\omega \geq 2 - 2\omega$; if $\omega \in (\frac{1}{3}, \frac{1}{2})$, $\sum S > (3 - \omega)/2 \geq 2 - 2\omega$; if $\omega \geq \frac{1}{2}$, $\sum S \geq s_1 + s_3 > 1$. In all ranges case (iii) is dominated by the formula’s value: it never furnishes the minimum.

Case (iv). Set $q = s_2 + s_3$. Blocking gives $q > 1$ and $q \geq 2s_3 > 2(s_1 - \omega)$, with $s_1 \leq \beta$, while $\rho \geq \max(s_1 + q, q/s_1)$. For fixed s_1 both bounds grow with q , so $q \downarrow Q(s_1) := \max(1, 2(s_1 - \omega))$. On the branch $s_1 \geq \frac{1}{2} + \omega$ ($Q = 2(s_1 - \omega)$) both $3s_1 - 2\omega$ and $2 - 2\omega/s_1$ increase in s_1 : the branch minimum is $\max(\frac{3}{2} + \omega, \frac{2}{1+2\omega})$ at the left endpoint. On the branch $s_1 \leq \frac{1}{2} + \omega$ ($Q = 1$), $h(s_1) = \max(s_1 + 1, 1/s_1)$ decreases until the crossing $s_1^2 + s_1 = 1$, i.e. $s_1 = 1/\varphi$ where $h = \varphi$, and increases after; the branch minimum sits at $s_1^* = \min(\beta, \frac{1}{2} + \omega, 1/\varphi)$. Comparing (exact crossings): for $\omega \leq \omega_\varphi := \frac{\sqrt{5}-2}{2}$ the infimum is $\frac{2}{1+2\omega}$ (the crossing $\frac{2}{1+2\omega} = \frac{3}{2} + \omega$ is $2\omega^2 + 4\omega - \frac{1}{2} = 0$, positive root ω_φ ; this local symbol is unrelated to the root ω_1 of Proposition 64); for $\omega_\varphi \leq \omega \leq 1 - 1/\varphi$ it is φ ; for $1 - 1/\varphi \leq \omega < \frac{1}{2}$ it is $\max(2 - \omega, 1/\beta) = 1/\beta \geq 2\beta$: dominated by case (i). For $\omega \geq \frac{1}{2}$ case (iv) is empty ($s_2 + s_3 \leq 2\beta < 1$). The lower bounds are approached by the stated families ($s_2 = s_3 = q/2$ with s_1 at each branch minimum satisfies all blocking conditions with margin ε).

Global minimum. $\frac{2}{1+2\omega} < 2(1 - \omega)$ on $(0, \frac{1}{2})$ (equivalent to $\omega(1 - 2\omega) > 0$) and $\varphi \leq 2(1 - \omega) \iff \omega \leq 1 - \varphi/2$: the case-wise minimum is the stated formula. $\square$

Proposition 57 (All profile sizes). $\rho_k^*(\omega) = \rho_3^*(\omega)$ for every $k \geq 3$ and every $\omega \in (0, 1)$.

Proof. ($\leq$) Dust: append $k - 3$ superdecreasing rings $\delta, \delta^2, \dots, \delta^{k-3}$ with $\delta \rightarrow 0$ to any blocked three-ring witness (superdecreasing, so that every dust tail also tends to 0 and ρ_{needed} tends to that of the witness). ($\geq$) Let S be non-reinsertable with k rings and $\rho := \rho_{\text{needed}}(S)$.

Step 0. The row of all of S in A fails: $\sum S > 1$, so $\rho > 1$; this closes $\omega \geq \frac{1}{2}$. Assume $\omega < \frac{1}{2}$ and $\rho < 2\beta$ (otherwise done).

Step 1. Two rings $> \beta$ would give $\sum S > 2\beta \geq \rho \geq \sum S$. If only $s_1 > \beta$: the placement “ $s_1 \rightarrow A$ alone, the rest in a row in B ” fails, and since each $s_i \leq \beta$ fits alone the failure is the sum: $\sum S - s_1 > \beta$, whence $\sum S > 2\beta$: impossible. So $s_1 \leq \beta$.

Step 2. “ $s_1 \rightarrow B$, the rest in a row in A ” fails: $Q := \sum S - s_1 > 1$. From $\rho \geq s_1 + Q$ and $\rho \geq Q/s_1$: $\rho \geq \min_{s \leq \beta} \max(1 + s, 1/s)$, which equals φ if $\beta \geq 1/\varphi$ and $1/\beta$ otherwise; in either case $\rho \geq \rho_3^*$ for all $\omega \geq \frac{\sqrt{5}-2}{2}$ (on the plateau $\rho_3^* \leq \varphi$; beyond it $\rho_3^* = 2\beta \leq \varphi$ resp. $2\beta \leq 1/\beta$ for $\beta \leq 1/\sqrt{2}$). Assume then $\omega < \frac{\sqrt{5}-2}{2}$, where $\rho_3^* = \frac{2}{1+2\omega}$, and towards a contradiction $\rho < \frac{2}{1+2\omega}$. From $\rho \geq Q/s_1 > 1/s_1$ we get $s_1 > \frac{1}{2} + \omega$, in particular $s_1 - \omega > \frac{1}{2}$.

Step 3. Let $p := \#\{i : s_i > s_1 - \omega\}$; all such rings exceed $\frac{1}{2}$ and $\sum S \leq \rho < 2$ allows at most three, so $p \in \{1, 2, 3\}$.

$p \geq 3$: $s_1, s_2, s_3 > s_1 - \omega > \frac{1}{2}$. No pair fits together in a root (sums $> 1 > \beta$, two-circle exact) and nobody nests ($s_j \leq s_i - \omega \leq s_1 - \omega < s_j$ is absurd): the prefix $\{s_1, s_2, s_3\}$ is already non-reinsertable, so $\rho \geq \rho_{\text{needed}}(\{s_1, s_2, s_3\}) \geq \rho_3^*$.

$p = 2$: $s_3, \dots, s_k \leq s_1 - \omega$. The placement “ $\{s_3, \dots, s_k\}$ in a row inside the hole of s_1 (capacity $s_1 - \omega$), $s_1 \rightarrow A$, $s_2 \rightarrow B$ ” fails, and the only piece that can fail is the row: $R := \sum_{i \geq 3} s_i > s_1 - \omega$. With $s_2 > s_1 - \omega$, the tail $Q = s_2 + R > 2(s_1 - \omega)$ and $Q > 1$ (Step 2): exactly the case-(iv) program of Theorem 56, whose value is $\geq \rho_3^*$. (This is the only biting branch: the dust of ($\leq$) lives here.)

$p = 1$: $s_2 \leq s_1 - \omega$. The placement “ s_2 in the hole of s_1 , $s_1 \rightarrow B$, $\{s_3, \dots, s_k\} \rightarrow A$ in a row” fails: $R' := \sum_{i \geq 3} s_i > 1$. Then $\sum S = s_1 + s_2 + R' > (s_2 + \omega) + s_2 + 1 = 1 + 2s_2 + \omega$ and the tail of s_2 gives $\rho \geq R'/s_2 > 1/s_2$, so

$$\rho \geq \min_s \max(1 + 2s + \omega, 1/s) = \frac{(1+\omega) + \sqrt{(1+\omega)^2 + 8}}{2} \geq 2,$$

contradicting $\rho < 2\beta < 2$. □

Corollary 58 (Exact combinatorial threshold). For $\omega < \omega_T = \frac{1}{T} - \frac{1}{2}$ no profile of any size blocks the exchange step with $\rho < T$; for $\omega > \omega_T$ the witness $\{\frac{1}{2} + \omega, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ blocks with ρ arbitrarily close to $\frac{2}{1+2\omega} < T$. □

C.2 The blocking frontier of the canonical trio

Work in the canonical template: the trio $\{\alpha, \sigma_1, \sigma_2\}$ with $0 < \sigma_2 \leq \sigma_1 \leq 1 < \alpha$ must pack in the pan of radius $R = \alpha + 1$. With the half-angle machinery of Appendix B (Lemma 47), set $f(x) = x/(R - x)$, $\theta(a, b) = 2 \arcsin \sqrt{f(a)f(b)}$ and $F(\sigma_1, \sigma_2) = \theta(\alpha, \sigma_1) + \theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)$; by the constructive direction (Lemma 48), $F \leq 2\pi$ implies the trio packs, so non-packability implies $F \geq 2\pi$ on the closure. Each θ is strictly increasing in both arguments, so $F = 2\pi$ defines the *blocking frontier* $\sigma_2 = h(\alpha, \sigma_1)$ where it meets the band. Two evaluations used repeatedly: $f(\alpha) = \alpha$ (since $R - \alpha = 1$) and $1 - \alpha f(\sigma) = R(1 - \sigma)/(R - \sigma)$.

Lemma 59 (Closed frontier). *Let $\alpha > 1$ and $\sigma := 1/\sqrt{\alpha(\alpha+1)}$. In the band $0 < \sigma_2 \leq \sigma_1 \leq 1$,*

$$F(\sigma_1, \sigma_2) = 2\pi \iff t(\sigma_1) + t(\sigma_2) = \sigma, \quad t(s) = \sqrt{\frac{1-s}{s}},$$

and $\sigma = t(b(\alpha))$. Consequently $h(\alpha, \sigma_1) = t^{-1}(\sigma - t(\sigma_1))$ with $t^{-1}(u) = 1/(1+u^2)$, defined and strictly decreasing exactly for $\sigma_1 \in [s^, 1]$ where $s^* = 4\alpha(\alpha+1)/(2\alpha+1)^2$ is the diagonal point ($t(s^*) = \sigma/2$); the range of h is $[b(\alpha), s^*]$, with $h(\alpha, 1) = b(\alpha)$.*

Proof. On a frontier point all three angles are interior: for the pairs (α, σ_i) , $P = \alpha f(\sigma_i) < 1$ since $\sigma_i < 1$ (and $\alpha f(1) = 1$); for (σ_1, σ_2) , $\sigma_1 + \sigma_2 \leq 2 < R$. Write $A = \theta(\alpha, \sigma_1)/2$, $B = \theta(\alpha, \sigma_2)/2 \in (0, \pi/2)$, so that $\sin^2 A = \alpha f(\sigma_1)$, $\cos^2 A = R(1 - \sigma_1)/(R - \sigma_1)$ (same for B with σ_2), and on the frontier $\theta(\sigma_1, \sigma_2)/2 = \pi - A - B \in (0, \pi)$.

($\Rightarrow$) Taking sines, $\sin(\theta(\sigma_1, \sigma_2)/2) = \sin(A+B) = \sin A \cos B + \cos A \sin B$. The cross terms factor:

$$(\sin A \cos B)^2 = \alpha f(\sigma_1) \cdot \frac{R(1 - \sigma_2)}{R - \sigma_2} = \alpha R t(\sigma_2)^2 f(\sigma_1) f(\sigma_2),$$

because $(1 - \sigma_2)/(R - \sigma_2) = t(\sigma_2)^2 f(\sigma_2)$; symmetrically with $t(\sigma_1)$. Since $\sin(\theta(\sigma_1, \sigma_2)/2) = \sqrt{f(\sigma_1)f(\sigma_2)} \neq 0$, dividing gives $1 = \sqrt{\alpha R} (t(\sigma_1) + t(\sigma_2))$, the stated equation with $\sigma = 1/\sqrt{\alpha R}$. The identity $(1 - b)/b = 1/(\alpha(\alpha+1))$ for $b = b(\alpha)$ gives $t(b(\alpha)) = \sigma$.

($\Leftarrow$) For fixed σ_1 , both $F(\sigma_1, \cdot)$ (increasing) and $t(\sigma_1) + t(\cdot)$ (decreasing) are strictly monotone in σ_2 ; by ($\Rightarrow$) their level sets coincide where the frontier meets the band, and by monotonicity each has a unique solution. Existence: on the diagonal, $F(s, s)$ increases from 0^+ to $F(1, 1) > 2\pi$, so there is a unique s° with $F = 2\pi$, and ($\Rightarrow$) forces $t(s^\circ) = \sigma/2$, i.e. $s^\circ = s^*$. For $\sigma_1 \in [s^*, 1]$, $F(\sigma_1, 0^+) \leq \pi < 2\pi \leq F(\sigma_1, \sigma_1)$: the frontier crosses, and $t(\sigma_1) \leq \sigma/2$ keeps $h(\sigma_1) \leq \sigma_1$. For $\sigma_1 < s^*$ it would need $\sigma_2 > \sigma_1$. Endpoints: $h(s^*) = s^*$ and $h(1) = t^{-1}(\sigma) = b(\alpha)$. $\square$

Theorem 60 (The κ identity). *On the frontier, for every $\alpha > 1$ and $\sigma_1 < 1$,*

$$\kappa := -\frac{\partial h}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \quad g(s) := s^3(1-s),$$

where g is local to this subsection, and $\kappa \geq 1$, with equality iff $\sigma_1 = \sigma_2 = s^$.*

Proof. The identity: $t'(s) = -1/(2\sqrt{g(s)})$, and differentiating $t(\sigma_1) + t(h(\sigma_1)) = \sigma$ gives $\kappa = t'(\sigma_1)/t'(\sigma_2) = \sqrt{g(\sigma_2)/g(\sigma_1)}$.

The inequality, in the t -coordinate: since t reverses order, $\sigma_2 \leq \sigma_1 \iff t_2 \geq t_1$ on the segment $t_1 + t_2 = \sigma$, and the claim is $G(t_2) \geq G(t_1)$ for

$$G(t) := g(1/(1+t^2)) = \frac{t^2}{(1+t^2)^4}, \quad G'(t) = \frac{2t(1-3t^2)}{(1+t^2)^5} :$$

G increases on $[0, 1/\sqrt{3}]$ and decreases after. Note $\alpha > 1 \iff \alpha(\alpha+1) > 2 \iff \sigma < 1/\sqrt{2}$. If $\sigma \leq 1/\sqrt{3}$, both $t_i \leq \sigma$ lie in the increasing range and $t_2 \geq t_1$ gives the claim, with equality only at $t_1 = t_2$. If $\sigma > 1/\sqrt{3}$ and $t_2 > 1/\sqrt{3}$ (otherwise the previous case applies), then $t_1 = \sigma - t_2 < 1/\sqrt{2} - 1/\sqrt{3}$, so $G(t_1) \leq G(1/\sqrt{2} - 1/\sqrt{3}) = 0.01574\dots$, while $t_2 \in (1/\sqrt{3}, \sigma)$ lies in the decreasing range, so $G(t_2) > G(\sigma) > G(1/\sqrt{2}) = \frac{8}{81}$: here $\kappa^2 > \frac{8/81}{0.0158} > 6$, far from tight. $\square$

Corollary 61 (Minimum of $\sigma_1 + \sigma_2$; Tribonacci closure). *For every $\alpha > 1$, the minimum of $\sigma_1 + \sigma_2$ over the region $\{F \geq 2\pi, 0 < \sigma_2 \leq \sigma_1 \leq 1\}$ – the closure of the non-packable locus $F > 2\pi$; on the boundary $F = 2\pi$ the corona closes with exact tangencies – is $1+b(\alpha)$, attained at $(\sigma_1, \sigma_2) = (1, b(\alpha))$ (equivalently, $1+b(\alpha)$ is the unattained infimum over $F > 2\pi$). Consequently a blocking satisfying the witness wall $\sigma_1 + \sigma_2 \leq \alpha - \omega$ forces $1+b(\alpha) \leq \alpha - \omega$, i.e. $\alpha \geq T_{1+\omega}$ (the root of $\alpha^3 = (1+\omega)(\alpha^2 + \alpha + 1)$); in particular no blocking exists for $\alpha \leq T$, at any width.*

Proof. The region is non-empty ($F(1, 1) > 2\pi$) and a minimum cannot have $F > 2\pi$ (decrease σ_2 : the sum drops, F stays $\geq 2\pi$ by continuity until the frontier is crossed, and $F(\sigma_1, 0^+) < 2\pi$ guarantees it is crossed). On the frontier, $d(\sigma_1 + h)/d\sigma_1 = 1 - \kappa \leq 0$ (Theorem 60): the minimum sits at $\sigma_1 = 1$ and equals $1 + b(\alpha)$. The sign of $1 + b(\alpha) - (\alpha - \omega)$ is governed by the polynomial identity $1 + b(\alpha) - \alpha = -(\alpha^3 - \alpha^2 - \alpha - 1)/(\alpha^2 + \alpha + 1)$ and its deformation $b(\alpha) - (\alpha - 1 - \omega) = -[\alpha^3 - (1 + \omega)(\alpha^2 + \alpha + 1)]/(\alpha^2 + \alpha + 1)$, whose numerator has the single positive root $T_{1+\omega}$ (Descartes' rule); $T_1 = T$. $\square$

C.3 Positive width: the lower-bound curve and the 13/7 corner

The full blocking program of the canonical template adds the remaining resources to the trio wall. With capacities $c_2 := 1 - \omega$ (the hole H_m) and $c_4 := \alpha - \omega - 1$ (σ_2 back in u next to m , two-circle exact), a realizable blocking satisfies (closures for infima)

$$(B1) \text{ trio non-packable, } (B2) \sigma_2 \geq c_2, \quad (B4) \sigma_2 \geq c_4, \quad (W) \sigma_1 + \sigma_2 \leq \alpha - \omega,$$

plus the band $\sigma_2 \leq \sigma_1 \leq 1$ and $\alpha - \omega \geq 1$, and $T_{\text{can}}(\omega)$ is the infimum of $\rho = \max(\sigma_1 + \sigma_2, (1 + \sigma_1 + \sigma_2)/\alpha)$ over blockings. (At $\omega = 0$ the program is empty – (B2) would force $\sigma_2 \geq 1$ – which is why the limit value T uses only (B1)+(W).)

Proposition 62 (Witness branch). *The infimum of ρ over (B1)+(W) alone is $\Phi(\omega) = T_{1+\omega} - \omega$, attained in the limit $\alpha = T_{1+\omega}$, $\sigma_1 \rightarrow 1$, $\sigma_2 \rightarrow b(\alpha) = \alpha - 1 - \omega$. Moreover Φ is strictly increasing and concave, with*

$$\Phi'(\omega) = \frac{2\alpha + 1}{\alpha^2(\alpha^2 + 2\alpha + 3)} \Big|_{\alpha=T_{1+\omega}} > 0, \quad \Phi'(0) = \frac{2T + 1}{7T^2 + 4T + 3} = 0.1374\dots,$$

and on $[0, \frac{1}{7}]$ the chord-tangent bounds $T + (13 - 7T)\omega \leq \Phi(\omega) \leq T + \Phi'(0)\omega$ hold, the chord endpoint being exact: $T_{8/7} = 2$ and $\Phi(\frac{1}{7}) = \frac{13}{7}$.

Proof. By Corollary 61, blocking at a given α forces $\rho \geq \sigma_1 + \sigma_2 \geq 1 + b(\alpha)$ and (W) forces $\alpha \geq T_{1+\omega}$; b is strictly increasing ($b' = (2\alpha + 1)/(\alpha^2 + \alpha + 1)^2$), so the infimum over admissible α is $1 + b(T_{1+\omega}) = T_{1+\omega} - \omega$ (the cubic identity with equality), approached by $\sigma_1 = 1 - \varepsilon$, $\sigma_2 = b(\alpha) + \varepsilon'$ – a genuine blocking: the rigidity of Proposition 51 at $\sigma_1 = 1$ and the propagation of Lemma 52(3), written for the corner $\omega = \frac{1}{7}$ in the proof of Theorem 30, apply verbatim at $\alpha = T_{1+\omega}$ for every $\omega \in [\frac{1}{7}, 0.30]$ ($\alpha \geq 2$ keeps the disk diametrically full). The derivative: implicit differentiation of $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$ and substitution of $1 + \omega = \alpha^3/(\alpha^2 + \alpha + 1)$ reduce the numerator of $d\alpha/d\omega - 1$ to $2\alpha + 1$ (symbolic identity). Concavity: $\frac{d}{d\alpha} \log \Phi' = \frac{2}{2\alpha+1} - \frac{2}{\alpha} - \frac{2\alpha+2}{\alpha^2+2\alpha+3} < 0$ and α increases with ω . $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7} \cdot 7$. $\square$

Adding (B2) gives the second branch: every full blocking has $\rho \geq \sigma_1 + \sigma_2 \geq 2(1 - \omega)$. The maximum of the two branches is minimized at their crossing $T_{1+\omega} = 2 - \omega$, i.e. at the root $\omega_{\times} = 0.0754315\dots$ of $2\omega^3 - 10\omega^2 + 14\omega - 1$, giving the uniform floor $T_{\text{can}}(\omega) \geq 2(\alpha_{\times} - 1) =$

$T+0.00985\dots$ for $\omega \in (0, 0.30]$ – already enough for “width only raises the floor”. The full curve is finer. For fixed α , pass to the coordinate $t_i = t(\sigma_i)$, where (B1) becomes *linear* (Lemma 59): $t_1 + t_2 \leq \Theta(\alpha) := t(b(\alpha))$, and the walls read $t_2 \leq \tau_c := t(c)$, $c := \max(c_2, c_4)$. If $\alpha > 2 + \omega$ there is no blocking at all ((B4) and the band give $\sigma_1 + \sigma_2 \geq 2c_4 > \alpha - \omega$, violating (W)); so $\alpha - \omega \in [1, 2]$ effectively.

Lemma 63 (Conditional minimum). *Let $\alpha \in (1, 2 + \omega)$ and $c = \max(1 - \omega, \alpha - \omega - 1) < 1$. The infimum of $S_0 = \sigma_1 + \sigma_2$ over non-packable trios with $\sigma_2 \geq c$ and $\sigma_2 \leq \sigma_1 \leq 1$ is*

(a) $1 + b(\alpha)$ if $c \leq b(\alpha)$; (b) $c + t^{-1}(\Theta(\alpha) - t(c))$ if $b(\alpha) < c \leq s^*(\alpha)$; (c) $2c$ if $c > s^*(\alpha)$,

where $t^{-1}(u) = 1/(1 + u^2)$ as in Lemma 59. Moreover each case is non-decreasing in α on its domain.

Proof. $t^{-1}(\cdot)$ is decreasing, so S_0 drops when either t_i rises. In the truncated box $\{t_1 \leq t_2 \leq \tau_c, t_1 + t_2 \leq \Theta\}$: if $2\tau_c \leq \Theta$ (case (c)) the box corner (τ_c, τ_c) is admissible and optimal, $S_0 = 2t^{-1}(\tau_c) = 2c$. Otherwise the minimum lives on the segment $t_1 + t_2 = \Theta$, where

$$\frac{dS_0}{dt_2} = -(t^{-1})'(\Theta - t_2) + (t^{-1})'(t_2) = 2[\sqrt{G(t_1)} - \sqrt{G(t_2)}] \leq 0$$

by the G -lemma inside Theorem 60 ($((t^{-1})'(x) = -2x/(1 + x^2)^2, ((t^{-1})'/2)^2 = G)$: push t_2 to its cap. If $\tau_c \geq \Theta$ (case (a)): $t_2 = \Theta$, $t_1 = 0$, i.e. $(\sigma_1, \sigma_2) = (1, b(\alpha))$. If $\tau_c < \Theta$ (case (b)): $t_2 = \tau_c$, $t_1 = \Theta - \tau_c$. Monotonicity in α : (a) is $1 + b(\alpha)$ with b increasing; (b) with $c = c_2$ fixed has $dS_0/d\alpha = (t^{-1})'(\Theta - \tau_c) \Theta'(\alpha) \geq 0$ ($(t^{-1})' < 0$, $\Theta' < 0$); (b)/(c) with the moving cap $c = c_4$ ($\alpha \geq 2$, $\tau_4 = t(c_4)$) has $dS_0/d\alpha = \tau_4'(-2)[\sqrt{G(\tau_4)} - \sqrt{G(t_1)}] + (t^{-1})'(t_1) \Theta' \geq 0$, since $\tau_4' < 0$, $G(\tau_4) \geq G(t_1)$ ($\tau_4 \geq t_1$ on the segment, same G -lemma) and the last two factors are both negative. $\square$

Proposition 64 (The lower-bound curve; exact on the witness branch). *For $\omega \in (0, 0.30]$ (the cap 0.30 is a convenient working restriction of the statement, not a structural obstruction: $T_{1+\omega} < 2 + \omega$ holds identically), the proved lower bound for T_{can} is*

$$T_{\text{can}}(\omega) \geq \begin{cases} 2(1 - \omega) & \omega \in (0, \omega_1], \\ \alpha_m(\omega) - \omega & \omega \in [\omega_1, \frac{1}{7}], \\ \Phi(\omega) & \omega \in [\frac{1}{7}, 0.30], \end{cases}$$

where $\omega_1 = 0.0413570\dots$ is the root in $(\frac{1}{25}, \frac{1}{14})$ of $4\omega^3 - 20\omega^2 + 25\omega - 1$ and $\alpha_m(\omega) \in (1, 2]$ is the unique root of $t(\alpha - 1) + t(1 - \omega) = \Theta(\alpha)$, an algebraic function of degree 6 (its polynomial $P(\alpha, \omega)$, of bidegree $(6, 2)$, is irreducible over $\mathbb{Q}[\alpha, \omega]$). Equality holds on the witness branch (the family is genuine as in Proposition 62) and at the corner; on the other two branches the matching upper bound inherits the angular-criterion caveat.

Proof. Fix ω and minimize the value of Lemma 63 over α subject to (W).

H_m branch ($\omega \leq \omega_1$). Case (c) gives $S = 2(1 - \omega)$, constant in α , admissible iff (W): $\alpha \geq 2 - \omega$, and iff $c_2 > s^*(\alpha)$, i.e. $\alpha \leq \bar{\alpha}(\omega)$. The window is non-empty iff $s^*(2 - \omega) \leq 1 - \omega$, and $(1 - \omega) - s^*(2 - \omega)$ has numerator $-(4\omega^3 - 20\omega^2 + 25\omega - 1)$; the cubic is strictly increasing on $(0, \frac{5}{6})$ (derivative $(5 - 2\omega)(5 - 6\omega)$), so the window exists exactly for $\omega \leq \omega_1$. The other cases do not undercut $2(1 - \omega)$ (monotonicity in Lemma 63 plus continuity across case junctions).

Mixed branch ($\omega_1 < \omega \leq \frac{1}{7}$). For $\alpha \leq 2$, case (c) is empty (it would need $1 - \omega \geq s^*(\alpha) \geq s^*(2 - \omega)$, against the cubic) and case (a) needs $b(\alpha) \geq 1 - \omega$, impossible ($b(2) = \frac{6}{7} \leq 1 - \omega$). Case (b) with $c = c_2$ has value $S_0^b(\alpha) = 1 - \omega + t^{-1}(\Theta - t(1 - \omega))$, increasing in α , and (W) is equivalent to $\Xi(\alpha) := \Theta(\alpha) - t(\alpha - 1) \geq t(1 - \omega)$ with Ξ strictly increasing on $(1, 2]$ (the claim reduces to the polynomial inequality $(\alpha(\alpha + 1))^3 > (2\alpha + 1)^2(\alpha - 1)^3(2 - \alpha)$, which has no root in $(1, 2)$), $\Xi(1^+) = -\infty$ and $\Xi(2) = t(\frac{6}{7}) \geq t(1 - \omega) \iff \omega \leq \frac{1}{7}$: the admissible α form $[\alpha_m, 2]$ and the minimum is $S_0^b(\alpha_m) = \alpha_m - \omega$ ((W) with equality). For $\alpha \in (2, 2 + \omega)$: if $b(\alpha) \leq 1 - \omega$, relax (B4) to (B2) and use monotonicity, $S_0 \geq S_0^b(\alpha) > S_0^b(\alpha_m)$; if $b(\alpha) > 1 - \omega$, $S_0 \geq 1 + b(\alpha) > 2 - \omega > \alpha_m - \omega$. Squaring the defining equation twice yields the polynomial P (necessary condition; the right branch is pinned by $\alpha_m \in (1, 2]$), with the junction identities $P(2 - \omega, \omega) = (\omega - 1)^3(4\omega^3 - 20\omega^2 + 25\omega - 1)$ (so $\alpha_m(\omega_1) = 2 - \omega_1$) and $\alpha_m(\frac{1}{7}) = 2$.

Witness branch ($\omega \geq \frac{1}{7}$). Now $b(T_{1+\omega}) = T_{1+\omega} - 1 - \omega \geq 1 - \omega \iff T_{1+\omega} \geq 2 \iff \omega \geq \frac{1}{7}$: the optimum of case (a) satisfies (B2), (W) forces $\alpha \geq T_{1+\omega}$ (Corollary 61) and the minimum is $\Phi(\omega)$; larger α and the cases (b)/(c₄) are increasing and start at $\Phi(\omega)$; smaller α admit no blocking (for $\alpha \leq 2$, $\Xi(\alpha) \leq \Xi(2) = t(\frac{6}{7}) \leq t(1 - \omega)$ with equality only at the corner; case (c) would need $\omega \leq \omega_1$). $\square$

Proof of Theorem 30. Lower bound. Take any blocking at width ω . If $\alpha \geq 2$: by Corollary 61 alone (no (B2), (B4), (W) needed), $\rho \geq 1 + b(\alpha) \geq 1 + b(2) = \frac{13}{7}$. If $\alpha < 2$: for $\omega \geq \frac{1}{7}$, by Corollary 61 and (W) blocking forces $\alpha \geq T_{1+\omega} \geq T_{8/7} = 2$, contradicting $\alpha < 2$: empty. For $\omega \leq \omega_1$: $\rho \geq 2(1 - \omega) \geq 2(1 - \omega_1) > \frac{13}{7}$ (since $\omega_1 < \frac{1}{14}$). For $\omega_1 < \omega < \frac{1}{7}$: $\rho \geq V(\omega) := \alpha_m(\omega) - \omega$, and $V > \frac{13}{7}$ there: if $V(\omega') = \frac{13}{7}$ then $(\alpha, \omega) = (\frac{13}{7} + \omega', \omega')$ satisfies $P = 0$, but

$$P(\frac{13}{7} + \omega, \omega) = \frac{4(7\omega - 1)Q_5(\omega)}{7^6},$$

$$Q_5 = 33614\omega^5 + 235298\omega^4 + 620830\omega^3 + 766066\omega^2 + 397831\omega - 289,$$

and Q_5 has no root in $[\frac{1}{25}, \frac{1}{7}]$ (exact root isolation; its only positive root is ≈ 0.0007), so the only root of the product there is $\omega = \frac{1}{7}$; since $V(\omega_1) - \frac{13}{7} = \frac{1}{7} - 2\omega_1 > 0$ and V is continuous, $V > \frac{13}{7}$ on all of $[\omega_1, \frac{1}{7}]$. Uniqueness of the infimum configuration: equality in the branch $\alpha \geq 2$ forces $\alpha = 2$, $\sigma_1 = 1$, $\sigma_2 = \frac{6}{7}$; then (B2) gives $\omega \geq \frac{1}{7}$ and (W) gives $\omega \leq \frac{1}{7}$.

The approximating family is genuine. At exactly $\alpha = 2$, $\omega = \frac{1}{7}$ there is no family: $\kappa > 1$ gives $\sigma_1 + h(2, \sigma_1) > \frac{13}{7} = \alpha - \omega$ for $\sigma_1 < 1$, violating (W); the corner is approached by *opening* (B4). Fix $\omega = \frac{1}{7}$ and let $\delta \downarrow 0$:

$$\alpha = 2 + \delta, \quad \varepsilon = \delta^2/4, \quad \sigma_1 = 1 - \varepsilon, \quad \sigma_2 = (\alpha - \omega - 1) + \varepsilon/2.$$

All walls hold ((B4) with slack $\varepsilon/2$; (B2) since $\alpha - \omega - 1 \geq 1 - \omega$ for $\alpha \geq 2$; (W) since $\sigma_1 + \sigma_2 = \alpha - \omega - \varepsilon/2$). Genuineness: since $\alpha > T_{8/7} = 2$, Corollary 61 gives $\sigma_2 > b(\alpha)$ strictly, so the trio $\{\alpha, 1, \sigma_2\}$ is non-packable by the *rigidity* of Proposition 51 (the disk $R = \alpha + 1$ is diametrically full; rescale by $t = 1/\alpha$) – not merely by the angular criterion – and the infeasibility propagates from $\sigma_1 = 1$ to $\sigma_1 = 1 - \varepsilon$ by Lemma 52(3) for all small ε (shrinking ε below the propagation radius δ_0 of that lemma if needed; any smaller ε serves). Then $\rho = \alpha - \omega - \varepsilon/2 \rightarrow \frac{13}{7}$. $\square$

Remark 65 (Fine structure). The curve is not monotone on $(0, \frac{1}{7}]$: differentiating the defining equation of α_m at the junction (where $\alpha = 2 - \omega_1$, so the leading terms $(\alpha - 1)^3(2 - \alpha)$ and $(1 - \omega_1)^3\omega_1$ agree by direct substitution) gives $V'(\omega_1^+) > 0$ (numerically +0.0721, by implicit differentiation of P): ω_1 is a *local minimum* of T_{can} , followed by a local maximum at $\omega_{\text{peak}} =$

0.0444700 (the unique root in $(\omega_1, \frac{1}{7})$ – by Sturm – of an explicit degree-8 factor R_8 of the resultant of P and $P_\alpha + P_\omega$), a bump of height $1.1 \cdot 10^{-4}$, before descending to the corner. T_{can} has two local minima, ω_1 and $\frac{1}{7}$, and the global one is the corner. (The computational claims of this appendix – the exact root isolation of Q_5 , the degree-8 factor R_8 and the bump height – are reproduced by `code/esquina.py` and `code/grosor.py`.)

D Complete proofs for generic containers

This appendix proves every statement of Section 8; its final subsection (D.6) proves the golden floor of the pan exchange for pair profiles, in full. Normalize $r_m = 1$, $\omega = w/r_m$. The template: v is the pan, of radius R , containing $\{\alpha\} \cup O \cup \{m\}$ with $O = \{o_1 \geq \dots \geq o_j\}$, $j \geq 0$, $o_i \geq m = 1$ (rings larger than m agree in F and P); u is the hole of α , capacity $\alpha - \omega \geq 1$; the witness placed $S = \{\sigma_1 \geq \sigma_2\}$ in u , $\sigma_2 \leq \sigma_1 \leq 1$. Holes may be occupied arbitrarily: contents of a hole at any depth are parametrized by $X_y := \sum \text{children}(y)$. The exchange builds a placement P' that preserves the container assignments of all rings $\geq m$; *positions* inside each container are existential (a placement is a nesting forest whose sibling groups pack). A *node* is any ring of radius ≥ 1 that is an occupant of v other than α and m , or is nested (at any depth) inside such an occupant; nodes and their descendants live outside m . Unless stated otherwise, blocked-exchange walls are closures of strict inequalities, as in Appendix C. Throughout this appendix and Appendix E we use the *first-copy convention*: ties among equal radii are broken by index, so the tail of a ring collects all later copies.

D.1 The universal frontier and the corona criterion

Lemma 66 (Universal frontier). *Let $\{A, x, y\}$ be wall-tangent in a disk of arbitrary radius R , in the interior domain $A + x, A + y, x + y < R$. Set $c := R - A$, $T_c(x) := \sqrt{(c-x)/x}$ (a function symbol local to the corona criterion, unrelated to the deformed Tribonacci root $T_{1+\omega}$ of Appendix C) and $\tau_R := c/\sqrt{AR}$. If $A \geq \min(x, y)$, then*

$$F = \theta(A, x) + \theta(A, y) + \theta(x, y) \geq 2\pi \iff T_c(x) + T_c(y) \leq \tau_R.$$

The direction $(\Rightarrow)$ is unconditional; the converse fails without the hypothesis ($R = 1$, $A = 0.01$, $x = y = 0.45$ is a counterexample).

Proof. The cross term factors for arbitrary R : with $\sin^2(\theta_{Az}/2) = f(A)f(z)$, $f(z) = z/(R-z)$, one has the identity

$$f(A)f(x)(1 - f(A)f(y)) = \frac{AR}{(R-A)^2} \cdot \frac{c-y}{y} \cdot f(x)f(y),$$

so $\sin(\theta_{Ax}/2)\cos(\theta_{Ay}/2) = \frac{\sqrt{AR}}{c} T_c(y)\sqrt{f(x)f(y)}$ and, adding the symmetric term, $\sin s = \frac{\sqrt{AR}}{c}(T_c(x) + T_c(y))\sin w$ with $s = (\theta_{Ax} + \theta_{Ay})/2$, $w = \theta_{xy}/2$. The linear form is thus $\sin s \leq \sin w$, while $F \geq 2\pi$ is $s + w \geq \pi$. Since $w \leq \pi/2$: $s + w \geq \pi \Rightarrow \sin s \leq \sin w$ always; the converse needs $s > w$, which the hypothesis provides: if $A \geq y$, monotonicity gives $\theta(A, x) \geq \theta(y, x) = 2w$, so $s > w$. $\square$

Uniform companions (same computations as in Appendix C, now for general R): the general pocket $b_R(A) = ARc/(AR + c^2)$, equal to Descartes' b_2 when $R = A + B$, increasing in R and always $\leq A$; and the slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$ with $g_c(s) = s^3(c-s)$, since T_c is a primitive of $-c/(2\sqrt{g_c})$.

Lemma 67 (Gap system). *A corona of $\{a_1, \dots, a_k\}$ in the disk R (all k circles wall-tangent; all pairs admissible, $a_i + a_j \leq R$, so that every θ_{ij} is defined) with cyclic order π exists iff the linear system*

$$\theta_{ij} \leq S_j - S_i \leq 2\pi - \theta_{ij} \quad (1 \leq i < j \leq k, S_1 = 0)$$

is feasible, where θ_{ij} is the angle of the pair in positions i, j of π .

Proof. The angular separation of positions $i < j$ is $\min(S_j - S_i, 2\pi - (S_j - S_i))$, and $\min(\Delta, 2\pi - \Delta) \geq \theta \iff \theta \leq \Delta \leq 2\pi - \theta$ because $\theta \leq \pi$; wall-tangent circles are contained in the disk and pairwise disjointness is exactly the angular condition. $\square$

Lemma 68 (Subset certificates). *If $\{a_1, \dots, a_k\}$ has a corona with cyclic order π , then every subset T , $|T| \geq 3$, satisfies $\sum_{\text{consecutive pairs of } T} \theta \leq 2\pi$ in the induced cyclic order. Consequently, if some T exceeds 2π in all its cyclic orders, no corona exists in any order – the unconditional direction used by all lower bounds.* $\square$

The naive criterion (“some order with consecutive sum $\leq 2\pi$ implies a corona”) is *false* for $k \geq 4$: the conditional triangle inequality $\theta(a, c) \leq \theta(a, b) + \theta(b, c)$ holds when $b \geq \min(a, c)$ but fails badly for small b ($R = 1$, $a = c = 0.45$, $b \rightarrow 0$), and $\{0.47, 0.47, 0.47, 0.02\}$ has a small cyclic sum while its top trio is already infeasible. The correct statement:

Theorem 69 (Exact criterion per order, $k = 4$). *For arbitrary symmetric $\theta_{ij} \in (0, \pi]$, the system of Lemma 67 for the order $(1, 2, 3, 4)$ is feasible iff the four trio certificates and the total $\theta_{12} + \theta_{23} + \theta_{34} + \theta_{14} \leq 2\pi$ hold.*

Proof. Necessity: Lemma 68. Sufficiency, by difference-constraint duality: each upper bound is an up-edge $i \rightarrow j$ of weight $2\pi - \theta_{ij}$, each lower bound a down-edge $j \rightarrow i$ of weight $-\theta_{ij}$; the system is feasible iff the digraph has no negative directed cycle, and it suffices to exclude *simple* cycles. A simple cycle Γ with U up-edges has weight $w(\Gamma) = 2\pi U - \sum_{e \in \Gamma} \theta_e$, so $w < 0$ requires $\#\text{edges}(\Gamma) > 2U$ (as $\theta \leq \pi$). With $k = 4$, a simple cycle has ≤ 4 edges, so only $U = 1$ can go negative; a $U = 1$ cycle is an up-edge $i \rightarrow j$ plus a decreasing chain from j to i , and its negativity condition $\theta_{ij} + \sum_{\text{chain}} \theta > 2\pi$ is exactly the subset certificate of $\{i, \text{chain}, j\}$: enumerating, sizes 3 give the four trios and size 4 gives the total. $\square$

Proposition 70 (The zigzag minimizes the total). *For $a_1 \geq a_2 \geq a_3 \geq a_4$, the three cyclic orders have totals $\sum_{i < j} \theta_{ij}$ minus one perfect matching, and the maximum-sum matching is $\theta_{12} + \theta_{34}$; hence $\min_{\pi} \text{TOT}(\pi) = \sum_{i < j} \theta_{ij} - \theta_{12} - \theta_{34}$, achieved by the zigzag order (a_1, a_3, a_2, a_4) .*

Proof. With $\sigma_i := \log f(a_i)$ (decreasing) one has $\theta(a_i, a_j) = g(\sigma_i + \sigma_j)$ for $g(s) = 2 \arcsin(e^{s/2})$ on $s \leq 0$ (a notation local to this proof), with $g''(s) = e^{-s} / (2(e^{-s} - 1)^{3/2}) > 0$: g is convex and increasing (continuity at $s = 0$ by passage to the limit). The three matchings have pair-sums $\{\sigma_1 + \sigma_2, \sigma_3 + \sigma_4\}$, $\{\sigma_1 + \sigma_3, \sigma_2 + \sigma_4\}$, $\{\sigma_1 + \sigma_4, \sigma_2 + \sigma_3\}$ with equal totals, and the first majorizes the other two; convexity turns majorization into the claimed maximality. $\square$

Theorem 71 (The U_4 criterion: global, two inequalities). *$\{a_1 \geq a_2 \geq a_3 \geq a_4\}$ admits a corona in the disk R (all pairs admissible) iff*

$$\theta_{12} + \theta_{13} + \theta_{23} \leq 2\pi \quad \text{and} \quad \sum_{i < j} \theta_{ij} - \theta_{12} - \theta_{34} \leq 2\pi.$$

Proof. Trio certificates do not depend on the order, so a corona exists iff all trios pass and the minimal total does (Theorem 69 + Proposition 70). By monotonicity of θ , every trio is dominated pairwise by the top trio $\{a_1, a_2, a_3\}$, so one trio inequality suffices. Neither inequality is redundant (both are active on the frontier). The top-trio condition linearizes by Lemma 66 with automatic hypothesis ($A = a_1$ is the global maximum): it reads $T_c(a_2) + T_c(a_3) \geq \tau_R$ with $c = R - a_1$. $\square$

For $k = 5$ the subset-only criterion fails for exactly one pattern, the *pentagram*: of the 84 simple cycles, all with $U \geq 2$ are count-dominated or LP-dominated except the cycle through the five diagonals, and adding its certificate $\sum_D \theta \leq 4\pi$ restores an exact criterion for arbitrary θ ; for *geometric* θ – separation matrices with $\sin^2(\theta_{ij}/2) = f_i f_j$ – the identity $\prod_{(i,j) \in D} f_i f_j = \prod_{(i,j) \in S} f_i f_j = (\prod_i f_i)^2$ (over the diagonal set D and the side set S of the cycle) obstructs the pattern and the best value found is 7.556 against the $4\pi = 12.57$ needed (the 84-cycle enumeration and the dominance checks are reproduced by script `code/corona.py`, a computational check); its redundancy for geometric θ is stated as a conjecture in the repository and is *not used*: lower bounds only ever use the unconditional direction of Lemma 68. For $k \geq 5$ the subset quantifier does not commute with the order quantifier (explicit five-ring counterexamples in the repository), so the per-order criterion or the LP is the correct tool.

D.2 Occupant walls and metallic floors

Lemma 72 (Walls, free-hole template). *Suppose the holes of m , σ_1 and all o_i are free. If the exchange is blocked, then*

$$\begin{aligned} (B2) \ \sigma_2 &> 1 - \omega; & (B3) \ \sigma_2 &> \sigma_1 - \omega; & (B4) \ \sigma_2 &> \alpha - \omega - 1; & (Bo) \ \sigma_2 &> o_k - \omega \ \forall k; \\ (D) \ \sigma_1 + \sigma_2 &> 1; & (W) \ \sigma_1 + \sigma_2 &\leq \alpha - \omega. \end{aligned}$$

In particular $o_k < \sigma_2 + \omega \leq 1 + \omega$ for all k : every extra occupant stays within one width of m .

Proof. Each line is the contrapositive of an explicit placement whose two resources are disjoint and whose feasibility criteria (pair, nesting, row) are exact: $\sigma_2 \rightarrow H_m$ with $\sigma_1 \rightarrow D_m$; $\sigma_2 \subset \sigma_1 \rightarrow D_m$; σ_2 next to m in u ($1 + \sigma_2 \leq \alpha - \omega$, pair-exact) with $\sigma_1 \rightarrow D_m$; $\sigma_2 \subset o_k$ with $\sigma_1 \rightarrow D_m$; the whole pair into D_m (a free unit disk). (W) is the legality of the witness's placement. $\square$

Proposition 73 (Occupant price). *Blocking in the free-hole template with $j \geq 1$ extra occupants implies*

$$\rho > 2 + \frac{j - 2\omega}{o_1} \geq \frac{j + 2}{1 + \omega} \quad (\omega \leq j/2), \quad \text{and for } j = 1, \ \omega \geq \frac{1}{2}: \quad \rho > \frac{4}{1 + 2\omega}.$$

Proof. The tail of o_1 contains $\{o_2, \dots, o_j, m, \sigma_1, \sigma_2\}$ (the first-copy convention of the preamble). By (Bo), $\sigma_1 \geq \sigma_2 > o_1 - \omega$, and each $o_i \geq 1$: $\rho \geq ((j-1) + 1 + \sigma_1 + \sigma_2)/o_1 > (j + 2(o_1 - \omega))/o_1 = 2 + (j - 2\omega)/o_1$. If $\omega \leq j/2$ the expression decreases in o_1 and $o_1 < 1 + \omega$ (Lemma 72). For $j = 1 < 2\omega$, add (D): $\sigma_1 + \sigma_2 > \max(1, 2(o_1 - \omega))$, so $\rho > (1 + \max(1, 2(o_1 - \omega)))/o_1$, whose infimum over o_1 is $4/(1 + 2\omega)$ at $o_1 = \frac{1}{2} + \omega$. $\square$

Corollary 74 (Crossings; canonical optimality). *For $j = 1$: $\rho > 3/(1 + \omega) > T$ for $\omega \leq \frac{1}{2}$ (the range where Proposition 73 provides that branch; on it $3/(1 + \omega) > T$ holds throughout since $\frac{1}{2} < 3/T - 1 = 0.6310\dots$), and combining with the fine branch $\rho > 4/(1 + 2\omega)$ for $\omega \geq \frac{1}{2}$,*

$\rho > T$ for all $\omega < 2/T - \frac{1}{2} = 2T^2 - 2T - \frac{5}{2} = 0.5873\dots$, with $\rho \geq \frac{13}{7}$ for $\omega \leq \frac{15}{26}$. For $j \geq 2$: $\rho > 4/(1+\omega) > 2 > T$ for all $\omega < 1$. Moreover every extra-occupant blocking has $\rho > 2$ for $\omega < \frac{1}{2}$, while the entire canonical curve lies strictly below 2 – the witness branch by the exact identity $(2+\omega)^3 - (1+\omega)((2+\omega)^2 + (2+\omega) + 1) = 1 > 0$, i.e. $T_{1+\omega} < 2+\omega$, hence $\Phi(\omega) < 2$ – so on $\omega < \frac{1}{2}$ adding occupants is strictly suboptimal for the adversary and the canonical template is the optimal v in this class (for $j = 1$ and $\omega \geq \frac{1}{2}$ the proven bound $4/(1+2\omega)$ may dip below the canonical curve, so there the comparison is not settled by these inequalities; for $j \geq 2$ the bound $4/(1+\omega) > 2$ covers all widths). $\square$

D.3 Occupied holes: the opposite-disk lemma and the silver floor

Lemma 75 (Opposite disk). *Let a disk have capacity c , let $\sigma \leq c$ be the largest piece and C a multiset of pieces $\leq \sigma$. If $\sum C \leq c - \sigma$, then $\{\sigma\} \cup C$ packs in the disk.*

Proof. Place σ wall-tangent with center $(-(c-\sigma), 0)$. The disk of radius $r = c - \sigma$ centered at $(c-r, 0)$ is wall-tangent and exactly tangent to σ (center distance $(c-\sigma) + (c-r) = c = \sigma + r$): a free disk of radius $c - \sigma$, in which C fits in a row (Lemma 3). $\square$

Lemma 76 (General blocking wall Bo''). *Blocking implies, for every node y : $y < \sigma_2 + \omega + X_y$.*

Proof. Otherwise $\sum \text{children}(y) \leq (y - \omega) - \sigma_2$ and Lemma 75 packs $\{\sigma_2\} \cup \text{children}(y)$ in the hole of y (if some child exceeds σ_2 , apply the lemma with that child as the largest piece; the inequality is the same), with $\sigma_1 \rightarrow D_m$: unblocked. Positions are re-eligible, and inserting σ_2 (a ring $< m$) only changes the container of σ_2 : legal. $\square$

Theorem 77 (Silver floor; m without children). *Blocking with $j \geq 1$, holes occupied arbitrarily at any depth, m childless, implies $\rho > \Psi(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + 1}$, the positive root of $u^2 - 2(1 - \omega)u - 1$. In particular $\Psi(0) = 1 + \sqrt{2}$ (the silver ratio), $\Psi(\frac{1}{4}) = 2$, and $\Psi > T \iff \omega < (T - 1)^2/2 = 0.35220\dots$*

Proof. Let y^* be a node of minimal radius; its children are all < 1 (a child ≥ 1 would be a smaller node), so $X := X_{y^*}$ is a sum of rings smaller than m , living in the tail of m . With $\sigma := \sigma_2 > 1 - \omega$ (wall (B2), valid since H_m is free): the tail of m gives $\rho \geq \sigma_1 + \sigma_2 + X \geq 2\sigma + X$, and the tail of y^* with Lemma 76 gives $\rho \geq (1 + \sigma_1 + \sigma_2 + X)/y^* > (1 + 2\sigma + X)/(\sigma + \omega + X)$. Minimize $F(\sigma, X) = \max(2\sigma + X, (1 + 2\sigma + X)/(\sigma + \omega + X))$ over $\sigma \geq 1 - \omega$, $X \geq 0$: for fixed σ the first term grows in X and the second decreases, so $F(\sigma, \cdot)$ is bounded below by the crossing value $u = 2\sigma + X$ with $u(u + \omega - \sigma) = 1 + u$, i.e. the positive root of $u^2 - (1 + \sigma - \omega)u - 1 = 0$ (when the crossing falls at $X < 0$, the same algebra shows $F(\sigma, 0) = 2\sigma$ already exceeds the root). The root increases in σ , so the program minimum is at $\sigma = 1 - \omega$, giving $\Psi(\omega)$. $\square$

Theorem 78 (Evacuation dichotomy; m with children). *The conclusion of Theorem 77 holds without the hypothesis that m is childless.*

Proof. Evacuation to D_m : place σ_1 and all children of m in a row inside D_m (possible if $\sigma_1 + M \leq 1$, $M := \sum \text{children}(m)$; children of σ_1 travel inside σ_1) and σ_2 in the emptied H_m (possible if $\sigma_2 \leq 1 - \omega$). Contrapositive: blocking implies $\sigma_2 > 1 - \omega$ or $\sigma_1 + M > 1$. In branch A the proof of Theorem 77 applies verbatim (it used only $\sigma_2 > 1 - \omega$, Bo'' and the two tails; the mass $M \geq 0$ only fattens them). In branch B, let y^* be the minimal node, $s := \sigma_2 + X > 1 - \omega$ (Bo'' and $y^* \geq 1$) and $A := \sigma_1 + \sigma_2 + M + X > 1 + s$; the two tails give $\rho \geq A > 1 + s$ and $\rho \geq (1 + A)/y^* > (2 + s)/(s + \omega)$, whose optimization yields the metallic mean $\Psi_B(\omega)$, the positive

root of $u^2 - (2 - \omega)u - 1$. Since the positive root of $u^2 - bu - 1$ increases in b and $2 - \omega \geq 2(1 - \omega)$, $\Psi_B \geq \Psi$: branch B is dominated. (The crossing $\Psi_B = T$ happens at $\omega = (T - 1)^2$ – twice the branch-A threshold – via the *polynomial* identity $(T - 1)^2 T - (2T - T^2 + 1) = T^3 - T^2 - T - 1$.) $\square$

Corollary 79. *In the canonical template ($j = 0$) with m arbitrary: branch A gives the full curve $T_{\text{can}}(\omega)$ (its program uses (B2) only as an inequality), branch B gives $\rho \geq \Phi(\omega)$ (Proposition 62 uses only (B1) and (W)); in both, $\rho > \Phi(\omega) > T$ for all $\omega > 0$.* $\square$

D.4 The double-pocket wall, the golden line, and Ψ_j

Throughout this subsection $j \geq 1$ and the walls above are in force; write $X := X_{o_1}$, $M := \sum \text{children}(m)$, $X_\sigma := \sum \text{children}(\sigma_1)$, and add the tariffed-nesting wall (B3'): $\sigma_2 + X_\sigma > \sigma_1 - \omega$ (Lemma 75 applied to the hole of σ_1).

Lemma 80 (Double pocket). *Blocking (template $j = 1$, arbitrary occupancy) implies $\sigma_1 > b_2(\alpha, o_1)$, where $b_2(A, B) = AB(A + B)/(A^2 + AB + B^2)$.*

Proof. Blocking implies $\{\alpha, o_1, \sigma_1, \sigma_2\}$ does not pack in the pan (full repacking is a legal resource: children travel inside their parents, positions are existential). Since $\alpha + o_1 \leq R$, non-packability is inherited by containment in the disk $\bar{R} = \alpha + o_1$, where the pair $\{\alpha, o_1\}$ is diametrically rigid: $|c_\alpha| \leq o_1$, $|c_{o_1}| \leq \alpha$, $|c_\alpha - c_{o_1}| \geq \alpha + o_1$ force equality throughout – the rescaled rigidity of Proposition 51. By the exact necessity of that proposition (rescaled), any third circle disjoint from both has radius $\leq b_2(\alpha, o_1)$, and there are *two* pockets, one on each side of the diameter, mutually disjoint with slack: their centers are at $(x_0, \pm y_0)$ with $y_0^2 - b_2^2 = 3b_2^2$, i.e. $y_0 = 2b_2$ exactly, so the mirror circles are $4b_2$ apart. If $\sigma_1 \leq b_2$ then also $\sigma_2 \leq b_2$, and placing each σ concentric in a pocket packs the quadruple in $\bar{R} \subseteq$ the pan: contradiction. $\square$

Theorem 81 (Golden line, both branches). *Blocking in the template ($j = 1$, arbitrary occupancy) implies $\rho > \varphi^2 - \frac{\varphi}{2}\omega$ for all ω in every case except the corner $\{\text{branch B, } \alpha < o_1, o_1 > \bar{o} = 1.2956, \text{ node-child, } \omega < \frac{1}{2}\}$, where the proved bound is $\geq 2 > T$. In particular $\rho > T$ for all $\omega < \omega_A = 2(\varphi^2 - T)(\varphi - 1) = 2 - 2(T - 1)(\varphi - 1) = 0.962585 \dots$*

Proof. Write $g := \sqrt{5} - 1$. *Tail chain.* The tail of o_1 contains $\{m, \sigma_1, \sigma_2\} \cup \text{children}(o_1) \cup \text{children}(m) \cup \text{children}(\sigma_1)$, so $\rho o_1 \geq 1 + \sigma_1 + \sigma_2 + X + M + X_\sigma$; eliminating X with Bo'' ($X > o_1 - \sigma_2 - \omega$):

$$(*) \quad \rho > 1 + \frac{1 + \sigma_1 - \omega + M + X_\sigma}{o_1}.$$

Branch A ($\sigma_2 \geq 1 - \omega$). From (W), $\alpha \geq \sigma_1 + \sigma_2 + \omega \geq 1 + \sigma_1$. Lemma 80 with b_2 increasing in α gives $\sigma_1 > b_2(1 + \sigma_1, o_1)$, i.e. $o_1 < N(\sigma_1) := (1 + \sigma_1)(\sqrt{1 + 4\sigma_1} - 1)/2$ (the condition $b_2(1 + \sigma_1, N) = \sigma_1$ is the quadratic $N^2 + (1 + \sigma_1)N - \sigma_1(1 + \sigma_1)^2 = 0$). With (*) and $M, X_\sigma \geq 0$: $\rho > 1 + (1 + \sigma_1 - \omega)/N(\sigma_1) =: 1 + h(\sigma_1)$, and h is decreasing in σ_1 : the comparison of derivatives reduces to the polynomial identity $(1 + 4\sigma_1) - (1 + 2\sigma_1 - 2\sigma_1^2)^2 = 4\sigma_1^3(2 - \sigma_1) > 0$. The infimum is at $\sigma_1 = 1$: $N(1) = g$, $1/g = \varphi/2$, and $\rho > 1 + (2 - \omega)\varphi/2 = \varphi^2 - \frac{\varphi}{2}\omega$.

Branch B ($\sigma_1 + M > 1$). Using $\sigma_1 + M > 1$, Bo'', (B3'), (W) and Lemma 80:

- (I) $\rho o_1 \geq 1 + \sigma_1 + \sigma_2 + M + X + X_\sigma > 2 + o_1 - \omega + 2b_2(\alpha, o_1) - \alpha$,
- (II) $\rho \alpha \geq o_1 + 1 + \sigma_1 + \sigma_2 + M + X + X_\sigma > 2o_1 + 2 - \omega + 2b_2(\alpha, o_1) - \alpha$ [if $\alpha \geq o_1$],

(in (I): $X > o_1 - \omega - \sigma_2$, $X_\sigma > \sigma_1 - \omega - \sigma_2$, $-\sigma_2 \geq \sigma_1 + \omega - \alpha$ by (W), $\sigma_1 > b_2$; (II) is the tail of α , which contains o_1 and its children only when $\alpha \geq o_1$). Set $f_1 := 1 + (2 - \omega + 2b_2 - \alpha)/o_1$ and $f_2 := (2o_1 + 2 - \omega + 2b_2 - \alpha)/\alpha$. If $o_1 \leq g$, the short chain (only $\sigma_1 + M > 1$ and Bo'', valid in any order) gives $\rho > 1 + (2 - \omega)/o_1 \geq 1 + (2 - \omega)/g = \varphi^2 - \frac{\varphi}{2}\omega$. If $o_1 > g$, the admissible α -region is $[1 + \omega, A_{\max}(o_1)]$ where $b_2(A_{\max}, o_1) = 1$, and two exact facts drive the optimization: b_2 is concave in α ($\partial^2 b_2 / \partial \alpha^2 = -6\alpha o_1^3(\alpha + o_1)/D^3 < 0$, D the denominator), so f_1 is concave in α and minimized at the endpoints; and $D^2 - \alpha o_1^2(o_1 + 2\alpha) = (\alpha + o_1)(\alpha^3 + \alpha^2 o_1 + o_1^3) > 0$ gives $\alpha \partial b_2 / \partial \alpha < o_1$, so f_2 is strictly decreasing in α . For $o_1 \in (g, \tilde{o}]$ ($\tilde{o} = 1.29556\dots$, the root of c_{20} below; any order of α, o_1 , only (I) used): at $\alpha = A_{\max}$, $f_1 - G_\omega = c_{10}(o_1) + \omega(1/g - 1/o_1)$ with the ω -coefficient ≥ 0 and $c_{10} \geq 0$ on $[g, o^*]$, $o^* = 1.59557\dots$ the root of c_{10} , with the exact identity $f_1 \equiv G_\omega$ at $o_1 = g$ (the golden corner, for all ω); at $\alpha = 1 + \omega$, $f_1 \geq G_\omega$ on $[g, o^*] \times [0, 1]$ with contact only at $(g, \omega \rightarrow 1)$. For $o_1 > \tilde{o}$ with $\alpha \geq o_1$ (forcing $o_1 \leq \frac{3}{2}$, since $A_{\max}$ is decreasing with $A_{\max}(\frac{3}{2}) = \frac{3}{2}$ exactly; also $A_{\max}(g) = 2$, $A_{\max}(2) = g$ – the corner is self-dual): $f_2 \geq f_2(A_{\max})$ and $f_2(A_{\max}) - G_\omega = c_{20}(o_1) + \omega c_{21}(o_1)$ with $c_{21} \geq 0$ for $o_1 \leq 2$ and $c_{20} \geq 0$ on $[\tilde{o}, \frac{3}{2}]$ (the sign certificates $c_{10} \geq 0$, $c_{20} \geq 0$ are *exact*: after isolating the radical $\sqrt{o_1^2 + 2o_1 - 3}$ the numerators are linear in it, their zero sets rationalize to polynomials in $\mathbb{Q}[o_1]$, and exact Sturm isolation shows the only roots in range are the golden corner g and the endpoints themselves – so o^* and $\tilde{o}$ are exact algebraic numbers, the sign is constant between roots, and one exact rational sample settles it; $c_{21} \geq 0$ is the linear-in- ω coefficient, elementary; script `goldencert`, 5/5). For $o_1 > \tilde{o}$ with $\alpha < o_1$ (a region requiring $\omega < \frac{1}{2}$: it needs $1 + \omega \leq \alpha < o_1 < N_1(\omega)$, and $N_1(\frac{1}{2}) = \frac{3}{2}$ exactly with N_1 decreasing, where $b_2(1 + \omega, N_1) = 1$): if the children of o_1 are all < 1 , the two α -free bounds $\rho > 1 + o_1 - \omega$ (tail of m : $\sigma_1 + M > 1$ and Bo'') and $\rho > 1 + (2 - \omega + 2b_2(1 + \omega, o_1))/o_1$ (tail of o_1 , which now contains α ; Bo'', (B3'), (W) cancel α , and Lemma 80 applies with $\alpha \geq 1 + \omega$) have $\max \geq G_\omega + \frac{1}{4}$ on $[\tilde{o}, 3.5] \times [0, \frac{1}{2}]$, box-certified by subdivision with corner minorants, while for $o_1 \geq 3.5$ the first bound alone already gives $1 + o_1 - \omega \geq 4 > G_\omega + \frac{1}{4}$ – together they cover the whole region, whose o_1 -extent $N_1(\omega)$ is unbounded as $\omega \rightarrow 0$ (script `goldencert`; the companion bound $f_1 \geq G_\omega$ on $[g, o^*] \times [0, 1]$ at $\alpha = 1 + \omega$ is box-certified as well, its single tangency at $(g, \omega \rightarrow 1)$ handled by certified local monotonicity around the contact); if o_1 has a node child we are in branch B with $\omega < \frac{1}{2}$, and Theorem 78 gives $\rho > \Psi_B(\omega) > \Psi_B(\frac{1}{2}) = 2 > T$ ($\Psi_B(\frac{1}{2}) = 2$ exact). $\square$

Proposition 82 (The Ψ_j ladder, unconditional). *Blocking with $j \geq 1$ extra occupants and arbitrary nested occupancy implies*

$$\rho > \Psi_j(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + j} \quad (\text{root of } u^2 - 2(1 - \omega)u - j),$$

with $\Psi_j > T \iff \omega < 1 - (T^2 - j)/(2T)$: 0.3522 ($j = 1$), 0.6240 ($j = 2$), 0.8959 ($j = 3$), and all $\omega \in (0, 1)$ for $j \geq 4$ ($\Psi_j(\omega) > \Psi_j(1) = \sqrt{j} \geq 2$ there; for $\omega > 1$ the $(1 - \omega)$ term is negative and Ψ_j drops below $\sqrt{j}$).

Proof (leaf argument). Call a node a *leaf* if it has no node children. The node subtree of each occupant o_i is finite and non-empty, hence contains a leaf ℓ_i ; the subtrees are disjoint, so there are j distinct leaves $\ell_1, \dots, \ell_j \geq 1$. Let L be the largest, $X_L = \sum \text{children}(L)$ (all < 1 , as L is a leaf), $M = \sum \text{children}(m)$, and W the total mass of rings < 1 other than σ_1, σ_2 (so $X_L \leq W$, $M \leq W$, and X_L, M are disjoint). Three facts: (1) *wall at the leaf* (Lemma 76): $L < \sigma_2 + \omega + X_L \leq \sigma_2 + \omega + W$; (2) *tail of L* : the other $j - 1$ leaves, m , σ_1 , σ_2 and all mass < 1 are $\leq L$ (first-copy convention), so $\rho L \geq (j - 1) + 1 + \sigma_1 + \sigma_2 + W \geq j + 2\sigma_2 + W$; (3) *tail of m* : $\rho \geq \sigma_1 + \sigma_2 + W \geq 2\sigma_2 + W$. In branch A of the evacuation dichotomy ($\sigma_2 \geq 1 - \omega$), facts

(1)–(3) give $\rho > \max(2\sigma + W, (j + 2\sigma + W)/(\sigma + \omega + W))$ at $\sigma = \sigma_2$; minimizing over $\sigma \geq 1 - \omega$, $W \geq 0$, the crossing $u^2 + u(\omega - \sigma - 1) - j = 0$ ($u = 2\sigma + W$) increases in σ , with minimum at $\sigma = 1 - \omega$: exactly $\Psi_j(\omega)$. (If the crossing falls at $W < 0$ – possible for $\omega > \frac{1}{2}$, $j = 1$, $\sigma \rightarrow 1 - \omega$ – the value 2σ at $W = 0$ satisfies $(2\sigma)^2 - 2(1 - \omega)(2\sigma) - j \geq j > 0$, so $2\sigma > \Psi_j$ anyway.) In branch B ($\sigma_1 + M > 1$), set $s := \sigma_2 + X_L$: by fact (1) and $L \geq 1$, $s > 1 - \omega$; the tail of L also contains M , disjoint from X_L , so $\rho L \geq j + \sigma_1 + M + \sigma_2 + X_L > j + 1 + s$ with $L < s + \omega$, and the tail of m gives $\rho \geq \sigma_1 + M + \sigma_2 + X_L > 1 + s$; minimizing $\max(1 + s, (j + 1 + s)/(s + \omega))$ yields, with $u = 1 + s$, the metallic mean $u^2 - (2 - \omega)u - j = 0$, whose root dominates Ψ_j ($2 - \omega \geq 2(1 - \omega)$, and the root increases in the linear coefficient). $\square$

The leaf argument is what closes the case caught by the adversarial verification (a node child inside the largest occupant): towers of nested nodes inflate X_{o_1} without paying the tail of m , but they always leave a leaf at the bottom of each subtree, and the largest leaf carries both the j -bonus in its tail and the leaf ceiling $\sigma_2 + \omega + W$.

Corollary 83 (Small rings are free for the combinatorial walls). *Proposition 73, Theorems 77 and 78 and Proposition 82 hold unchanged if the instance contains additional rings smaller than m anywhere compatible with each theorem’s template (with S still the pair $\{\sigma_1, \sigma_2\}$).*

Proof. Every unblocking placement used by those walls is *local*: it lives in D_m (freed by the exchange), in H_m , in the holes of nodes or of σ_1 , or next to m inside u – never in the free space of v . An extra ring smaller than m , wherever it sits, intersects none of these resources (hole contents are already counted in the X ’s; rings in v proper stay put, which is legal since no other position changes), and tails can only grow. (The geometric wall of Lemma 80 is *not* covered: it uses the global repacking of the pan, which does see small rings in v ; this is the “gap lemma” left open.) $\square$

D.5 Three-piece profiles in the canonical template

Here v is the canonical pan of radius $R = \alpha + 1$ with $\{\alpha, m\}$ diametral, u is the hole of α , and the witness placed $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\} \subset (\omega, 1)$ in u ; the hole of m may carry mass $M \geq 0$. The reinsertion resources are D_m (rows of sum ≤ 1 , with α in place), H_m (capacity $1 - \omega$; its content M can be evacuated in a row to D_m), the slot next to m in u (row $1 + \Sigma \leq \alpha - \omega$), nesting inside S , and the repack of v : placing $\{\alpha\} \cup (\text{roots of } S)$ as a corona in the pan – incompatible with using D_m or the evacuation, both of which pin α . The witness gives (W) $\sigma_1 + \sigma_2 \leq \alpha - \omega$.

Theorem 84 (Three pieces; the zigzag pays). *Blocking with S a triple in $(\omega, 1)$ (as in the preamble above) and arbitrary M implies $\rho > \max(\Phi(\omega), \frac{13}{7})$ for all $\omega \in (0, 1)$; in the zigzag branch below, $\rho > \frac{17}{7}$.*

Proof. Branch 1 ($\sigma_3 \leq \sigma_1 - \omega$): the dust rides the pair. Nest σ_3 in σ_1 inside each pair placement; their failures give five walls: (B1) “ $\sigma_3 \subset \sigma_1$; corona $\{\alpha, \sigma_1, \sigma_2\}$ in v ” fails, so the trio has no corona and $F(\sigma_1, \sigma_2) > 2\pi$ (corona $\iff F \leq 2\pi$ is exact for three circles); (B2/dichotomy) “ $\sigma_1(\sigma_3) + M$ in a row in D_m ; $\sigma_2 \rightarrow$ emptied H_m ” fails, so $\sigma_2 > 1 - \omega$ or $\sigma_1 + M > 1$; (B4) “ $\sigma_2 \rightarrow u$ next to m ; corona $\{\alpha, \sigma_1(\sigma_3)\}$ ” fails, so $\sigma_2 > \alpha - \omega - 1$; (BH) “ $\sigma_1(\sigma_3) \rightarrow D_m$; $\sigma_2 + M$ in a row in H_m ” fails, so $\sigma_2 + M > 1 - \omega$; and (W). From (B1)+(W), by Corollary 61, in *both* branches of the dichotomy $\sigma_1 + \sigma_2 \geq 1 + b(\alpha)$, $\alpha \geq T_{1+\omega}$ and $\rho \geq \sigma_1 + \sigma_2 \geq \Phi(\omega) > T$. Moreover: if $\sigma_2 > 1 - \omega$, the program (B1)+(B2)+(B4)+(W) is exactly that of Theorem 30, whose lower bound gives $\rho \geq \max(\sigma_1 + \sigma_2, \frac{1+\sigma_1+\sigma_2+\sigma_3}{\alpha}) \geq \max(\sigma_1 + \sigma_2, \frac{1+\sigma_1+\sigma_2}{\alpha}) \geq T_{\text{can}}(\omega) \geq \frac{13}{7}$; if instead

$\sigma_1 + M > 1$ and $\sigma_3 > \omega$, then (BH) gives $\rho \geq \sigma_1 + \sigma_2 + \sigma_3 + M > \sigma_1 + (1 - \omega) + \omega = 1 + \sigma_1 \geq 1 + \Phi(\omega)/2 \geq 1 + T/2 = 1.9196 > \frac{13}{7}$.

Branch 2 ($\sigma_3 > \sigma_1 - \omega$): *nothing nests* (also $\sigma_2, \sigma_3 > \sigma_2 - \omega$ and the star needs $\sigma_2 + \sigma_3 \leq \sigma_1 - \omega$, impossible). Split by the trio.

2A: the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona. Then $F > 2\pi$ and, as above, $\rho \geq \sum S > \sigma_1 + \sigma_2 \geq \Phi(\omega)$. Furthermore $\sigma_1 \geq (\sigma_1 + \sigma_2)/2 \geq \Phi/2$, so $\sigma_3 > \sigma_1 - \omega \geq \Phi/2 - \omega$ and $\rho > \frac{3}{2}\Phi - \omega$ when $\omega \leq \Phi/2$; by the chord bound $\Phi \geq T + (13 - 7T)\omega$ this exceeds 2.64 on $(0, \frac{1}{7}]$, while for $\omega \geq \frac{1}{7}$, $\Phi \geq \frac{13}{7}$: in all cases $\rho \geq \frac{13}{7}$.

2B: the trio has a corona. Three mixed placements fail: “ $\sigma_3(+M)$ in a row in H_m + trio corona” gives (m1) $\sigma_3 + M > 1 - \omega$; “ $\sigma_3 \rightarrow u$ next to m + trio corona” gives (m2) $\alpha < 1 + \omega + \sigma_3$; and “corona of the quadruple” fails, so by Theorem 71 – the top trio passing its certificate, by Lemma 68 – the *zigzag* fails: $\theta(\alpha, \sigma_2) + \theta(\alpha, \sigma_3) + \theta(\sigma_1, \sigma_2) + \theta(\sigma_1, \sigma_3) > 2\pi$. By monotonicity, $2[\theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)] > 2\pi$, so $A + B > \pi/2$ with $A = \theta(\alpha, \sigma_2)/2$, $B = \theta(\sigma_1, \sigma_2)/2$, both $< \pi/2$ ($A \geq \pi/2$ would force $\sigma_2 \geq 1$; $\sigma_1 + \sigma_2 < R$). Hence $\sin A > \cos B$, i.e. $\sin^2 A + \sin^2 B > 1$: $f(\sigma_2)[f(\alpha) + f(\sigma_1)] > 1$ with $f(\alpha) = \alpha$ and $f(\sigma_1) \leq 1/\alpha$, so $f(\sigma_2)(\alpha + 1/\alpha) > 1$, which is *exactly* $\sigma_2 > b(\alpha)$. With (W): $2b(\alpha) < \sigma_1 + \sigma_2 \leq \alpha - \omega$, and since $\alpha \mapsto \alpha - \omega - 2b(\alpha)$ is strictly increasing ($1 - 2b' > 0$ for $\alpha \geq 1$), $\alpha > \gamma_\omega$, the root of $2b(\alpha) = \alpha - \omega$ – equivalently of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$, with $\gamma_0 = \varphi$. The tails: by (m1), $\rho \geq \sum S + M > 2b(\alpha) + (1 - \omega) \geq \gamma_\omega + 1 - 2\omega$, and by (m2), $\rho > 2b(\alpha) + \alpha - 1 - \omega \geq 2\gamma_\omega - 1 - 2\omega$. The first bound decreases in ω and the second increases ($1 < \gamma' < 2$), and they cross *exactly* at $\gamma = 2$, $\omega = \frac{2}{7}$ ($8 = 4 + 2 + 2$ in the cubic), where both equal $3 - \frac{4}{7} = \frac{17}{7}$: $\rho > \frac{17}{7}$ throughout branch 2B. $\square$

Proposition 85 (Reduction branch with occupants). *In the templates with $j \geq 1$ extra occupants and S of $k \geq 3$ pieces with $\sum_{i \geq 3} \sigma_i \leq \sigma_1 - \omega$, all pair walls and bounds (Proposition 73, Theorems 77 and 78, Proposition 82) hold unchanged: blocking implies $\rho > \Psi_j(\omega)$.*

Proof. Nest $\sigma_3, \dots, \sigma_k$ in a row inside the hole of σ_1 ; every unblocking placement of those walls moves σ_1 (with its cargo travelling inside) and σ_2 exactly as in the pair case, and the tails only grow. $\square$

D.6 Golden floor for pan exchanges

Here u is the pan itself: F places m at top level, the witness P keeps m in the hole of a ring y ($1 + X_y^{\text{rest}} \leq y - \omega$), the pan holds occupants $O = \{\sigma_1 \geq \dots \geq \sigma_j\}$ (> 1 , top level, shared by F and P , arbitrarily occupied) and, per P , the pair $S = \{\sigma_1 \geq \sigma_2\}$. Resources: D_m (free unit disk in y 's hole), H_m , hole tariffs (Lemma 75), nesting, and the pan repack. F 's pan gives the pair necessities $\sigma_i + 1 \leq R$ and $\sigma_i + \sigma_\ell \leq R$ for all $i \neq \ell$. Write $S_0 := \sigma_1 + \sigma_2$; the row placement of the pair in D_m fails, so (D) $S_0 > 1$; and the joint placement of the pair in y 's hole beside its remaining content X_y^{rest} fails, which by Lemma 75 gives the wall

$$\sigma_1 + \sigma_2 + X_y^{\text{rest}} > y - \omega. \quad (\text{Ry})$$

Strict leaf: a leaf whose hole is not y 's; with $j \geq 2$ at least $j - 1$ exist (j if y is not a leaf).

Theorem 86 (Golden floor, pan exchange, pair profiles). *Blocking implies $\rho > \varphi$ in each of the following cases, which together exhaust the template: (i) $j = 1$, every width $\omega > 0$ (including the solid-pivot regime $\omega \geq 1$: the proof uses no width) and every occupancy; (ii) $j = 2$, likewise every $\omega > 0$ and occupancy; (iii) the evacuation branch $\sigma_1 + M > 1$, every $j \geq 2$ and every $\omega \in (0, 1)$; (iv) $j \geq 3$, every $\omega \in (0, 1)$. (Note $j \geq 1$ always: the carrier y has $y \geq 1 + \omega > 1$, so*

y is an occupant or lies inside one.) The infimum is exactly φ , realized by the golden family of Theorem 19 inside case (i).

Proof. $j = 1$. The placement “ $\sigma_i \rightarrow \text{pan}$, the other to D_m ” fails, so $\{o_1, m, \sigma_i\}$ does not pack in R , hence (antitone containment, $R \geq o_1 + 1$) not in the disk $o_1 + 1$, where $\{o_1, 1\}$ is rigid: $\sigma_i > b(o_1)$ by Proposition 51. The tails give $\rho > \max(2b(o_1), (1 + 2b(o_1))/o_1)$; b is increasing, $g(A) = (1 + 2b)/A$ (notation local to this proof) decreasing (the certificate $(3A^2 + 3A + 1)(A^2 + A + 1) - 2A(2A + 1)$ has positive coefficients), and the crossing $2b(A)A = 1 + 2b(A)$ factors as $(A^2 - A - 1)(2A + 1)$: unique positive root $A = \varphi$, value φ ($2b(\varphi) = \varphi$, $g(\varphi) = (1 + \varphi)/\varphi = \varphi$).

$j = 2$: *the mirror-pocket wall*. The same placement fails, so $\{o_1, o_2, m, \sigma_2\}$ does not pack in R , hence not in the disk $o_1 + o_2$, where $\{o_1, o_2\}$ is rigid and the quadruple packs iff $m \leq b_2(o_1, o_2)$ and $\sigma_2 \leq b_2$ (necessity by Proposition 51 per circle; sufficiency by placing them concentric in the two mirror pockets, disjoint since $y_0 = 2b_2$, as in Lemma 80). Thus $b_2(o_1, o_2) < 1$, i.e. $o_2 < \bar{A}(o_1)$, the decreasing root of $b_2(o_1, \cdot) = 1$ with $\bar{A}(2) = \sqrt{5} - 1$ and $\bar{A}(\frac{3}{2}) = \frac{3}{2}$. The tails of o_1 and o_2 , with (D) $\sigma_1 + \sigma_2 > 1$ (row in D_m), give $\rho > \max((o_2 + 2)/o_1, 2/o_2)$ on $o_2 \in (1, \bar{A}(o_1))$: for $o_1 \leq \frac{3}{2}$, $3/o_1 \geq 2$; for $o_1 \in (\frac{3}{2}, 2)$ the crossing $o_2^* = \sqrt{1 + 2o_1} - 1$ is interior ($b_2(o_1, o_2^*) < 1$ there, with exact equality at $o_1 = 2$) and its value $2/o_2^* > 2/(\sqrt{5} - 1) = \varphi$; for $o_1 \geq 2$, $2/o_2 > 2/\bar{A}(2) = \varphi$. The crossing at $o_1 = 2$ is the golden corner $b_2(2, \sqrt{5} - 1) = 1$ once more.

Evacuation branch, $j \geq 2$. A strict leaf L' with $q := \sigma_2 + X_{L'} > 1 - \omega$ satisfies (Lemma 75) $L' < q + \omega$, and its tail collects $\{m, \sigma_1, \sigma_2, M\} \cup \text{children}(L')$: with $\sigma_1 + M > 1$, exactly the branch-B program of Theorem 78, so $\rho > \Psi_B(\omega)$, and $\Psi_B(1) = \varphi$ *exactly* with Ψ_B decreasing: $\rho > \varphi$ for all $\omega < 1$.

$j \geq 3$, *remaining branch*. The strict-leaf ladder of Proposition 82 needs only k *distinct* strict leaves, each a node with the wall (Bo) at it (not one per occupant): it gives $\rho > \Psi_k(\omega)$, and here $k \geq j - 1$ always ($k \geq j$ if y is not a leaf). The golden crossings are exact: $\Psi_1(\frac{1}{2}) = \varphi$, $\Psi_2(\varphi/2) = \varphi$, $\Psi_3(1) = \sqrt{3} > \varphi$ – covering $j \geq 4$ outright and $j = 3$ up to $\varphi/2$ (always if y is not a leaf). The rest of $j = 3$ is a case tree: $o_2 < 3/\varphi$ (tail of o_2 exceeds $3/o_2$), or $o_1 < 3$ (tail of o_1 exceeds $(3/\varphi + 3)/o_1$, and $(3/\varphi + 3)/\varphi = 3$ exactly), or $o_1 \geq 3$ with the node/dust dichotomy on o_1 ’s hole content: a node makes y ’s subtree or o_1 ’s contribute a third strict leaf (Ψ_3), dust lands in m ’s tail. In detail, for $o_1 \geq 3$: if $y = o_1$, (Ry) gives $X_1 \geq o_1 - \omega - S_0 > 0$ (as $S_0 < 2 \leq o_1 - \omega$); a node inside makes y a non-leaf, whence $k = 3$ and Ψ_3 applies, while pure dust lives in m ’s tail and $\rho \geq S_0 + X_1 \geq o_1 - \omega \geq 2$. If y lies in another occupant’s subtree and is not a leaf, its subtree supplies the third strict leaf ($k = 3$); if y is a leaf and $\omega < \varphi/2$, $\Psi_2 > \varphi$; and if y is a leaf, $\omega \geq \varphi/2$ and $\sigma_2 > \omega$, then $\rho \geq \sigma_1 + \sigma_2 > 2\omega \geq \varphi$. If y lies *strictly inside* o_1 ’s subtree and is a leaf, two sub-cases close it: if some node of that subtree has two node children, one of its two disjoint subtrees avoids y and contributes a strict leaf, and o_2 ’s and o_3 ’s subtrees (neither containing y) contribute the other two: Ψ_3 . Otherwise the subtree is a chain $o_1 \supset v_1 \supset \dots \supset y$ and the pincer below runs with V defined as there: if $v^* \neq y$ the argument is verbatim — (Bo) holds at every node of the chain other than y , and m , now inside v^* ’s subtree (it sits in y ’s hole), is counted through $T_{v^*} \geq X_{v^*} + 1$ instead of externally, which reassembles (C1) unchanged; if $v^* = y$, the tail of y contains $o_2, o_3, m, \sigma_1, \sigma_2$, so $\varphi y > o_2 + 3 \geq 3/\varphi + 3 = 3\varphi$ gives $y > 3$, while (Ry) plus the tail of m give $y - \omega < \sigma_1 + \sigma_2 + X_y^{\text{rest}} \leq \rho \leq \varphi$, so $y < \varphi + \omega < \varphi^2 < 3$: contradiction. What survives is y a leaf outside o_1 ’s subtree, closed by the pincer below.

The pincer (closes y a leaf outside o_1 ’s subtree; the interior-chain case above reuses it as noted). Here $o_2 \geq 3/\varphi$, and no width, solid-disk or o_1 hypothesis is needed: it suffices that $s := \sigma_2 + \omega < 2$, which holds since $\sigma_2 \leq 1$ (the profile consists of rings smaller than m) and

$\omega < 1$. Suppose $\rho \leq \varphi$. Write D for the total radius of the rings < 1 *other than* σ_1, σ_2 , and for a node v let X_v be its hole content and $T_v \geq X_v$ its whole subtree. (a) The tail of m collects σ_1, σ_2 and all of D , whence $\rho > 1 + D$ and so $D < \varphi - 1$. (b) If some node of $\{o_1\} \cup o_1$'s subtree had two node children, their subtrees are disjoint and each contains a leaf, and the occupant of $\{o_2, o_3\}$ not containing y contributes a third; all three are strict (their holes are not y 's, as y lies outside o_1 's subtree), so $\rho > \Psi_3(\omega) > \Psi_3(1) = \sqrt{3} > \varphi$ in branch A (branch B is case (iii)). (c) Hence $\{o_1\} \cup o_1$'s subtree is a chain. Let V be the set of its nodes whose tail contains both o_2 and o_3 (so $o_1 \in V$, by the first-copy convention even if $o_1 = o_2$), and let $v^* = \min V$. The pieces $o_2, o_3, m, \sigma_1, \sigma_2$ lie outside v^* 's subtree — o_2, o_3 are top-level occupants, m sits in y 's hole, and P keeps the profile in the pan — so the tail of v^* counts them and T_{v^*} without repetition. Now (Bo) applies at every node of o_1 's subtree (its hole is not y 's, and $\sigma_1 \leq 1$), giving $X_{v^*} > v^* - s$, so

$$o_2 + 3 + (v^* - s) < \varphi v^* \implies o_2 < (\varphi - 1)v^* - 3 + s, \quad (\text{C1})$$

using $o_3 > 1$ and $\sigma_1 + \sigma_2 > 1$. With $o_2 \geq 3/\varphi$ and $3/\varphi + 3 = 3\varphi$ this forces $v^* > \varphi(3\varphi - s)$ (C2). Moreover v^* has a node child: otherwise $X_{v^*} \leq D < \varphi - 1$ and (Bo) would give $v^* < s + \varphi - 1$, against (C2) whenever $s \leq (2\varphi + 4)/\varphi^2 = 2.7639\dots$; and that child is $\leq o_2$ by minimality of v^* , so $X_{v^*} < o_2 + (\varphi - 1)$ and (Bo) gives $o_2 > v^* - s - (\varphi - 1)$ (C3). Combining (C1) and (C3) with $1/(2 - \varphi) = \varphi^2$,

$$v^*(2 - \varphi) < 2s + \varphi - 4 \implies v^* < \varphi^2(2s + \varphi - 4). \quad (\text{C4})$$

(C2) and (C4) are incompatible exactly when $6\varphi - 1 \geq s(2\varphi + 1)$, that is when

$$s \leq \frac{6\varphi - 1}{2\varphi + 1} = 11 - 4\sqrt{5} = 15 - 8\varphi = 2.05572\dots,$$

and $s < 2$ here. Contradiction; so $\rho > \varphi$. At the extreme $s \rightarrow 2$ the two bounds are $\varphi + 3 = 4.618\dots$ and $\varphi^3 = 2\varphi + 1 = 4.236\dots$, a margin of $2 - \varphi$. The constant is sharp for the argument: for $s > 11 - 4\sqrt{5}$ the pincer no longer closes, which is why it says nothing about a solid pivot. The adversarial reports (three rounds, including two repaired steps and a hostile audit of this pincer) and the verification scripts are in the repository (`code/batalla2.py`, 6/6; `code/microcelda.py`, 5/5). $\square$

For larger profiles the walls depend on how the rest of the profile is stashed; write $W := \sum_{i \geq 3} \sigma_i$, $\Sigma := \sigma_1 + \sigma_2 + W$, $M := \sum \text{children}(m)$ and X_{σ_1} for the prior content of σ_1 's hole.

Theorem 87 (Golden floor, pan exchange, larger profiles). *Let the witness keep $p \geq 3$ rings $\sigma_1 \geq \dots \geq \sigma_p$ (all < 1) in the pan. Blocking implies $\rho > \varphi$ in each of: (i) light, $\sigma_1 + W \leq 1$; (ii) nested, $W \leq \sigma_1 - \omega - X_{\sigma_1}$ (each with the ω domains of Theorem 86); (iii) heavy large, $\sigma_1 + W > 1$ and $\sigma_2 > \varphi - 1$, every j and every $\omega > 0$; (iv) heavy Ψ_B , $\sigma_1 + W > 1$, $\sigma_1 + M > 1$ and some strict leaf exists (in particular whenever $j \geq 2$), $\omega \in (0, 1)$; (v) heavy mirrors: for $p = 3$, $j = 1$, $\sigma_2 \leq \varphi - 1$, blocking is impossible – with (iii), $p = 3$ profiles with one occupant are closed at every $\omega > 0$; (vi) heavy swap, $p = 3$, $j = 2$, $\sigma_1 + M \leq 1$, $\sigma_2 \leq \varphi - 1$, except the sliver $\{\sigma_2, \sigma_3 > 1 - \omega\}$ (which forces $\omega > 2 - \varphi$). Regions outside these particular constructions are discussed in Appendix E; the uniform theorem does not rely on them.*

Proof. (i) Every placement of the proof of Theorem 86 sends one of σ_1, σ_2 to a target and the other to a row in D_m ; adding W to that row is legal ($\sigma_j + W \leq \sigma_1 + W \leq 1$), so every wall – (D_p): $\Sigma > 1$; (G) with either σ_i ; (Ry); (Bo); the evacuation dichotomy, now $\sigma_2 > 1 - \omega$ or

$\sigma_1 + M + W > 1$ – ports verbatim, and every tail only grows by W . In branch B the bookkeeping is $A = \sigma_1 + \sigma_2 + W + M + X = (\sigma_1 + M + W) + (\sigma_2 + X) > 1 + q$ with $q := \sigma_2 + X$: the same Ψ_B program. The pincer is unchanged ($s = \sigma_2 + \omega < 2$). (ii) By Lemma 75, W rows inside σ_1 's hole beside X_{σ_1} , and $\{\sigma_1 \text{ carrying } W\}$ is a single piece of radius $\sigma_1 \leq 1$ *wherever* σ_1 goes – to D_m in (G_{σ_2}) , (Bo) and the evacuation, to y 's hole in (Ry) , to the pan in (G_{σ_1}) : Theorem 86 ports again. (iii) The tail of m collects the whole profile: $\rho \geq \Sigma = \sigma_2 + (\sigma_1 + W) > (\varphi - 1) + 1 = \varphi$. (iv) The row $\{\sigma_2, \dots, \sigma_p\}$ into the strict leaf's hole ($\sigma_1 \leq 1$ to D_m) gives the wall $L' < (\sigma_2 + W) + \omega + X_{L'}$; with $\tilde{q} := \sigma_2 + W + X_{L'} > 1 - \omega$ (from $L' \geq 1$) and $A := (\sigma_1 + M) + \tilde{q} > 1 + \tilde{q}$, the two tails give $\rho \geq A$ and $\rho > (2 + \tilde{q})/(\tilde{q} + \omega)$: the branch-B program, $\rho > \Psi_B(\omega) > \Psi_B(1) = \varphi$. (v) $\sigma_2, \sigma_3 \leq \varphi - 1 < \frac{2}{3} = b(1) \leq b(o_1)$ fit concentrically in the two mirror pockets of $\{o_1, m\}$ in the disk $o_1 + 1 \leq R$ (disjoint, Lemma 80), and $\sigma_1 \leq 1$ rows in D_m : the exchange is never blocked there. (vi) The placement “ $\sigma_2 \rightarrow H_m, \sigma_1 + M \rightarrow D_m$ (row ≤ 1 by hypothesis), $\sigma_3 \rightarrow \text{pan}$ ” and its $\sigma_2 \leftrightarrow \sigma_3$ mirror force either $\{o_1, o_2, m, \sigma_3\}$ not to repack – by containment to the disk $o_1 + o_2$ and Proposition 51, $b_2(o_1, o_2) < 1$, the mirror-pocket wall of case (ii) of Theorem 86, whose optimization ports with fatter tails ($1 + \Sigma > 2 + \sigma_2$; the mixed case, $\{o_1, o_2, m, \sigma_3\}$ repacking while $\{o_1, o_2, m, \sigma_2\}$ does not, lands on the same wall, since $\sigma_2 \leq 1$ forces $b_2(o_1, o_2) < 1$) – or $\sigma_2, \sigma_3 > 1 - \omega$, and then $\rho \geq \Sigma > 1 + \sigma_2 > 2 - \omega$ closes $\omega < 2 - \varphi$. (Only $j = 2$: non-packability is not inherited by subsets, so the containment step fails for $j \geq 3$ – the same reason case (ii) of Theorem 86 is a pair statement.) The case tree, the two coverage cells missed by a first version and repaired, and the verification script are in the repository (`docs/drafts/perfilp.md`; `code/perfilp.py`, 5/5; adversarially verified). $\square$

The remaining heavy region shrinks further. The tool is a *mural-placement criterion* corrected for the pentagram phenomenon: circles $a_0, \dots, a_{k+1}$ tangent to the wall of a disk $\bar{R}$, with a_0, a_{k+1} diametral (so their own separation is exactly π on the far side), can be placed with *all* pairs disjoint if and only if, in the chosen order, the longest path

$$\max_{S \subseteq \{1, \dots, k\}} \sum_{\text{consecutive in } a_0, S, a_{k+1}} \theta(a_i, a_j)$$

is at most π (difference-constraint scheduling over Lemma 47; summing adjacent arcs alone is the criterion that the pentagram refutes).

Theorem 88 (The heavy region closes except one cell). *In the heavy non-nestable region of Theorem 87 ($\sigma_1 + W > 1$, $W + X_{\sigma_1} > \sigma_1 - \omega$, $\sigma_2 \leq \varphi - 1$, $\sigma_1 + M \leq 1$ where applicable), blocking implies $\rho > \varphi$ in each of: (i) $p = 3$, $j \geq 3$: the $j = 3$ case tree of Theorem 86 ports with the row wall $s' := \sigma_2 + \sigma_3 + \omega$ and the profile mass Σ in every tail, and the pincer closes whenever $s' \leq (\varphi - 1)\Sigma + (16 - 9\varphi)$ – which holds on the whole cell: $\sup(s' - (\varphi - 1)\Sigma) = 5\varphi - 7$, a margin of exactly $23 - 14\varphi > 0$, and at $\Sigma = 1$ the frontier degenerates to the $11 - 4\sqrt{5}$ of Theorem 86(iv); (ii) $p = 3$, $j = 2$: three branches – $\sigma_2 + \sigma_3 \leq 1$ resurrects the mirror wall of Theorem 86(ii); $\sigma_2 + \sigma_3 > 1$, $\omega \leq \frac{1}{2}$ runs a Ψ -program with interior crossing $\sqrt{1 + (1 - \omega)^2}$ and $\Psi(\frac{1}{2}) = \varphi$ exactly; and for $\omega > \frac{1}{2}$ the cell is empty of blockings: the tails of $\rho \leq \varphi$ force $o_2 \geq (1 + \Sigma)/\varphi$, $o_1 \geq (o_2 + 1 + \Sigma)/\varphi$, and there the mural placement of $\{o_2, m, \sigma_2, \sigma_3, o_1\}$ in the disk $o_1 + o_2$ always fits (longest path $\leq 2.648 < \pi$, computer-assisted maximization); (iii) $p \geq 4$, $j = 1$: mural coronas with margins ≥ 0.54 (computer-assisted); (iv) $p \geq 4$, $j = 2$: the supremum of the longest path over the closed domain is exactly π , attained only at the excluded boundary corner $\{\sigma_1 = 1, W = 0\}$, where $o_2 = 2/\varphi$, $o_1 = 2$, $\bar{R} = 2\varphi$ and $\sin^2 \frac{\theta(o_2, m)}{2} = \frac{1}{2} - \frac{\sqrt{5}}{10}$,*

$\sin^2 \frac{\theta(m, o_1)}{2} = \frac{1}{2} + \frac{\sqrt{5}}{10}$ sum to 1 exactly; the open domain stays strictly below π . The cell not covered by these particular constructions is

$$\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$$

(the containment to the disk $o_1 + o_2$ does not re-place o_3 , and the pan-level cyclic corona is outside this construction), together with extra small rings and the solid-pivot regime $\omega \geq 1$ for $j \geq 3$.

Proof sketch and verification. The chains and identities are exact (sympy); the two maximizations are grid-plus-refinement sweeps over compact four-parameter domains with the boundary behaviour controlled by the exact identities above (script `code/rstar.py`, 6/6, and a hostile adversarial round that refuted and repaired two steps: the naive adjacent-arc criterion – the pentagram – and a $j \geq 3$ corona that failed to re-place o_3 , whence the surviving cell; report in the repository). $\square$

E The corona-versus-tails campaign: closure of the residual cells

This appendix records the machinery and the results that close the cells left open by Theorems 86–88 and by the Status paragraph of Section 8. Epistemic labels are stated with each claim: *theorem* means a written proof (with exact identities delegated to `sympy/Lean`); *computational closure* means per-instance certificates over swept parameter ranges, with the certificate soundness proved but the extension to unswept (j, p, k) ranges supported numerically. Throughout, *certified maximization* historically meant the standard of Theorem 88 – directed grid-plus-refinement with exact per-instance criteria, not interval arithmetic. Following an external review, the *central* such maximization (the corona wall of cell C3.3, the supremum that closes the heaviest DPr regime) has been upgraded to a genuine box-by-box certificate: an exact corner majorant (each f -factor is coordinatewise monotone, and the products against f_{o_1} are capped by their monotone limits x/o_2 , so no sweep ceiling is needed) drives a branch-and-bound that certifies $\sup g < \pi$ over the *continuous* unbounded domain in 71 boxes plus two analytic tail regimes (script `rstarcert`, 5/5); the remaining carriers of the historical label have since been upgraded as well – see item (v) of the honest residue below, now closed throughout. *Certified* without qualification is reserved for branch-and-bound bounds valid box by box. Every script below is paired with an adversarial verification report in the repository (`docs/drafts/VEREDICTOS.md`). Instance and family totals attributed below to adversarial rounds (e.g. the $\sim 45\text{k}$ -family, 55k-instance, 20 000-fuzz and 1 460-test figures) are *report figures* from those rounds; a default run of the published scripts uses smaller budgets and prints correspondingly smaller totals.

Small rings adjoin to the profile.

Corollary 89 (Pan exchange, extra small rings). *In the pan-exchange template, every extra ring $e < r_m$ that P keeps in the pan adjoins the reinsertion profile: writing $S^+ := S \cup \{\text{extras}\}$, re-indexed decreasingly, the enlarged template is the pan-exchange template with profile S^+ , so Theorem 86 (for $|S^+| = 2$) and Theorems 87–88 (for $|S^+| \geq 3$) apply to S^+ with their stated case assignments and ω domains, and the residual cell of the assignment over S^+ is again $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ with $p = |S^+|$ (closed, for every small-ring count, by Theorem 92).*

Proof. The profile of the template is by definition the multiset of sub-unit rings the witness keeps in the pan, ordered decreasingly, and the hypotheses of the cited theorems use no property of the σ_i beyond $\sigma_i < 1$ and that placement; the extras satisfy both ($e < r_m = 1$). The enlarged template is therefore literally the template with profile S^+ : blocking in the enlarged template *is* blocking for S^+ , and every witness placement in the cited proofs places all of S^+ , so no implication between blocking for S and blocking for S^+ is invoked. The case assignment over $(|S^+|, j, \sigma_1 + W, \sigma_2, \sigma_1 + M)$ is the finite trichotomy already recorded: the cited cases cover $|S^+| = 3$ for every j , $|S^+| \geq 4$ with $j \leq 2$, and $|S^+| \geq 4$ with $\sigma_1 + M > 1$, and the complement is exactly the residual cell. $\square$

Certified computational closures

The remaining cells are closed by a necessity/sufficiency pair sharing one wall-tangency function: $\sin^2(\theta(a, b)/2) = f(a)f(b)$, $f(x) = x/(R - x)$, as in Lemma 47. *Necessity* (a true lower bound R_{lb} for the radius of any disk packing a given family): for any pair in any packing, the angular separation of centers satisfies $\cos \gamma \leq h(d_a, d_b) = (d_a^2 + d_b^2 - (a + b)^2)/(2d_a d_b)$, whose maximum over the admissible box of center distances is attained at a corner; for a non-stackable pair ($R < \max + 2\min$) without confinement the binding corner is mural and $h(R - a, R - b) = 1 - 2f(a)f(b)$ exactly, so $\gamma \geq \theta(a, b)$; stackable pairs are handled by subset certificates and by confinement ($|c| \geq 2g + r - R$ when a giant g is present), and the cyclic sum of separations in any packing is 2π . *Sufficiency* (a constructive placement at $R \geq R_{\text{lb}}$): the cyclic corona by longest-path positions with all pairs validated and small rings assigned to Descartes pockets ($k_p = k_a + k_b - 1/R + 2\sqrt{k_a k_b - (k_a + k_b)/R}$) as row bins. Its soundness is a theorem, by maximality of the longest path: every skipped ring s between consecutive spine members a, b satisfies $\theta(a, s) + \theta(s, b) \leq \theta(a, b)$, and this margin vanishes *exactly* at the Descartes pocket radius (the three mutually tangent mural circles have adjacent wall angles summing to the pair's), so each skipped ring sits murally inside its gap; positions satisfy $\alpha_j - \alpha_i \geq \theta(i, j)$ for all $i < j$ by construction, so only wrap-around pairs need (and receive) explicit validation. At the golden instance the margin is critically zero: $p(\varphi, 1; \varphi + 1) = \varphi/2$ with vanishing Descartes discriminant, and $f(\varphi)f(\varphi/2) + f(1)f(\varphi/2) = 1 = f(\varphi)f(1)$, so the counterexample of Theorem 19 is the exact critical point of the certificate system and is never unblocked by it (script `zigzag`, 5/5; the adversarial round refuted a stronger induction – consecutive triple margins do not imply global pair legality – and the construction therefore validates all pairs explicitly, as it always did).

Closure of the residual cells. With tails in cascade ($\rho \leq \varphi$ forces $o_k \geq (\sum_{i>k} o_i + 1 + \Sigma)/\varphi$, and $o \geq \varphi(1 + \Sigma)$ at near-ties), the constructive corona at $R = R_{\text{lb}}$ succeeds with *uniform tangent duality* (deficit 0.0 within the $2 \cdot 10^{-3}$ bisection tolerance of R_{lb} , with the complementary probe that the deficit strictly decreases in R ; the boundary placement is tangent, hence legal) across: the heavy cell $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ of Theorem 88 ($j \leq 5, p \leq 6$ swept); the solid-pivot regime $\omega \geq 1, j \geq 3$ (no step uses the width); and, in the nested template of Section 8, the three golden-level gaps declared in the Status paragraph – the $j = 2$ tip $\omega \in [\varphi/2, 1)$, profiles of four or more pieces beyond the reduction branch, and the quantitative gap lemma (extra smalls adjoin as in Corollary 89; the reduction re-routes through the corona where the light-cell inheritance fails, and sweeps directly with the floor $\alpha \geq \max(1 + \omega, \text{true pair of } u + \omega)$ where transport is illegitimate). Scripts `coronacolas` (5/5) and `coronanidada` (5/5), each with an adversarial round that repaired the routing (six repairs in the pan campaign, four in the nested

one; reports in the repository). These are computational closures: the per-instance certificate is theorem-backed as above, the swept ranges are stated, and the extension in (j, p, k) rests on the monotonicity evidence recorded by the scripts.

Written closure theorems

From computational closure to written proof: the insertion lemma. Two later results upgrade the central pan cells from computational closure to written mathematics.

Lemma 90 (Mural compaction). *If a family C packs in the disk R and every pair of C is non-stackable ($R < \max + 2\min$), then C packs murally in R : pushing every circle to the wall at the angle of its true center is legal.* Proof. Angular separations are unchanged; for any pair in any packing, $\cos \gamma \leq h(d_a, d_b)$ with the box maximum at the mural corner $h(R - a, R - b) = 1 - 2f(a)f(b) = \cos \theta$ (the corners with a vanishing coordinate give $h \rightarrow -\infty$ for non-stackable pairs, whose centers live in an annulus of positive margin $d_a \geq a + 2b - R$), so $\gamma \geq \theta(a, b)$, which is exactly mural disjointness. The hypothesis is tight in two directions (a central ring family, and a family non-stackable only against its largest member, both packable but not murally). $\square$

Lemma 91 (Insertion by shadows). *Let P pack $F = \{x_i\}$ in R and let s satisfy the regime $R > 2s + x_i$ for every i . If $\sum_i 2\arcsin((s + x_i)/(R - s)) < 2\pi$, then s inserts murally into P without moving any piece.* Proof. The forbidden arc of x_i at any depth d_i has half-width $\arccos h(R - s, d_i)$, and on $v \in (0, \infty)$ the map $v \mapsto h(R - s, v)$ has the single interior minimum $v^* = \sqrt{u^2 - w^2}$ ($u = R - s$, $w = s + x_i$) with $h(u, v^*) = \sqrt{u^2 - w^2}/u = \cos \arcsin(w/u)$ exactly: the shadow $\arcsin((s + x_i)/(R - s))$ bounds every arc uniformly in depth. If the shadows sum below 2π , a free angle exists. Without the regime the statement fails (a deep piece of a non-stackable pair forbids arcs up to π); the two regimes are mutually exclusive. $\square$

Theorem 92 (The heavy pan cells, written). *In the cell $\{j \geq 3, \sigma_2 \leq \varphi - 1\}$ of the pan exchange – for every profile size p , every small-ring count and every width, including the solid pivot $\omega \geq 1$ (where the branch $\sigma_2 > \varphi - 1$ is also closed – Step 6 of the proof) – $\rho \leq \varphi$ implies the exchange does not block.*

Proof. Throughout suppose $\rho \leq \varphi$. Write T_s for the total mass of the sub-unit rings that P keeps in the pan – the profile $\sigma_1 \geq \sigma_2 \geq \dots$, the extra small rings and the pan dust; this is exactly the mass the exchange displaces, since hole contents ride inside their parents. All of it lies in the tail of m , so $T_s \leq \varphi$.

Step 1 (the pan). The witness adopts F 's pan verbatim: the occupants at the positions F certified when it placed m , each carrying its P -subtree (hole contents are interior to the parent disk, so outer disjointness depends only on the outer radii, which are the ones F certified), and m at F 's position with M riding inside. The unit disk D_m is vacated in the carrier's hole, whose remaining content is untouched.

Step 2 (the row). Fill D_m greedily: process the displaced pieces other than σ_2 in decreasing order, adding each piece while the running mass stays ≤ 1 . The rowed collection, of mass $B \leq 1$, packs inside the vacated unit disk without disturbing anything (Lemma 3); $\sigma_1 < 1$ enters first.

Step 3 (the spillover bound). If nothing spills, the witness is complete and the exchange is performed. Otherwise let W' be the unrowed mass. Every unrowed piece π was rejected while the running mass was at most B , so $\pi > 1 - B$; and, after the extras and dust adjoin the profile and it is re-indexed decreasingly (Corollary 89), $\pi \leq \sigma_3 \leq \sigma_2$, since σ_1 is rowed and σ_2 is

reserved. Hence $B > 1 - \sigma_3$, and the tail of m gives

$$W' = T_s - \sigma_2 - B < \varphi - \sigma_2 - (1 - \sigma_3) = (\varphi - 1) - (\sigma_2 - \sigma_3) \leq 1/\varphi.$$

(When $\sigma_1 + \sigma_2 > 1$ this recovers the earlier bound $W' \leq \varphi - \sigma_1 - \sigma_2$ through the wall (D); the greedy row makes that wall unnecessary, which is what removes the cell hypothesis $\sigma_1 + \sigma_2 > 1$ of a previous version – the gap was found by an external review and is closed by the bound above.) The remainder packs as a single row-disk of radius W' (Lemma 3).

Step 4 (regime of the insertions). Two mural insertions remain: $\sigma_2 \leq \varphi - 1$ and the W' -disk, $W' \leq 1/\varphi = \varphi - 1$. Both go into the pan of Step 1 by Lemma 91, whose hypotheses we verify on the whole cell. Since the insertions are needed only in the spill case, $T_s \geq B + \pi > 1$; under $\rho \leq \varphi$ the tails then cascade: $o_3 \geq (1 + T_s)/\varphi > 2/\varphi$, $o_2 \geq (o_3 + 1 + T_s)/\varphi \geq (2/\varphi + 2)/\varphi = 2$ and $o_1 \geq (o_2 + o_3 + 1 + T_s)/\varphi \geq 2\varphi$ (the identity $(2/\varphi + 2)/\varphi = 2$ once more). The shadows $2 \arcsin((s + x_i)/(R - s))$ decrease in R , and the pan packs the disjoint pair o_1, o_2 , so evaluating them at the floor $R = o_1 + o_2$ majorizes. For every pan member $x_i \leq o_1$ and either insertand $s \leq \varphi - 1$,

$$R - x_i \geq o_2 \geq 2 > 2(\varphi - 1) \geq 2s, \quad o_2 - 2s \geq 4 - 2\varphi > 0 \text{ exactly,}$$

so the regime $R > 2s + x_i$ of Lemma 91 holds at every member, including, for the second insertion, the first insertand ($R - s_1 \geq o_1 + o_2 - (\varphi - 1) > 2s_2$ a fortiori).

Step 5 (the budget; the delimited certificate). It remains to check that the shadow sums stay below 2π for both insertions, over every occupant count, every profile and dust configuration and the whole cell. This is the single computational step of the proof, and it is box-certified: grouping the $k \geq 3$ occupants by their mass m and count $n \leq m$, the convexity chord of $\arcsin(z)/z$ (increasing) bounds their total shadow by $C(z_{\max})(1+s)m/u$, the polytope collapses to four aggregated variables, and a branch-and-bound with exact corner majorants and monotone-limit caps (no sweep ceiling) certifies $\sup < 2\pi$ for both insertions over every occupant count and unbounded slack (149 and 173 boxes plus closed tail cells; script `insercioncert`, 5/5, with an adversarial round that added one coupled-corner gate; it supersedes the directed maximization of script `insercion`, 7/7, whose adversarial report had refuted and repaired the naive statement of Lemma 91 with mural angles in place of shadows). The occupant-count direction is removed by the geometric-tail lemma (suffix masses decay geometrically, $S_i \geq \varphi S_{i+1}$, and the leader's own cascade $t_1 \geq (t_2 + S_3)/\varphi$ forbids a tight pair with a full tail; script `colageometrica`, 5/5). The supremum is attained at the cascade corner $j = 3$, $(o_1, o_2, o_3, m) = (2\varphi, 2, 2/\varphi, 1)$ – the floors of Step 4 with $T_s = 1$ – where the two sums are 4.7225 and 5.2644 against 2π , margins 1.56 and 1.02; the o_2 -direction is exactly monotone and the o_1 -direction an exact bathtub with limit $\pi < 2\pi$. With both insertions placed, all of the displaced mass is reinserted: the exchange is not blocked.

Step 6 (the solid pivot). For $\omega \geq 1$ the branch $\sigma_2 > \varphi - 1$ also closes, so no width hypothesis remains. If $\sigma_2 > \varphi/2$, the tail of m gives $\rho \geq \sigma_1 + \sigma_2 \geq 2\sigma_2 > \varphi$, against $\rho \leq \varphi$: the branch is empty by mass. If $\sigma_2 \in (\varphi - 1, \varphi/2]$, run the same witness with insertand $\sigma_2 \leq \varphi/2$: now $T_s \geq \sigma_1 + \sigma_2 \geq 2\sigma_2$ fattens the cascade to $o_2 \geq (1 + 2\sigma_2)(1 + \varphi)/\varphi^2 = 1 + 2\sigma_2 > 2\sigma_2$, so the regime holds with margin ≥ 1 , and the budget for insertands up to $\varphi/2$ under $T_s \geq 2\sigma_2$ is part of the certified parametric block of `insercioncert`. No step of the witness uses the width: D_m is a unit disk and every cap is a mass bound. $\square$

This supersedes the computational closure of $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ and of the solid-pivot regime: both cells now carry written proofs, with the budget maximization box-certified

by `insercioncert` (as the C3.3 wall is by `rstarcert`): no optimization asterisk remains in these cells.

Theorem 93 (Nested exchange, two or more occupants, written). *In the nested template (u the hole of a top-level α , v the pan) with $j \geq 2$ occupants, $\rho \leq \varphi$ implies the exchange does not block – for every width, every σ_2 , every profile size and any extra small rings.*

Proof. Suppose $\rho \leq \varphi$. Here the exchange sends m from the pan into u ; the displaced mass is the sub- m content that P keeps in u – the profile, extras and dust – of total mass Λ , all of it in the tail of m , so $\Lambda \leq \varphi$.

Step 1 (m into u by F 's certificate). The exchange is taken at the largest disagreement, so F and P place every ring above m identically; F processes decreasingly, so when it placed m into u the hole contained exactly those shared rings. The witness starts from P , removes u 's sub- m content, and moves m – with M riding inside – to F 's certified position in u : disjointness inside u is F 's certificate on identical outer radii. The unit disk D_m is vacated at top level.

Step 2 (the row and its caps). Fill D_m greedily with the loose pieces in decreasing order (all are $< m = 1$). If nothing spills the witness is complete. Otherwise let s' be the first non-fitting piece and B the mass already rowed, so $B + s' > 1$ and in particular $\Lambda > 1$. Three caps are exact. The largest loose piece rows first, so s' is at most the second largest; the two largest sum within Λ and within the tail of m , whence

$$s' \leq \min(\Lambda/2, \varphi/2).$$

The remainder $w^* = \Lambda - B - s'$ satisfies $w^* < 1/\varphi$ ($B + s' > 1$ and $\Lambda \leq \varphi$), and packs as one row-disk (Lemma 3).

Step 3 (automatic regime). With $j \geq 2$ the top level holds $j + 1 \geq 3$ pieces besides m ; order them $t_1 \geq t_2 \geq t_3 \geq \dots$. Under $\rho \leq \varphi$ the cascade gives $t_3 \geq (1 + \Lambda)/\varphi$ and

$$t_2 \geq \frac{t_3 + 1 + \Lambda}{\varphi} \geq \frac{(1 + \Lambda)(1 + \varphi)}{\varphi^2} = 1 + \Lambda \geq 2 \quad (\varphi^2 = 1 + \varphi).$$

The pan packs the disjoint pair t_1, t_2 , so $R \geq t_1 + t_2$, and for every pan member $x \leq t_1$,

$$R - x \geq t_2 \geq 2 > \varphi \geq 2s' \quad \text{and} \quad 2 > 2/\varphi \geq 2w^*,$$

with margin $2 - \varphi$: the regime of Lemma 91 holds for both insertions at every member (including, for the second, the first insertand: $R - s' \geq t_1 + t_2 - \varphi/2 > 2w^*$), uniformly in ω – no step of the witness uses the width.

Step 4 (insert). Insert s' murally into the pan as P placed it (only m has left; Lemma 91 accepts arbitrary positions, since the shadow bounds every forbidden arc uniformly in depth), then the w^* -disk likewise.

Step 5 (the budget; the delimited certificate). The shadow sums stay below 2π across the domain: this is the single computational step, a certified maximization with deterministic corners and margin ≥ 1.07 (script `optimizacion`, 5/5 — the uniform closure over $t_2 \geq 1 + \Lambda$, unbounded slack and both insertands — with `insercionanidada`, 6/6, as numerical contrast, whose large-slack block sweeps pieces up to 10^4 above the cascade floors — one and two inflated at a time — and widths up to $\omega = 1.35$, with the worst budget attained at the floor and the $t \rightarrow \infty$ limit closing by formula at $\pi < 2\pi$: the certified domain admits arbitrary values above the floors); the occupant-count caps drop by the geometric-tail lemma (script `colageometrica`, 5/5). With

both insertions placed the displaced mass is reinserted: the exchange is not blocked. For $j \leq 1$ the second insertion dies exactly on the golden razor ($o_1 \rightarrow 2/\varphi$ against $2w^* = 2/\varphi$, shadow ratio identically 1), which is why the hypothesis $j \geq 2$ is stated: that strip is Theorem 94. $\square$

Theorem 93 supersedes the computational closures of the golden-level nested gaps with two or more occupants – in particular the $j = 2$ tip $\omega \in [\varphi/2, 1)$ entirely, and profiles of four or more pieces beyond the reduction branch (profile size is mass here).

Theorem 94 (Gap lemma, nested $j \leq 1$, written). *In the nested template with $j \leq 1$ occupants, $\rho \leq \varphi$ implies the exchange does not block, for every width (solid pivot included), every profile and any extra small rings.*

Proof. Steps 1–2 are those of Theorem 93: m moves into u by F 's certificate, D_m is filled greedily, and either nothing spills (done) or the spill produces s' and w^* with the exact ligatures

$$s' \leq \min(\Lambda/2, \varphi/2), \quad w^* < \min(1/\varphi, \Lambda - 1), \quad w^* \leq \Lambda - 2s',$$

the last two because $B + s' > 1$ and because the rowed mass contains the largest loose piece, which is $\geq s'$. With $j \leq 1$ the insertion route of Theorem 93 is not available (the golden razor), and the witness instead re-packs the whole top level from scratch as a corona of at most five circles: α , the optional occupant o_1 , a unit disk carrying the D_m row, and the two circles s' and w^* – a bounded family, decided exactly.

The available radius. The true pan radius R packs P 's top level, which contains $\{\alpha, o_1, m\}$ (or $\{\alpha, m\}$), so R is at least every pair sum; moreover any packing of a trio $\{a, b, c\}$ satisfies the written trio bound $R \geq \min(R_3, M)$, where R_3 solves $\theta(a, b) + \theta(b, c) + \theta(c, a) = 2\pi$ and $M = \min(\max + 2\min)$ over the pairs is the stackability threshold: if $R < M$, every pair is non-stackable, so by the corner lemma of the duality certificates each angular separation of centres is at least the mural angle θ , and the three central separations sum to at most 2π (twice the extreme separation when the three centres lie in a half-plane, exactly 2π in the surrounding case) – $\theta(a, b) + \theta(b, c) + \theta(c, a) \leq 2\pi$, i.e. $R \geq R_3$ (θ decreases in R). The certificate below therefore works throughout with the proved floor $R_0 = \max\{\text{pair sums}, \min(R_3, M)\}$; at the golden point $(\alpha, o_1) = (2, 2/\varphi)$, pair sums, R_3 and M all collapse to $1 + \sqrt{5}$ and the trio sum equals 2π exactly.

The corona fits. For $j = 0$ this is an exact algebraic theorem: at $R = \alpha + 1$ the Descartes discriminant of the tangent pair $\{\alpha, 1\}$ vanishes identically, the wall pocket is $p(u) = u(u + 1)/(u^2 + u + 1)$, increasing. Parametrize the cascade floor by $u = (1 + \Lambda)/\varphi$, so $\Lambda = \varphi u - 1$ and $2/\varphi < u \leq \varphi$ on the domain $\Lambda \in (1, \varphi]$. With the coupled mass caps ($s' \leq \Lambda/2$, $w^* \leq \Lambda - 1$) the sub-pocket condition for s' reads $2p(u) \geq \Lambda$, and

$$2p(u) - \Lambda = \frac{(\varphi - u)[\varphi u^2 + 2(\varphi - 1)u + (\varphi - 1)]}{u^2 + u + 1} \geq 0$$

since the bracket has positive coefficients: the condition holds on all of $\Lambda \leq \varphi$, with equality exactly at $\Lambda = \varphi$ and α on its cascade floor – tangent to the golden pocket $b_2(\varphi, 1) = \varphi/2$ of Theorem 86's counterexample. For w^* the condition is $p(u) \geq \varphi u - 2$, and $g(u) = p(u) - (\varphi u - 2)$ is decreasing on the domain with $g(\varphi) = (3 - \sqrt{5})/4 > 0$: it holds throughout. When α exceeds its floor the pocket only widens (p increasing) and both conditions relax; the non-consecutive pair (s', w^*) is separated by the unit disk in the cyclic order, so its own tangency imposes nothing new: the quartet cell is a theorem. For $j = 1$ the quintet is the single delimited computational

step: it is box-certified over the entire domain by script `code/quintetocert.py` (4/4): a branch-and-bound over (Σ, α, o_1) with the exact ligatures of this proof, the radius handled through the necessity bracket $R_0 = \max\{\text{pair sums}, \min(R_3, M)\}$ in *both* branches (the $R_3 \leq M$ safeguard is bracketed, not sampled), five closed per-box certificates – pocket assignments, a row-disk merge of the two smalls, the diametral band with a closed-form 4-cycle test and a range dichotomy for w^* , a *stacked witness* that closes the golden point $(\varphi, \varphi, \varphi)$ algebraically (exact margin $1/\varphi^3$ in $\mathbb{Q}[\sqrt{5}]$), and a rational *semiarc threshold* $x \leq R/(1 + f(a) + f(o))$ that collapses to exactly 1 at the critical corner (the configuration of Theorem 19 reappearing as a tangent-legal witness) – with every acceptance decided by exact rational comparisons (no tolerances; angle sums by outward directed intervals with every operation rounded outward, and the arcsines evaluated by high-precision arithmetic on exact rational arguments, so no hypothesis on the platform’s math library remains), together with a written trio lemma showing that the trio fits within the proved floor R_0 : directly at the pair-sum floor when $\max(\alpha, o_1) \geq 2$, and through $R_3 \leq M$ when $\max(\alpha, o_1) \leq 2$ – both anchored at the kernel-checked mirror corner $b_2(2, \sqrt{5} - 1) = 1$ – and the $\alpha \rightarrow \infty$ tail closed by the exact pocket infimum $ab/(\sqrt{a} + \sqrt{b})^2$; the binding tangent curve of the trio sum $= 2\pi$ from $(2, 2/\varphi)$ through $(3/2, 3/2)$ is covered by the same subdivision with no special neighbourhoods. It supersedes, for this quintet, the earlier arc-LP and pocket sweeps (scripts `code/arcstp.py` and `code/bolsillos.py`, 5/5 each, three adversarial rounds), which in turn superseded the directed maximization of script `code/gaplemma.py`. No step uses the width – the solid pivot is included – and the caps are mass bounds, so every profile and any extra small rings are covered. With the corona placed, the displaced mass is reinserted: the exchange is not blocked. $\square$

With Theorems 93 and 94, the *entire nested template* of case (b) carries written proofs for every j , every width, every profile and extras – at the same epistemic standard as the pan side (the central pan maximization is box-certified by `rstarcert`, the D1 budget by `insercioncert`). For $j = 0$ the domain sweep has since been upgraded to an *exact algebraic pocket certificate*: at $R = \alpha + 1$ the Descartes discriminant vanishes identically and the wall pocket is $p(u) = u(u + 1)/(u^2 + u + 1)$, increasing; with the coupled mass caps ($s' \leq \Sigma/2$, $w^* \leq \Sigma - 1$) the sub-pocket condition factors as $(\varphi - u)r(u) \geq 0$ with $r(u) = \varphi u^2 + 2(\varphi - 1)u + (\varphi - 1)$ positive-coefficient, so equality holds exactly at $\Sigma = \varphi$ with α on its cascade floor – where the cap $s' = \varphi/2$ is *tangent* to the golden pocket $b_2(\varphi, 1) = \varphi/2$ of Theorem 86’s counterexample; the second cap closes with exact minimum $(3 - \sqrt{5})/4$ (the quartet cell is a theorem, its critical corner the golden pocket itself). The remaining bounded-corona domains – the $j = 1$ quintet and the $k \leq 2$ hole cells – are certified by subdivision as well, on two new exact tools: an *arc-LP characterization* of small mural coronas (feasibility of the per-arc requirement system is equivalent to the corona, with the pairs-fit precondition; its disjoint-family dual serves as pruning and an exact base-enumeration primal as the criterion – closed inequalities, so it certifies *at tangency*), and a closed-form feasibility test for the critical 4-cycle ($-\sigma \geq \max(B_1, B_2)$: the two diagonal NS-2 deficits against the slack). The binding frontier turned out to be an entire *tangent curve*: the trio sum $\theta(a, o) + \theta(o, 1) + \theta(1, a)$ at $R = a + o$ equals 2π exactly along a curve running from the golden point $(2, 2/\varphi)$ through $(3/2, 3/2)$ – where $\theta(3/2, 1; 3) = \pi/2$ exactly – to its mirror image, the boundary of the trio-necessity band; on the curve the corona is handled *by construction* (the trio is exactly tangent and the small pieces insert sub-pocket at zero angular cost), with the golden endpoints – where pair sums, R_3 and the stackability threshold all collapse – excluded into certified neighborhoods. Behind it all sits one more exact golden identity whose algebraic core is kernel-checked in Lean: $\theta(\varphi, 1/\varphi) + \theta(1/\varphi, 1) + \theta(1, \varphi) = \pi$ at $R = 2\varphi$, which makes

the dangerous 4-cycle of the $j = 1$ cell exactly tangent and ties its 5-cycle to the shared-root corner constant $\pi + 4 \arcsin(1/\sqrt{3})$ (scripts `code/arcolp.py` and `code/bolsillos.py`, 5/5 each, three adversarial rounds – one of which refuted and repaired the arc-LP statement itself). For the $j = 1$ quintet this closure is now superseded by the tolerance-free box certificate of script `code/quintetocert.py` (see the proof of Theorem 94): the tangent curve is covered by the same subdivision with no special neighbourhoods, and the golden endpoints are no longer excluded; `code/arcolp.py` and `code/bolsillos.py` remain the certificates of the other small-corona cells they back.

The exchange assembly and the container port

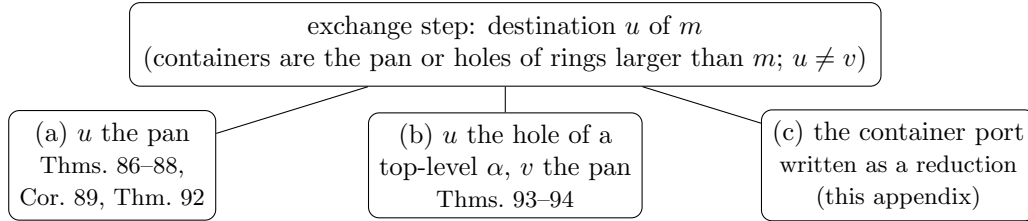


Figure 5: The case partition of the exchange assembly. Cases (a) and (b) are covered by the pan and nested theorems; case (c), the container port, is written as a reduction whose remaining domains are inventoried at the close of this appendix.

The exchange assembly. The exchange step of Theorem 9 is assembled into a case partition (a theorem: containers are the unique pan or holes of rings larger than m , and $u \neq v$): (a) u is the pan – the template of Theorems 86–88 plus the closures above; (b) u is the hole of a top-level α and v is the pan – the nested templates plus the closures above; (c) the *container port*. Assembly facts (theorems): $|S| \leq 1$ never blocks (a single $\sigma < r_m$ rows into D_m ; with $S = \emptyset$, m moves under F ’s own certificate, containers of larger rings being shared); feasibility is per-container and positions are existential (Section 2) – and this is now a proved *realization-and-repack lemma*, not a gloss: any forest whose containers each carry a placement of their children as balls with disjoint interiors is realized globally by composing translations along root-to-leaf paths (inductive invariant: a ring’s subtree stays in its ball, its strict descendants in its hole ball; the annulus material is disjoint from the hole interior; translations preserve distances, and any isometry of a container placement is just another placement), so replacing one container’s placement – subtrees rigid – yields another realization of the same forest, and a witness with a *new* assignment composes the per-container certificates (vacated ball, row, corona, pocket, shadow insertion) through the same lemma (script `repack`, 5/5, adversarial round: exact tangencies, solid rings and subtree isometries all verified, and the checker proved equivalent to material disjointness). Repacking any container is therefore a legal resource – the pan repack of Theorem 86 is the precedent; and every geometric wall descends by antitone containment to the intrinsic disk of its members, so walls port to any container with the corresponding capacity. Porting *programs* additionally needs their cohabitation hypotheses, which yields the closure split: if α is a direct member of v , case (b) is inherited verbatim with $R \mapsto Y - \omega$ (theorem); an extension lemma transports the (N) -heritage walls with the load travelling inside σ_1 (monolith for placements not using σ_1 ’s hole, exact absorption of the fattened B'_3 in the golden-line chain).

The genuinely new configurations – α outside v – were closed by a dedicated campaign (script `puertocii`, 7/7, two hostile rounds): exact pincers reduce the light profile to a window that is empty for $\omega_{\text{ef}} \leq 3/(2\varphi)$ (the corner degenerates at $\sigma_2 = \frac{1}{2}$ exactly), and inside the window the pan repack places σ_2 in the mirror pocket of the top-level pair, by $\alpha > 2$, $T > \sqrt{5} - 1$ and monotonicity of b_2 against the corner $b_2(2, \sqrt{5} - 1) = 1$; the heavy profile is closed by an exact partition pincer ($b_2(4/\varphi, 2/\varphi) = 12/(7\varphi) > 1$); towers, the $Y \geq \alpha$ branch and the shared-root configurations (where the top-level pair degenerates) are closed by a dedicated mural-trio analysis: the exchange lives in the hole of α with capacity $c = \alpha - \omega \geq \Sigma_S + Y$ (hole tariff), the trio sum decreases in c (exact), and its supremum over the legal window is the certified corner $\pi + 4 \arcsin(1/\sqrt{3}) \approx 5.60 < 2\pi$ (margin 0.68; lightness with $\sigma_1 \geq \sigma_2$ forces $\sigma_2 \leq \frac{1}{2}$, while the analytic danger threshold is $\sigma_2 = \frac{2}{3}$), with the heavy branch by an exact partition and the deeper/mirrored configurations by direct sweeps with derived tariffs.

The port itself is now *written as a reduction*: every case below is a theorem/corollary, lands in a computationally certified cell, or belongs to one of the residual domains inventoried honestly in the closing subsection. When v is the pan and α sits nested in a tower whose root t is top-level, the case is a *corollary* of Theorems 93–94: those theorems use, of the configuration, exactly four things – a top-level pan family $\{t, o_1, \dots, o_j, m\}$ in P , the vacated unit disk D_m at top level (v the pan makes m top-level in P), the legality of m 's destination by F 's certificate with rigid subtrees (agnostic to *which* hole receives m ; the tower is shared between F and P), and the mass ligatures – and all four hold with the tower root t in the role of “ α ”, the per-rank floors only rising (the tail of t contains the whole tower; $t \geq \alpha + \omega \geq 1 + 2\omega$ by two circles level by level; the hole-mass floor is automatic by an exact two-branch argument), while both certified domains admit arbitrary values above their floors (large-slack sweeps to 10^4 with the worst budget at the floor, and the $t \rightarrow \infty$, $\alpha \rightarrow \infty$ limits closing by formula – an adversarial round of this very reduction demanded and obtained the large-slack block). When v is the hole of a top-level Y , the case assembles into a theorem from the exact pincers above together with the two *hole-corona* theorems, which run the nested template *inside* the hole: the cascade chain among a hole's occupants yields the same automatic regime ($x_2 \geq 1 + \Sigma$ for $k \geq 3$ occupants, $\varphi^2 = 1 + \varphi$; tails are global), and for $k \leq 2$ the family is bounded and the armored trio bound applies, on the hole's own low floors (occupants ≥ 1 with no wall relation – a domain the pan boxes do not cover, swept separately with 10^4 slack and the $x_1 \rightarrow \infty$ limit by formula). The breathing branch $Y \geq \alpha$ needs a *strong breathing lemma* since the hole mass includes dust: the full cross-tail pincer plus the wall $\Sigma_S > 1$ force $X_Y + \omega > \varphi(3 - \varphi) = 2\varphi - 1 = \sqrt{5}$, dust is at most $\varphi - \Sigma_S < \varphi - 1$ by the tail of m , and $\sqrt{5} - (\varphi - 1) = \varphi$ exactly: the above- m mass satisfies $X_{>m} + \omega > \varphi$, so $k \geq 1$ – the golden threshold reappears after discounting dust. The α -corona branch is lighter: lightness gives $\Sigma_S - \sigma_2 < 1$ exactly (the rest of S rows into D_m and only σ_2 needs mural insertion), and the capacity window $[\Sigma_S + X_\alpha, 1 + \sigma_2 + X_\alpha]$ is exact with X_α cancelling (script `coronaagujero`, 5/5, adversarial round with zero counterexamples under 55k independent adversarial instances).

A *geometric-tail lemma* then makes every one of these shadow budgets *uniform in the occupant count*. For any cascade family $t_1 \geq t_2 \geq \dots \geq t_n$ (global tails, $\rho \leq \varphi$) with $R \geq t_1 + t_2$, regime $2s < t_2$ and $t_2 \geq 1 + \Sigma$, the budget is dominated term by term by an explicit finite series: the suffix masses S_i decay geometrically ($S_i \geq \varphi S_{i+1}$, again $1 + 1/\varphi = \varphi$), each tail piece obeys three exact caps ($t_{3+r} \leq \min(t_2, S_3/\varphi^r - (1 + \Sigma), (S_3 - 1 - \Sigma)/(r + 1))$), a piece at rank $3 + r$ drags a suffix of mass at least $1 + \Sigma + p_{\min}$ so the series has finitely many terms, and – the load-bearing link – the cascade of t_1 itself forces $t_1 \geq (t_2 + S_3)/\varphi$: a heavy tail forces a large

leader, so the tight pair and the full tail are *jointly infeasible* (allowing both would push the majorant to $6.37 > 2\pi$; with the link its supremum over the whole box is $5.2115 < 2\pi$, attained with *zero* gap at the real family $\{2\varphi, 2, 2/\varphi\}$ plus the unit disk – the very corner every sweep had found). The hypothesis $t_2 \geq 1 + \Sigma$ is forced by three-or-more occupants via $\varphi^2 = 1 + \varphi$ and is exactly the lemma’s frontier: at $n = 2$ the golden razor ($t_2 \rightarrow 2/\varphi$, shadow ratio identically 1) breaks the budget, which is the known boundary where the bounded-family closures take over. Consequently the sweep caps ($j \leq 6$, $k \leq 14$, $k \leq 12$) drop out of the written theorems’ statements: shadow coverage holds for *every* occupant count (script `colageometrica`, 5/5, adversarial round: domination unbroken under $\sim 45k$ families including the theorems’ own cascade generators). Finally, a *pocket- φ lemma* removes the profile- and dust-count directions of the scaling question exactly: at $R = o_1 + o_2$ the Descartes discriminant of the tangent pair vanishes identically, the pocket radius $1/(1/o_1 + 1/o_2 - 1/(o_1 + o_2))$ is monotone in o_1, o_2, R and equals φ *exactly* at the cascade minimum $(2\varphi, 2, 2\varphi + 2)$, while the total small mass is $\leq \varphi$ by the tail of m : the entire profile and dust row into a single pocket (p and k free; script `escala`, 5/5, with the nested template verbatim and a geometric non-invasion check).

The multipiece campaign and the vacuity closures

The shared-root *multipiece* sweeps are certified by a pessimistic-corner engine over the arc-LP’s sufficiency direction (valid for every k): a term-by-term majorant matrix (each wall angle bounded at its own corner, participants at their ceilings *including* their contribution to the capacity, the rest at their floors – valid since $\partial\theta/\partial a$ carries the sign of $R - a - b = \Sigma_S + \text{rest} > 0$), a maximum-slack LP whose returned gap vector is re-verified in plain floating point, and an *antipodal* criterion for the $\Sigma_S \rightarrow 1$ edge (Y and m at distance exactly π , strict by algebra, with the remaining pieces as an interval – hence totally unimodular, hence exact – path system on one half-circle, where the circular LP’s margin degenerates even at real points). This certifies the explicit extra occupants (1–3 corona pieces up to size Y , coronas of 4–6 pieces, entire $\sigma_2 < 1$) over the sweep box, and the mirrored branch beyond its $X = 0$ cut in the $X_Y = 0$ direction (shifted windows with $X_\alpha, X_z, X_m > 0$, dual per-box bounds coupling the diametral floor $c' \geq 1 + z$, certified at margins both 0.3 and 0.35). The heavy-profile partition – multipiece A with *unbounded* cardinality – is closed by a *reduction lemma with a golden threshold*, exact in $\mathbb{Q}(\sqrt{5})$: taking B^* the best subset of S of mass ≤ 1 (rowed into D_m) and $A = S \setminus B^*$ with $\beta = \Sigma B^*$, maximality gives $a + \beta > 1$ for every $a \in A$, $\beta \geq \sigma_1 \geq \max A$ (hence $a \leq \varphi/2$), and *at most four* pieces of A exceed $t_0 = (\varphi - 1)/4 = 1/(4\varphi)$ – a dichotomy at $\beta^* = (9 - \sqrt{5})/8$ whose critical identity $5t_0 = \varphi - \beta^*$ is exact, with the twin identity $4t_0 = \varphi - 1$ making four large pieces and dust *incompatible* – while the remaining dust (pieces $\leq t_0$, present only for $\beta > \beta^*$, mass $\mu < 5t_0$) is paid by *mass, not count*: convexity of arcsin gives $\theta(a, b) \leq \arcsin f(a) + \arcsin f(b)$, so a monotone dust chain costs at most $\pi\mu/(R - t_0)$ regardless of how many pieces carry the mass. The heavy mural is then $\{Y \text{ or } z, m\}$ plus at most four large pieces plus one aggregated dust block, and the two-sided antipodal engine certifies both heavy branches over their sweep boxes – the direct one entirely; the mirrored one on its $X = 0$ cut – and the slot-reduction closure **1emaA3** now re-certifies both heavy murals with Y , z and $|A|$ unbounded (the golden-chain lemma settling the tangency corner); and now, closing the mirrored cell, *beyond* that cut: within the dust convention the global pivot tail caps *every* extra mass at once, $\Sigma_S + X_m + X_\alpha + X_z + \Sigma X_Y \leq \varphi$, so each dust mass is below $\varphi - 1$ – the old sweep ceilings $X_\alpha \leq 1.5$, $X_z \leq 1$ cease to be ceilings – and a frozen-criterion sweep certifies the heavy mirrored branch with *all* dusts free: X_m eliminated as a dimension (its majorant only ever uses its lower endpoint), X_z eliminated by

credit segments, the dust block made *splittable* across the two half-circles (a greedy halving of any multiset of pieces $\leq \text{cap}$ leaves $|m_1 - m_2| \leq \text{cap}$, so two sub-blocks of mass $M/2 + \text{cap}/2$ cover every real split – the atomic block saturated one side), and the costly $\Sigma_S \rightarrow 1$ edge collapsed by a *degeneracy reduction*: there the heavy wall forces $\sigma_2 \leq \varepsilon_0$, the best-subset row fills $\beta \geq 1 - \varepsilon_0$, and both windows have width $\leq \varepsilon_0$, so fourteen dimensions reduce to four and the whole strip $\Sigma_S \in (1, 1.016]$ certifies in 85 boxes (six Σ_S -bands, $\sim 32\,100$ boxes, cover the rest). What remains sampled there: only the model-conditional occupant channel – the width $\text{cap } \omega \leq 1.6$ has since been removed: the tail $\omega > 1.6$ is box-certified at once for every width by monotone-limit caps (the fast pieces grow linearly in ω , the receiving hole satisfies $c \geq 1 + z$, and each corona term is bounded uniformly in ω ; script `espomegacola`, 5/5, adversarial round: confirmed with no soundness gap) (scripts `esppesada` and `espfinal`, 5/5 each, adversarial rounds; the latter’s referee found one surgical soundness gap – σ_1 can be excluded from the best subset, so the strip’s dust cap is $2\varepsilon_0$, not ε_0 – repaired with identical counts, and proved the greedy-halving invariant). Also settled: the $X_Y > 0$ mirrored corona – whose exact danger geometry is now settled by a two-round exploration: at the coexistence floor $c' = 1 + z$ the pair (z, m) is exactly diametral, and an extra piece x of X_Y fits beside m precisely when $x \leq p(z, 1; 1 + z) = z(z + 1)/(z^2 + z + 1)$, the *Descartes pocket of the diametral pair* (degenerate form, exact at $R = a + b$; the would-be sliver $x \in (p, 1)$ has width $1/(z^2 + z + 1) - 1/(2\varphi^2)$ at $z = \varphi$, where also $p(\varphi, 1; \varphi + 1) = \varphi/2$ exactly, the golden pocket of Theorem 86’s counterexample, already met as the critical point of the duality certificates above); a first adversarial round confirmed this algebra by hand, and a second one then proved the sliver *vacuous*: the global tail of the pivot counts every smaller piece, so $\Sigma_S + X_m + \Sigma X_Y \leq \varphi$, which is incompatible with the light-profile wall $\Sigma_S > 1$ and $x > p \geq 0.91$ (100% of the previously sampled “variety” violates it, minimum 1.96 against φ), and independently the floor itself is *rigid* – the diametral pair is forced tangent and the largest hole left in the container is exactly the pocket, so no legal configuration ever placed such an x murally at the floor. The positive consequence: every *legal* X_Y piece measures at most $\varphi - \Sigma_S < \varphi - 1 < p(z)$ throughout the domain – *always sub-pocket* – so the light mirrored $X_Y > 0$ cell *closes* by vacuity plus a certified k -piece corona: a nine-dimensional branch-and-bound over the sweep box with the dust mass $\mu \leq \varphi - \Sigma_S - X_m$ as its own dimension (each piece $\leq \mu$ by pigeonhole) certifies the corona $\{z, D_m, \sigma_2\}$ plus the dust block – paid by mass, monotone chain at most $\pi\mu/(c' - \mu)$, any k – throughout the domain (1 329 boxes; end-to-end sanity with explicit 1–4-piece partitions, 250/250 across seeds), all *within the dust convention* $X_Y < r_m$. The “occupant $\geq r_m$ inside Y ” channel, long model-conditional with its tariff underived, now has a *derived* tariff – two node walls: in the light profile blocking implies $x < \sigma_2 + \omega + X_x$ (Lemma 76 ported to the mirrored cell: the opposite disk packs $\{\sigma_2\} \cup \text{children}(x)$ into x ’s hole and the row $S \setminus \{\sigma_2\} < 1$ fills m ’s vacated disk; nothing else moves), and in *every* profile $x < \omega + \Sigma_S - 1 + \sigma_1 + X_x$ (a greedy partition of S sends B into x ’s hole and $A \leq 1$ to the row) – and the light channel is *closed entirely on v and on the depth-one tower cut*: the exact tie $x = r_m$ is ρ -illegal (the first-copy convention makes the first copy’s tail collect the twin *plus* all of S and the dust, $\geq 1 + \Sigma_S > 2 > \varphi$ – the referee’s find, which dissolved the residual “twin sheet” the campaign had first delimited: one more boundary vacuity), the band $x \in (r_m, 2/\varphi)$ is ρ -illegal by the tail of x itself, the high band certifies by subdivision (79 277 boxes, using the trio floor of $\{z, m, x\}$ from a new *crescent lemma* – exact angular windows around a near-mural giant, matching the classical three-circle optimum $1 + 2/\sqrt{3}$ to 10^{-13} – and a *dust-pooling* witness: the blocking walls with the X -masses in situ are only one assignment, and the witness repools σ_2 and the obstructing dust across the free disks of u , of z ’s hole and

of m 's vacated disk), and above the node wall the cell unblocks outright (script `espcanal`, 5/5, adversarial round run on rigor criteria alone); the heavy profile with x explicit is certified as well, by a *dimension reduction* that the extra ring itself makes possible: with $x \geq (1 + \Sigma_S)/\varphi$ forced by x 's tail, the capacity floor is far above `espfinal`'s $1 + z$, so the whole partition load $A = S \setminus B^*$ is paid *by mass* as a splittable block with the exact cap $\min(\beta, \varphi/2, \Sigma A)$ (every piece of A is bounded by each term: $\beta \geq \sigma_1 \geq \max A$ even when σ_1 is excluded from B^* , maximality plus $\Sigma_S \leq \varphi$ gives $a \leq \varphi/2$, and a piece cannot outweigh its multiset), the six explicit-partition dimensions collapse to β alone, and a new exact prune $\beta > \frac{1}{2}$ (the best-subset greedy stops only when every remaining piece exceeds $1 - \beta$, so $\sigma_1 > 1 - \beta$ with $\sigma_1 \leq \beta$) dissolves a phantom fine-dust band: five Σ_S -bands (144 337 boxes) plus the tower cut (15 571 boxes) certify the heavy channel entirely (script `espcanalp`, 5/5, adversarial round: the referee re-derived the reduction theorem independently and probed it with 20 000-instance fuzz, and found the uncovered tower cut and a mislabeled band edge, both repaired – the tower cut is now *certified*, not declared); what remains of the channel is the mid band at tower depth ≥ 2 and the structural exclusion of u 's content; and the heavy mirrored branch's dust is now *merged* with the reduction-lemma block as well – on the $X = 0$ cut, the X_Y dust (mass $\leq \varphi - \Sigma_S$) joins the partition dust in a single mass-paid chain, the hidden μ_Y range is handled by an eight-segment ceiling/tail-credit split (a full waiver of the dust's own tail term creates a *phantom* tangency exactly at $\Sigma_S \rightarrow 1$, where that tail is the rescue), and the heavy wall $\Sigma_S \geq 1 + \sigma_2$ prunes the light region already certified – whose exact boundary $\Sigma_S = 1 + \sigma_2$ is *vacuous* ($S \setminus \{\sigma_2\}$ has mass exactly 1, so $\beta = 1$ and the α -window is empty): the mirrored cell with $X_Y > 0$ is closed in both profiles on its declared cuts, certified over six Σ_S -bands ($\sim 34\,800$ boxes) (scripts `espxy`, `espvals`, `espkp` and `esppesada`, 5/5 each, four adversarial rounds; the second refuted a proposed vacated-ball witness – the two modes of the repack lemma, positional and fresh-certificate, must not be mixed in one container – and found the missing global-tail wall itself; the third confirmed the certificate with no soundness gap and forced the convention to be declared); the caps of the sweep boxes themselves have since been removed by the tail certificates `espomegacola` and `r2bcolas` (scripts `r2bmulti` and `areduccion`, 5/5 each, adversarial rounds: no soundness gap – 1 460 directed tests for the engine, hand-redone algebra and adversarial profiles for the lemma; the rounds corrected domain claims to the sweep boxes, replaced the solver-trusting margin by a float re-verification of the returned gap vector, fixed a degenerate half-circle that silently re-introduced the excluded antipodal pair, and contributed the $4t_0$ incompatibility). The near-equal-tops duality gap ($R_{\text{fit}}/R_{\text{lb}} \leq 1.0116$) was first re-delimited by a *conditional duality lemma*: for families whose pairs are all non-stackable at the radius in play, the arc-LP radius of Appendix E is the exact minimal mural-corona radius on both sides, by compactification for necessity and the closed arc-LP for sufficiency, reproducing the classical equal-circle optima $1 + 2/\sqrt{3}$ and $1 + \sqrt{2}$ at tangency; on the synthetic non-stackable domain the true load-versus-tops interval has corner supremum 1.082, evaluated over a deterministic corner mesh – now an *abstract* arc-LP statement: a subsequent global-tail audit showed that every sampled gap instance of the realistic generator is ρ -illegal (0/30 under the full tail legality), by a one-line *forbidden-triple* fact: three pieces whose second and third are $\geq (\varphi/2)$ -largest already violate $\rho \leq \varphi$ (their tail sum is $\geq 2 \cdot (\varphi/2) = \varphi$ times the largest), while the duality gap needs at least three near-equal tops (ratio $0.9 > \varphi/2$) – and the complementary case, a third piece too small for the forbidden triple, is closed by a *forced sub-pocket* lemma: full tail legality gives $t_3 < (\varphi - r_2)t_1 - 2$, the dominant pair's pocket at its pairs floor is $q(r_2)t_1$ with $q(r) = r(1 + r)/(1 + r + r^2)$ (the same degenerate-pocket function, scaled by the smaller piece), and $(\varphi - r_2)t_1 - 2 \leq q(r_2)t_1$ holds

throughout $t_1 \leq 2/(\varphi - r_2 - q(r_2))$ – at the cell’s edge $r_2 = 0.9$ the ceiling is 23.0, growing to infinity at the real root $r^* = 0.9637$ of the golden cubic $r^3 + (2 - \varphi)r^2 + (2 - \varphi)r = \varphi$, while the generator’s true domain is $t_1 \leq 6.64$ – so the third piece always inserts freely, the trio sits at its pairs floor, and (checked against the confinement strengthening of the lower bound, 100% of 5 300 cell instances) R certificates match: the cell has *no* legal instance of the phenomenon (0 gaps across $\sim 11\,000$ adversarial legal samples, including sub- m third pieces and four-top profiles). The old 1.0116 residue is therefore *withdrawn*: $\rho \leq \varphi$ itself forbids the configurations that carried it, and only the abstract arc-LP statements (the conditional duality lemma and the synthetic 1.082 corner) remain; and the converse “gap $\Rightarrow$ cell”, long declared open with only the empirical boundary 0.9, is now closed in an *exact* form: a certified-gap family, after an exact *peeling* step (any piece below the straight-wall pocket $p_\infty(a, b) = ab/(\sqrt{a} + \sqrt{b})^2$ of the core’s smallest pair inserts interior to every wall corona at every radius – the Descartes pocket *shrinks* as R grows, so the flat-wall limit is the uniform floor – leaving the wall radius unchanged), must either retain a *stackable* pair at the arc-LP radius (some $u \geq t$ with $R^* \geq u + 2t$: the cyclic certificate loses that pair’s angle) or, with four or more core pieces, have a non-adjacent pair dominating both complementary arcs of the minimal-sum cyclic order at an intermediate radius; the complement is gap-free by two small theorems (a three-piece non-stackable family reduces the arc-LP to boxes, $\Sigma\theta \leq 2\pi$ exactly the cyclic condition; and without stackable pairs the corner analysis of the cyclic engine gives $\gamma_{\min} = \theta_w$ exactly, so non-domination would make the system feasible below R^*). Across 4 000 sampled families every one of the 1 266 certified gaps (gap-certification threshold $2 \cdot 10^{-6}$; the band $(10^{-9}, 2 \cdot 10^{-6}]$ is declared unswept) lands in the exact cell – all of them via the stackable-pair route: the true anatomy of the duality gap is a stackable grain that fails to peel, not the near-equal tops themselves (a peeled four-equal-tops family has *no* gap; script `f3converso`, 5/5). The scripts `f3cierre`, `auditcolas` and `f3vacio` (5/5 each) passed adversarial rounds: the first refuted the original “closure” claim and its sampled 1.030 bound, the second refuted a partial-tail audit and supplied the forbidden-triple mechanism, the third corrected the declared generator domain (6.64, not 5.1) and verified the confinement attack. The shared-root *trio* itself is certified by subdivision over its entire legal domain in the direct orientation – light interior, both frontiers and every depth at once, via the free-leader reduction – and over the $X = 0$ cut of the mirrored one, where the same branch-and-bound also proves the mass wall $\Sigma_S \leq \varphi$ *necessary*: without it the corner $\sigma_2 \rightarrow 1$ is legal and its corona genuinely fails, the (z, m) pair going exactly diametral (script `r2bcert`, 5/5, adversarial round; the pan-repack legality behind the mirror-pocket pincers is likewise no longer on this list: it is the realization-and-repack lemma above). The optimization asterisk of the *bounded-family* domain sweeps is, for the shadow budgets themselves, closed: since every term of a shadow budget is monotone in each piece radius and in s , and antitone in R , corner evaluation gives a valid upper bound on any box, and an exhaustive branch-and-bound over the majorant’s hypothesis box (with the exact prunings $t_2 \geq 1 + \Sigma$, $u \leq \varphi t_2$ and the t_1 -link, the π -cap for $t_1 \rightarrow \infty$, and a normalized form with an analytic geometric remainder for $t_2 > 10^3$) *certifies* $\sup G \leq 5.25$ on the principal box and $< 2\pi - 0.05$ globally – margins 0.98 and 0.19 radians, twelve orders of magnitude above IEEE rounding (script `optimizacion`, 5/5, adversarial round with a repaired normalized-link bound and a certified-honesty control: tightening the target below the true supremum 5.2115 makes the branch-and-bound stall at exactly that value).

The extra-ring cells of the occupant channel

The occupant channel ($\geq r_m$ inside Y 's container) has its tariff *derived* (the two node walls) and is closed in both profiles on v and on the depth-one tower cut; the mid band $x \in [2/\varphi, \sigma_2 + \omega + X_x]$ at tower depth ≥ 2 is closed for chain towers in both profiles (scripts `code/esptorre.py`, `code/esptorrep.py`: the tower enters as mass on favourable floors only, and the naive reduction to depth one fails because t_1 's children are not dust). The inherited width ceiling $\omega \leq 1.6$ of the channel cells is removed: the six criteria are ω -generic, the mid range $\omega \in [1.6, 40]$ re-runs them over extended roots ($\sim 5M$ boxes), and the tail $\omega \geq 40$ closes by self-contained cap arguments with margins 0.4–0.8 rad (`code/espomegacanal.py`).

The $k \geq 2$ extra-ring cells are certified on their declared domains (scripts `code/lemaA4.py` and `code/lemaA5.py`; nine adversarial rounds, four intermediate refutations repaired): *light leaf extras* for $\omega \leq 1.25$ with the full nested-mass range – $W_z \leq 34$ by domain, and the $W_z > 34$ tail by a mode in which the nested mass counts toward z 's tail, so $\rho \leq \varphi$ both bounds the uncovered sector by infeasibility and couples $z \leq C_0 + W_z$; the piecewise-linear lower bound on the hosting capacity then has an upper piece of slope $2/\varphi > 1$ where the interior critical points of the pair bound are *maxima* – the reverse of the generic-slope case – with a closed-form critical point, so the supremum over the unbounded z -ray is computed from finitely many candidates. *Parent extras* (nested inside other extras of v) are certified on $\omega \leq 1.05$, $W_v \leq 8$, $W_z \leq 34$ (extended node walls $T + W_z$, positional caps for $j \geq 6$). The width range $\omega \in [1.6, 2]$ is certified for $j_v \leq 1$ configurations via a movable antipodal exemption in the placement engine (any single-clamped pair $(z, \cdot)$ is exemptible by coexistence), with the finding that the saturated-diametral obstruction family is *width-invariant*: the $j_v \geq 2$ residues of all width ranges are one object. The *heavy* profile ($\Sigma_S \geq 1 + \sigma_2$) with $k \geq 2$ extras is certified on $\omega \leq 1.15$ over a 16-band manifest (the A7 heavy node wall applied per-extra, the partition collapsed to its best-subset mass, exact ladder variants up to $j \leq 8$). The light-leaf width cut rose from 1.15 to 1.25 via a *necessity* wall: the floor capacity c^* of the mandatory corona set, computed by bisection over the repository's audited order-sum refuter (radial stacking handled). Every mirrored-cell closure stays within the dust convention (every extra piece is dust, of size $< r_m$; the extra *masses* are bounded by the global tail, not by r_m).

Historical coverage limits of the computational campaign

The list below records the coverage limits of this particular campaign, not remaining cases of the universal disk theorem. Theorem 22 now settles the golden guarantee by a different argument, independently of every computational closure below. Each item retains its original epistemic label; bounded or sampled claims are not upgraded to universal certificates by the new theorem.

- (i) *Swept*: the occupant-count direction j of the *computational* scaling closures, swept to $j \leq 9$ (pan) and $j \leq 8$ (nested) with uniform tangent duality and the exact geometric growth $T_{k-1} \geq \varphi T_k$ of the cascade; the *written* theorems' occupant counts are closed by the geometric-tail lemma.
- (ii) *Box-certified, on declared boxes*: every subdivision certificate is rigorous over its declared box; the former width, mass and piece-count ceilings are removed by the tail certificates and the slot-reduction engine (`code/espomegacola.py`, `code/r2bcolas.py`, `code/lemaA.py`–`code/lemaA3.py`). What remains: the engine's mirrored-ESP corona boxes hold on their swept ranges only.

- (iii) *Declared residues of the extra-ring cells:* the band $\omega \in [1.25, 1.6]$ and the width-invariant $j_v \geq 2$ family for $\omega > 1.6$; parent extras with $W_v > 8$ or $\omega > 1.05$ or $W_z > 34$; the heavy profile outside $\omega \leq 1.25$; branched towers (chain towers are closed in both profiles); and the structural exclusion of u 's content (the one model-conditional piece). The high-width, large- W_z leaf branch is certified on $\omega \in [1.6, 6]$, $W_z > 34$, excluding the declared $j_v \geq 2$ family; its $\omega > 6$ continuation remains open. The low- W_z , $j_v \leq 1$ branch is certified only through $\omega = 2$. Recorded truth probes found no violations; they do not establish universal coverage.
- (iv) *Sampled:* the supporting sweep of the exact “gap $\Rightarrow$ cell” converse (closed in exact form; the domination branch is an unexercised prediction; the realistic cell itself carries no legal instance).
- (v) *Exposition program:* analytic proofs replacing the remaining computational certificates. Their optimization budgets are already certified, as recorded in Appendix A. A rigorous certificate on its declared domain is already a proof; replacing it is not itself a missing mathematical case. This is an exposition direction of Open Problem 43, not a gap in the global golden proof.

Everything else in this appendix is a written proof, an exact identity in $\mathbb{Q}(\sqrt{5})$, a vacuity, or a subdivision certificate recorded in the verification map (Appendix A).

F Verification details for Theorem 14

All feasibilities in the twin instances are two-circle-exact except three facts, proved here; every numerical claim is also reproduced by the scripts in the repository (`code/gemelas.py`).

(V1) $\{10, 4.99, 4.50\}$ does not pack in a disk of radius 15. As in Lemma 12: $|c_A| \leq 5$, $|c_X| \leq 10.01$, $|c_Y| \leq 10.5$, $|c_A - c_X| \geq 14.99$, $|c_A - c_Y| \geq 14.5$, $|c_X - c_Y| \geq 9.49$; the first two give $a := |c_A| \geq 4.98$. With $u = -c_A/|c_A|$, $c_X \cdot u \geq (124.5 - a^2)/(2a) \geq 9.95$ and $c_Y \cdot u \geq (100 - a^2)/(2a) \geq 7.5$ (both decreasing in a , evaluated at $a = 5$), with transverse norms at most $\sqrt{10.01^2 - 9.95^2} \leq 1.095$ and $\sqrt{10.5^2 - 7.5^2} \leq 7.349$. Hence $|c_X - c_Y|^2 \leq 10.01^2 + 10.5^2 - 2(9.95 \cdot 7.5 - 1.095 \cdot 7.349) \leq 77.30$, so $|c_X - c_Y| \leq 8.80 < 9.49$.

(V2) With the pan pair $\{10, 5\}$ placed, no third circle of radius $z \in \{4.74, 4.76\}$ fits. From $|c_A| \leq 5$, $|c_B| \leq 10$ and $|c_A - c_B| \geq 15$ the triangle inequality forces equalities: the pair is diametrically rigid, $c_B = 10u$ for a unit vector u and $c_A = -5u$. For a third circle of radius z : $c_z \cdot u \geq ((10+z)^2 - (15-z)^2 - 25)/10$, which equals 8.8 for $z = 4.76$ and 8.7 for $z = 4.74$; then $|c_z - c_B|^2 = |c_z|^2 - 20 c_z \cdot u + 100 \leq (15 - z)^2 - 20 c_z \cdot u + 100$, giving $|c_z - c_B| \leq 5.38 < 9.76$ and $\leq 5.60 < 9.74$ respectively.

(V3) A feasible triple. The circles of radii 10, 4.76, and 4.74 pack in a disk of radius 15. An exact boundary configuration is $c_A = (-5, 0)$,

$$c_X = (8.8, \sqrt{27.4176}), \quad c_Y = (8.7, -\sqrt{29.5776}).$$

Then $|c_A| = 5$; $|c_X|^2 = 77.44 + 27.4176 = 104.8576 = 10.24^2$ and $|c_Y|^2 = 75.69 + 29.5776 = 105.2676 = 10.26^2$, so $|c_X| = 15 - 4.76$ and $|c_Y| = 15 - 4.74$ (both on the boundary; the abscissas

are the exact tangency values $10.24 \cdot \frac{55}{64} = 8.8$ and $10.26 \cdot \frac{145}{171} = 8.7$); $|c_A - c_X|^2 = 13.8^2 + 27.4176 = 217.8576 = 14.76^2$ and $|c_A - c_Y|^2 = 13.7^2 + 29.5776 = 217.2676 = 14.74^2$ (exact tangencies); and $|c_X - c_Y|^2 = 0.01 + (\sqrt{27.4176} + \sqrt{29.5776})^2 > 113.9 > 9.50^2 = 90.25$ with ample margin.

(V4) Hole facts. The hole of the 10 has radius 9.495; $4.99 + 4.50 = 9.49 \leq 9.495$ (the pair fits, two-circle exact) while $4.76 + 4.74 = 9.50 > 9.495$ and $5 + 4.50 = 9.50 > 9.495$ (blocked, two-circle exact).

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